MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MICROPROCESSOR, KEY CODE MJA

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic microprocessor module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specification and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirments. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 3.550 failures per million operating hours for class I or 12.316 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 6.4).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.4).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be MJA.

3.5.2 Size. The module size shall be 2A.

3.5.3 Weight. The module shall have a maximum weight of 70 grams.

3.5.4 Contact pins. The module shall have 80 contact pins positioned 1A1 through 1A40 and 2A1 through 2A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/306-1 for class I and M28787/306-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

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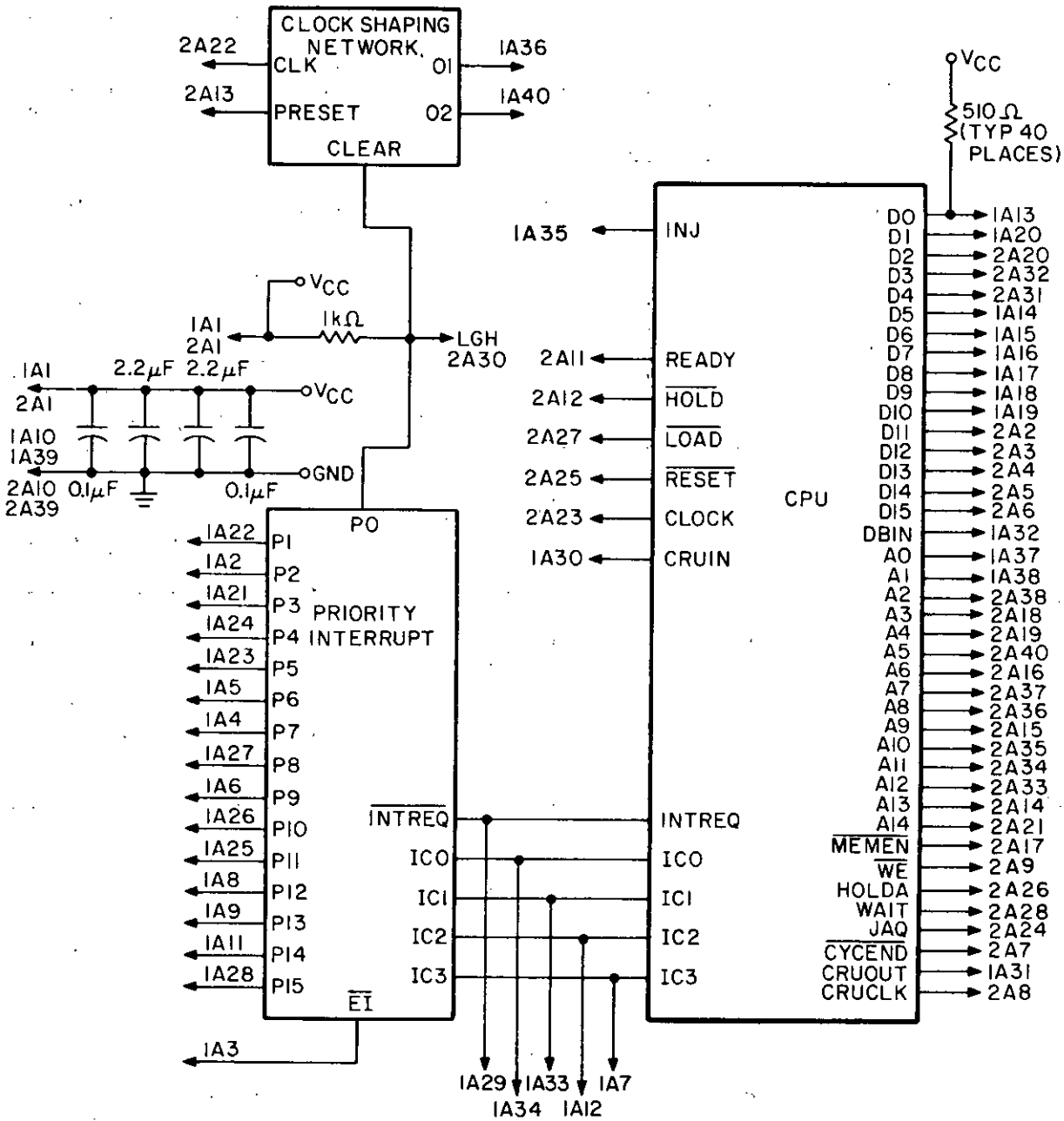


FIGURE 1. MJA module function diagram, microprocessor.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation													
	Initial	II	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	II	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B4	Continuity	III	X	0.0	0.1	0.0	0.1	0.0	0.1	0.0	0.1	0.0	0.1	Ω
C	Power supply stress check	IV	S	← Must pass all remaining tests →										
D1-D16	Input breakdown current	V	S	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	mA
D17	Input breakdown current	V	S	0.0	0.1	0.0	0.1	0.0	0.1	0.0	0.1	0.0	0.1	mA
D18	Input breakdown curret...	V	S	0.0	0.2	0.0	0.2	0.0	0.2	0.0	0.2	0.0	0.2	mA
E1, E3-E5	Output short circuit current	VI	X	-20	-55	-20	-55	-20	-55	-20	-55	-20	-55	mA
E2,E6,E7	Output short circuit current	VI	X	-20	-100	-20	-100	-20	-100	-20	-100	-20	-100	mA
E8-E47	Output short circuit current	VI	X	-9.1	-12.4	-9.1	-12.4	-9.1	-12.4	-9.1	-12.4	-9.1	-12.4	mA
F1-F40	Input clamp diode voltage	VII	X	-0.2	-1.5	-0.2	-1.5	-0.2	-1.5	-0.2	-1.5	-0.2	-1.5	V
G1	Injector supply voltage	VIII	X	0.71	1.32	0.71	1.32	0.71	1.32	0.71	1.32	0.71	1.32	V
G2	Injector supply voltage	VIII	X	0.74	1.37	0.74	1.37	0.74	1.37	0.74	1.37	0.74	1.37	V
G3	V_{CC} power supply current	VIII	X	20	175	20	175	20	175	20	175	20	175	mA
H1-H6,H8-H14,H16,H18	Input current, high level	IX	X	0	80	0	80	0	80	0	80	0	80	µA
H7,H15,H17	Input current, high level	IX	X	0	40	0	40	0	40	0	40	0	40	µA
H19	Input current, high level	IX	X	-2.6	-3.4	-2.6	-3.4	-2.6	-3.4	-2.6	-3.4	-2.6	-3.4	mA
H20-H23,H25	Input current, high level	IX	X	0	300	0	300	0	300	0	300	0	300	µA
H24	Input current, high level	IX	X	0	1000	0	1000	0	1000	0	1000	0	1000	µA
J1-J6,J8-J14, J16	Input current, low level	IX	X	-0.2	-3.50	-0.2	-3.50	-0.2	-3.50	-0.2	-3.50	-0.2	-3.50	mA
J7,J15,J18	Input current, low level	IX	X	-0.1	-1.75	-0.1	-1.75	-0.1	-1.75	-0.1	-1.75	-0.1	-1.75	mA
J17	Input current, low level	IX	X	-0.1	-0.88	-0.1	-0.88	-0.1	-0.88	-0.1	-0.88	-0.1	-0.88	mA
J19	Input current, low level	IX	X	-5.0	-10.8	-5.0	-10.8	-5.0	-10.8	-5.0	-10.8	-5.0	-10.8	mA
J20-J23,J25	Input current, low level	IX	X	0	-10	0	-10	0	-10	0	-10	0	-10	µA
J24	Input current, low level	IX	X	0	-20	0	-20	0	-20	0	-20	0	-20	µA

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
K1-K24	Output voltage, high level	X	X	5.22	5.55	5.22	5.55	5.22	5.55	5.22	5.55	5.22	5.55	V
K25-K40	Output voltage, high level	X	X	5.05	5.55	5.05	5.55	5.05	5.55	5.05	5.55	5.05	5.55	V
K41-K47	Output voltage, high level	X	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
L1-L47	Output voltage, low level	XI	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
M1-M28	Static functional test, priority interrupt and clock shaping	XII	X	← See tables XII and XIII →										
N1,N2	Dynamic functional test, minimum and maximum injector current	XIV	X	← See tables XIV and XV and 6.3 →										
P1-P40	Transition time (t_{THL}), C_L = 150 pF	XVI	Q	5	40	5	40	5	45	5	40	5	50	ns
P41,P43-P45	Transition time (t_{THL}), C_L = 50 pF	XVI	Q	4	50	4	50	4	55	4	50	4	60	ns
P42,P46,P47	Transition time (t_{THL}), C_L = 50 pF	XVI	Q	4	20	4	20	4	25	4	20	4	30	ns
Q1,Q3-Q5	Transition time (t_{TLH}), C_L = 50 pF	XVII	Q	10	80	10	80	10	90	10	80	10	100	ns
Q2,Q6,Q7	Transition time (t_{TLH}), C_L = 50 pF	XVII	Q	3	45	3	45	3	55	3	45	3	65	ns
R1	Propagation delay time (t_{PLH}), C_L = 50 pF	XVIII	Q	10	58	10	60	10	60	10	60	10	62	ns
R2	Propagation delay time (t_{PLH}), C_L = 50 pF	XVIII	Q	10	43	10	45	10	45	10	45	10	47	ns
R3-R18,P28, P42,R60	Propagation delay time (t_{PLH}), C_L = 50 pF	XVIII	Q	10	61	10	63	10	63	10	63	10	65	ns
R19,R34-R37,R47, R49,P52, R53,R64, R67,R70, R72	Propagation delay time (t_{PLH}), C_L = 50 pF	XVIII	Q	10	56	10	58	10	58	10	58	10	60	ns
R20-27	Propagation delay time (t_{PLH}), C_L = 50 pF	XVIII	Q	10	51	10	53	10	53	10	53	10	55	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
R29-R33, R38-R41, R43-R45, R50, R51, R54, R55, R58, R59, R61, R62, R65, R68, R71, R73, R75, R77, R79	Propagation delay time (t_{PLH}), C_L = 50 pF	XVIII	Q	10	46	10	48	10	48	10	48	10	50	ns
R46, R48, R56, R57, R63, R66, R69, R74, R76, R78	Propagation delay time (t_{PLH}), C_L = 50 pF	XVIII	Q	10	50	10	52	10	52	10	52	10	54	ns
R80-R83	Propagation delay time (t_{PLH}), C_L = 50 pF	XVIII	Q	5	29	5	31	5	31	5	31	5	33	ns
S1, S28, S42, S60	Propagation delay time (t_{PHL}), C_L = 50 pF	XIX	Q	10	68	10	71	10	71	10	71	10	74	ns
S2, S29, S43, S61	Propagation delay time (t_{PHL}), C_L = 50 pF	XIX	Q	10	38	10	41	10	41	10	41	10	44	ns
S3-S18	Propagation delay time (t_{PHL}), C_L = 50 pF	XIX	Q	10	48	10	51	10	51	10	51	10	54	ns
S19	Propagation delay time (t_{PHL}), C_L = 50 pF	XIX	Q	10	41	10	44	10	44	10	44	10	47	ns
S20-S27	Propagation delay time (t_{PHL}), C_L = 50 pF	XIX	Q	10	36	10	39	10	39	10	39	10	42	ns
S30-S33, S38-S41, S44, S45, S50, S51, S54, S55, S58, S59, S62, S65, S68, S71, S73, S75, S77, S79	Propagation delay time (t_{PHL}), C_L = 50 pF	XIX	Q	10	37	10	40	10	40	10	40	10	43	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
S34-S37, S47,S49, S52,S53, S64,S67, S70,S72	Propagation delay time (t_{PHL}), C_L = 50 pF	XIX	Q	10	63	10	66	10	66	10	66	10	69	ns
S46,S48, S56,S57, S63,S66, S69,S74, S76,S78	Propagation delay time (t_{PHL}), C_L = 50 pF	XIX	Q	10	42	10	45	10	45	10	45	10	48	ns
S80-S83	Propagation delay time (t_{PHL}), C_L = 50 pF	XIX	Q	5	44	5	47	5	47	5	47	5	50	ns
T1,T2	Dynamic functional test and propagation delay time check	XX	Q	← See tables XX and XXI and 6.3 →										
U1,U3	Maximum frequency check, clock shaping (PRP2)	XXII	Q	201	207	201	207	201	207	201	207	201	207	ns
U2,U4	Maximum frequency check, clock shaping (PWA2)	XXII	Q	52	69	52	69	52	69	52	69	52	69	ns
V1-V31	Output crosstalk, low level	XXIII	Q	0	400	0	400	0	400	0	400	0	400	mV
W1-W5	Output crosstalk, high level	XXIII	Q	0	400	0	400	0	400	0	400	0	400	mV
X	Power supply transient amplitude	XXIV	Q	0	400	0	400	0	400	0	400	0	400	mV
Y1	Power supply capacitance, V_{CC}	XXV	S	3.22	5.98	3.22	5.98	3.22	5.98	3.22	5.98	3.22	5.98	µF
Y2	Power supply capacitance, INJ	XXV	S	200	1500	200	1500	200	1500	200	1500	200	1500	pF
Z1-Z16, Z20-Z25	Input capacitance	XXV	Q	2	20	2	20	2	20	2	20	2	20	pF
Z17-Z19	Input capacitance	XXV	Q	2	40	2	40	2	40	2	40	2	40	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through Z shall be performed in any order with the following exceptions. If test conditions A through F are part of a sequence of tests, the order of testing shall be as follows: A through F where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through Z in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B and E through X shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 6. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition Y1, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests B and E through X shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C and E through Z shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 603) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and may be obtained upon request.

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6.4 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

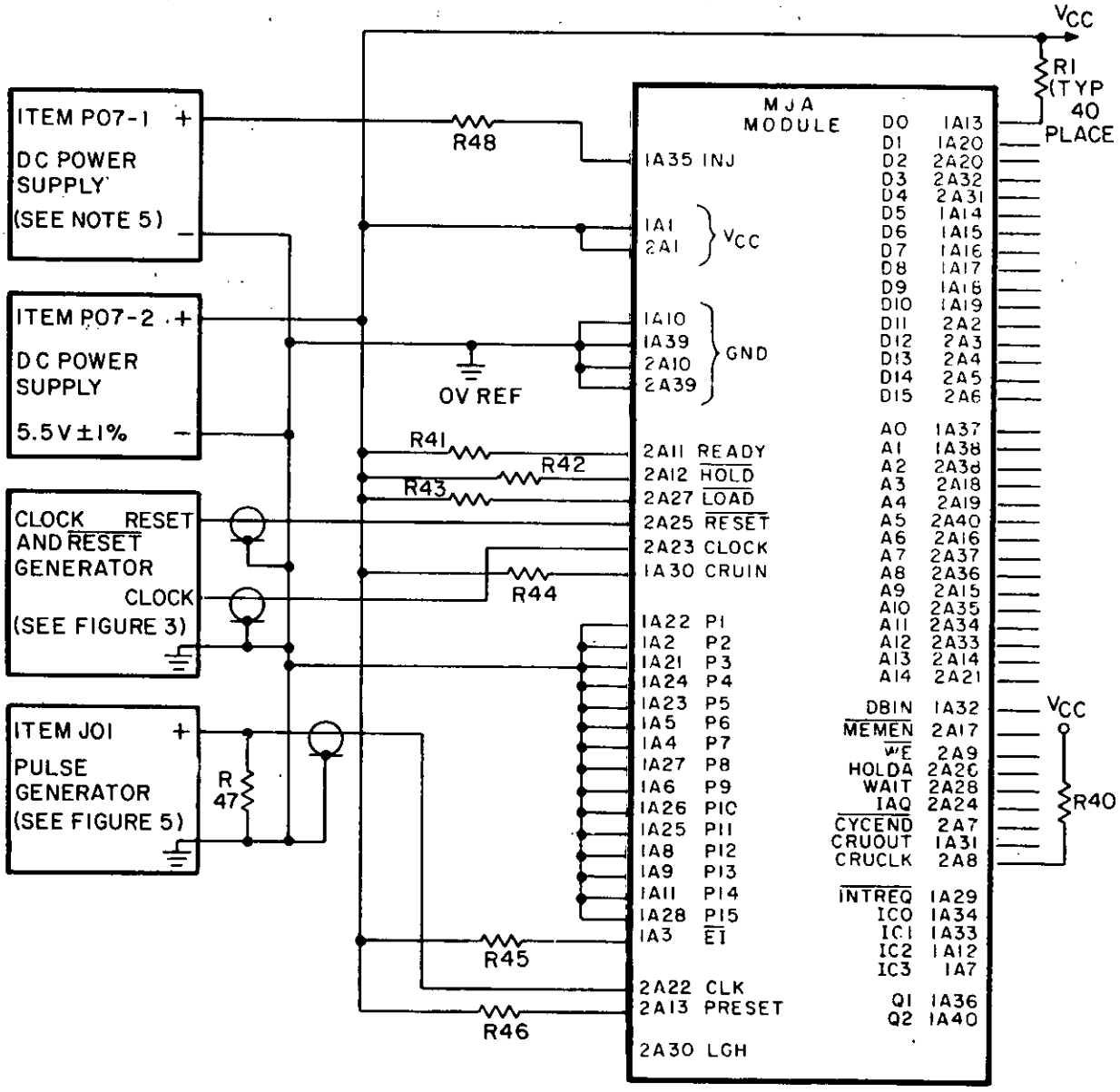
6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Agent:
DLA - ES

Preparing activity:
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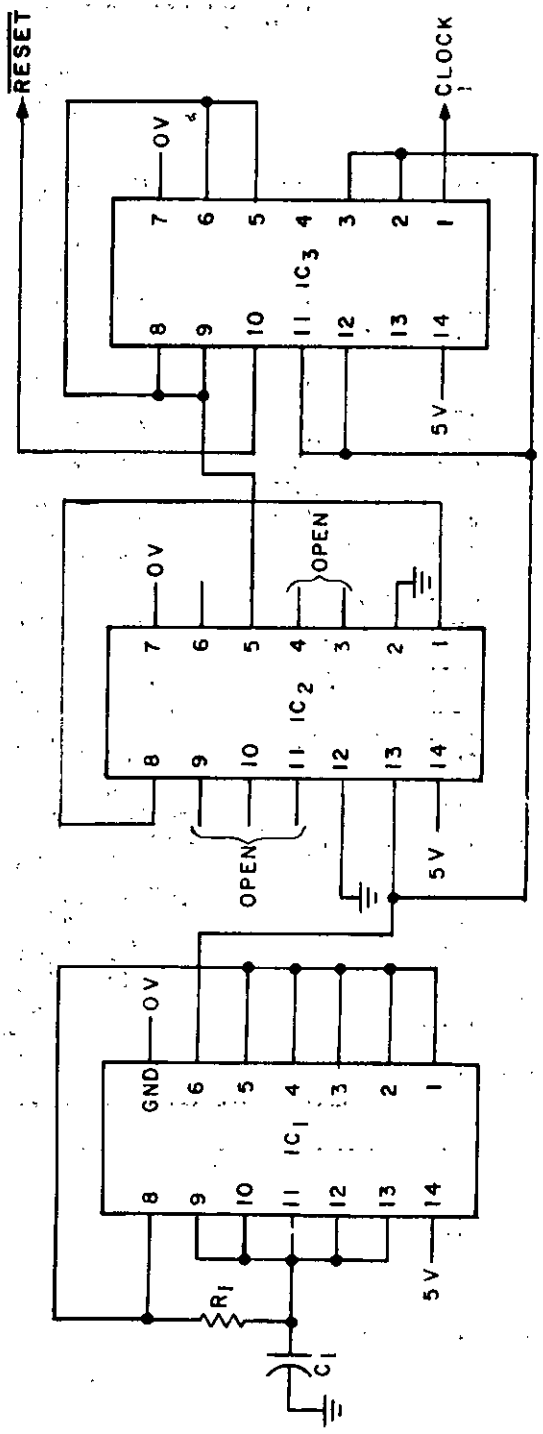
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NOTES:
1. R1-R40 = 536Ω ±1%, 1/8 W.
2. R41-R46 = 1.0 kΩ ±5%, 1/8 W.
3. R47 = 51Ω ±5%, 1/2 W.
4. R48 = 4Ω ±1%, 2 W.
5. Adjust item P07-1 voltage to limit pin 1A35 current to 550 mA ±1%.

FIGURE 2. Operating temperature and life test circuit.

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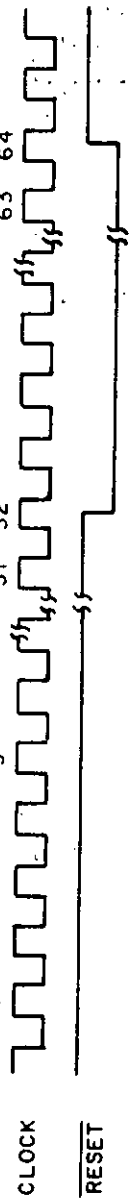
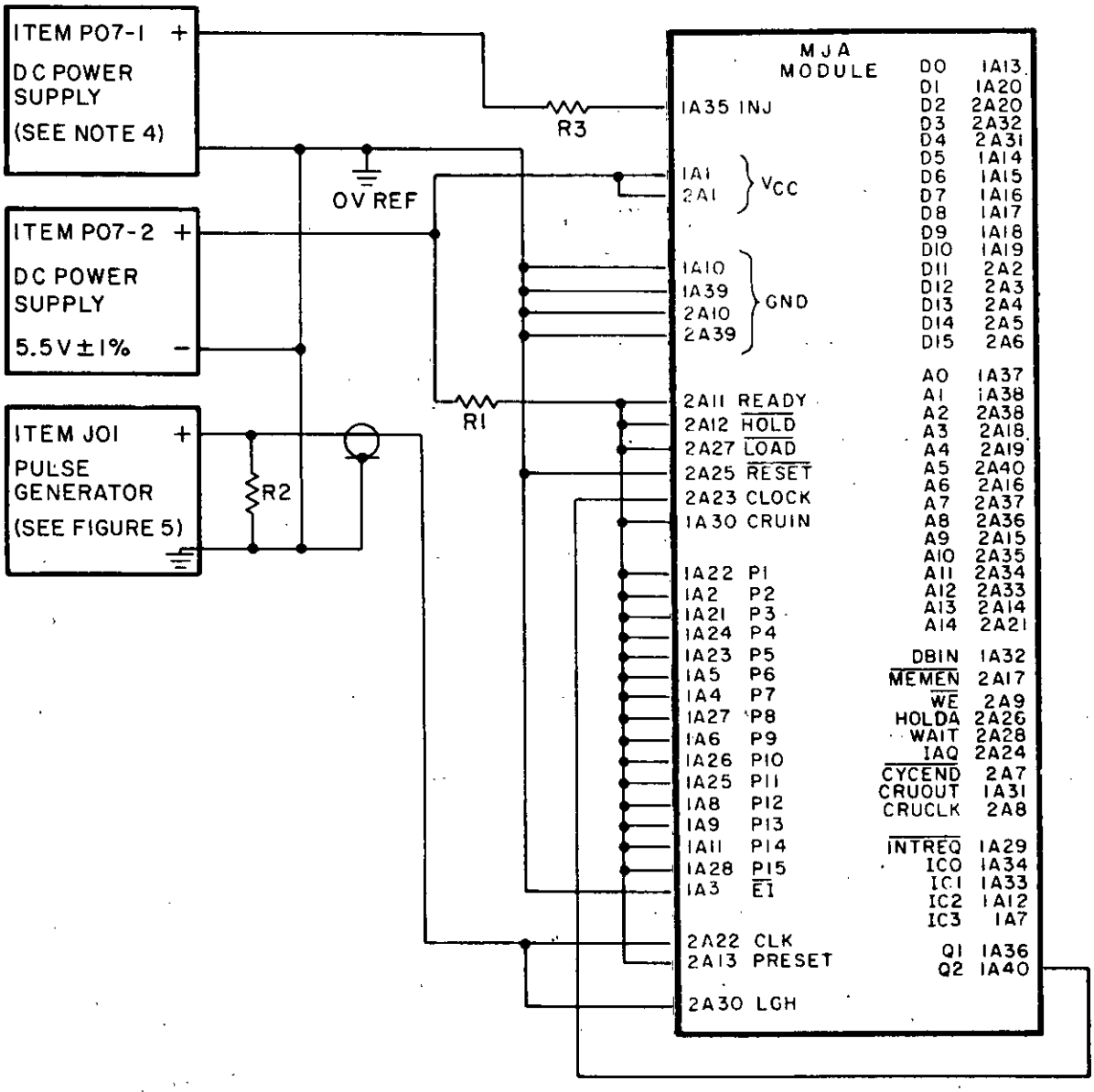


FIGURE 3. CLOCK and RESET generator circuit.

NOTES:
1. R1 = 390Ω ±5%, 1/4 W.
2. C1 = .001 µF, +20%, -10%.
3. IC1 = 5413 or equivalent.
4. IC2 = 54393 or equivalent.
5. IC3 = 54128 or equivalent.

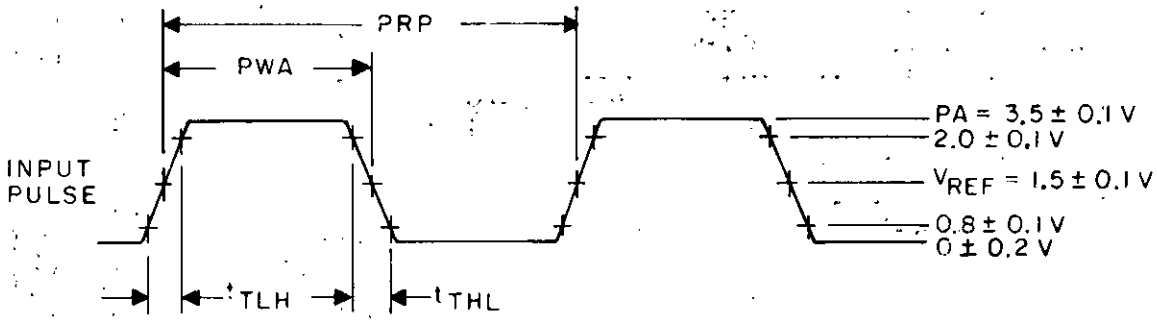
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NOTES:
1. R1 = 1.0 kΩ ±5%, 1/8 W.
2. R2 = 51Ω ±5%, 1/2 W.
3. R3 = 4Ω ±1%, 2 W.
4. Adjust item P07-1 voltage to limit pin 1A35 current to 550 mA ±1%.

FIGURE 4. Component temperature test circuit.

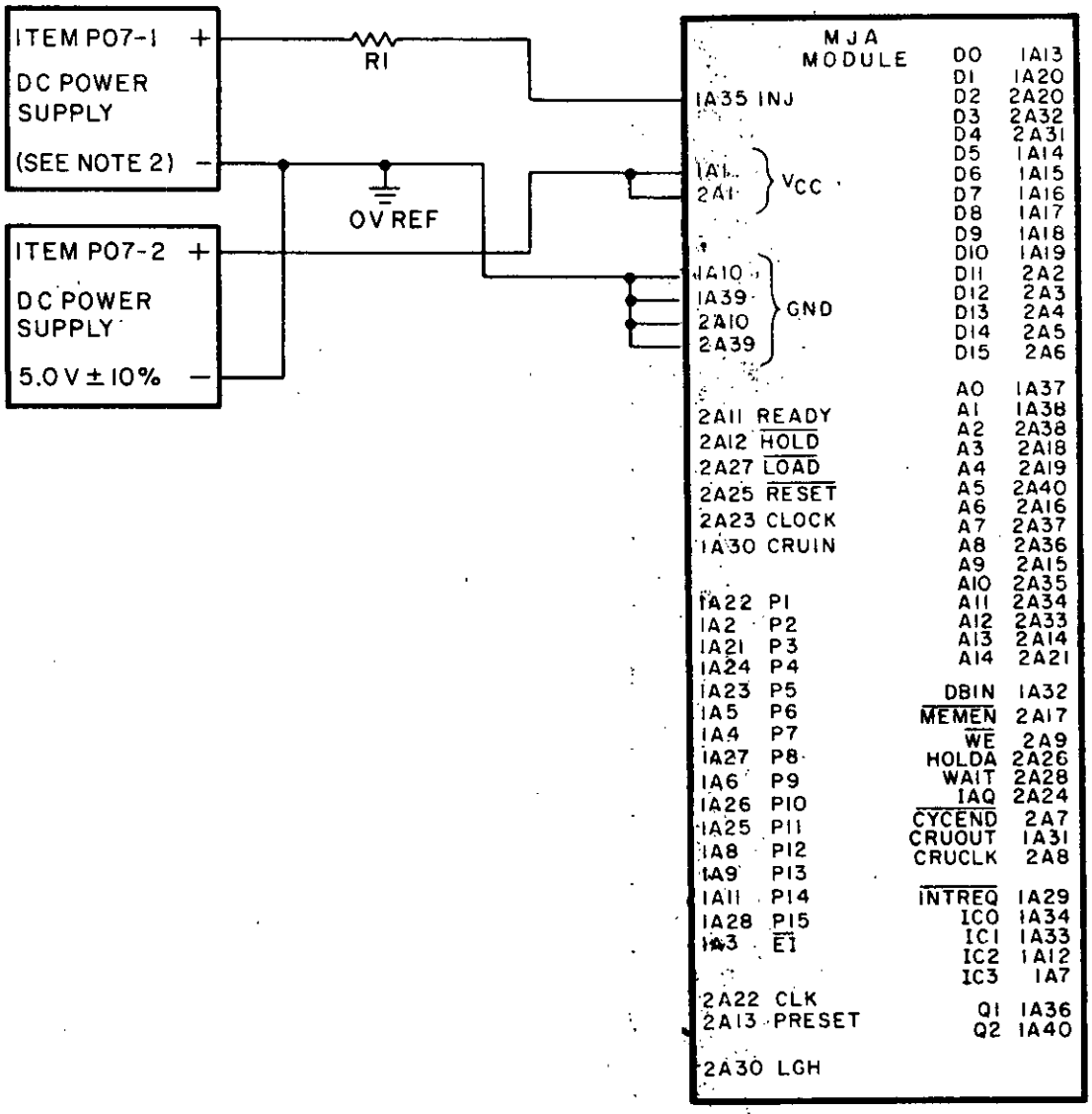
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NOTE: Input pulse parameters:
$t_{TLH} = t_{THL} = 15 \pm 5$ ns.
PWA = 45 ±2.5 ns.
PRP = 111 ±5.6 ns.

FIGURE 5. Operating and component temperature and life test circuit waveform for figures 2 and 4.

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NOTES:
1. R1 = 4Ω ±1%, 2 W.
2. Adjust item PO7-1 voltage to limit pin 1A35 current to 500 mA ±1%.

FIGURE 6. Vibration test circuit.

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TABLE II. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°+5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	C	Frame	10	∞	MΩ	7

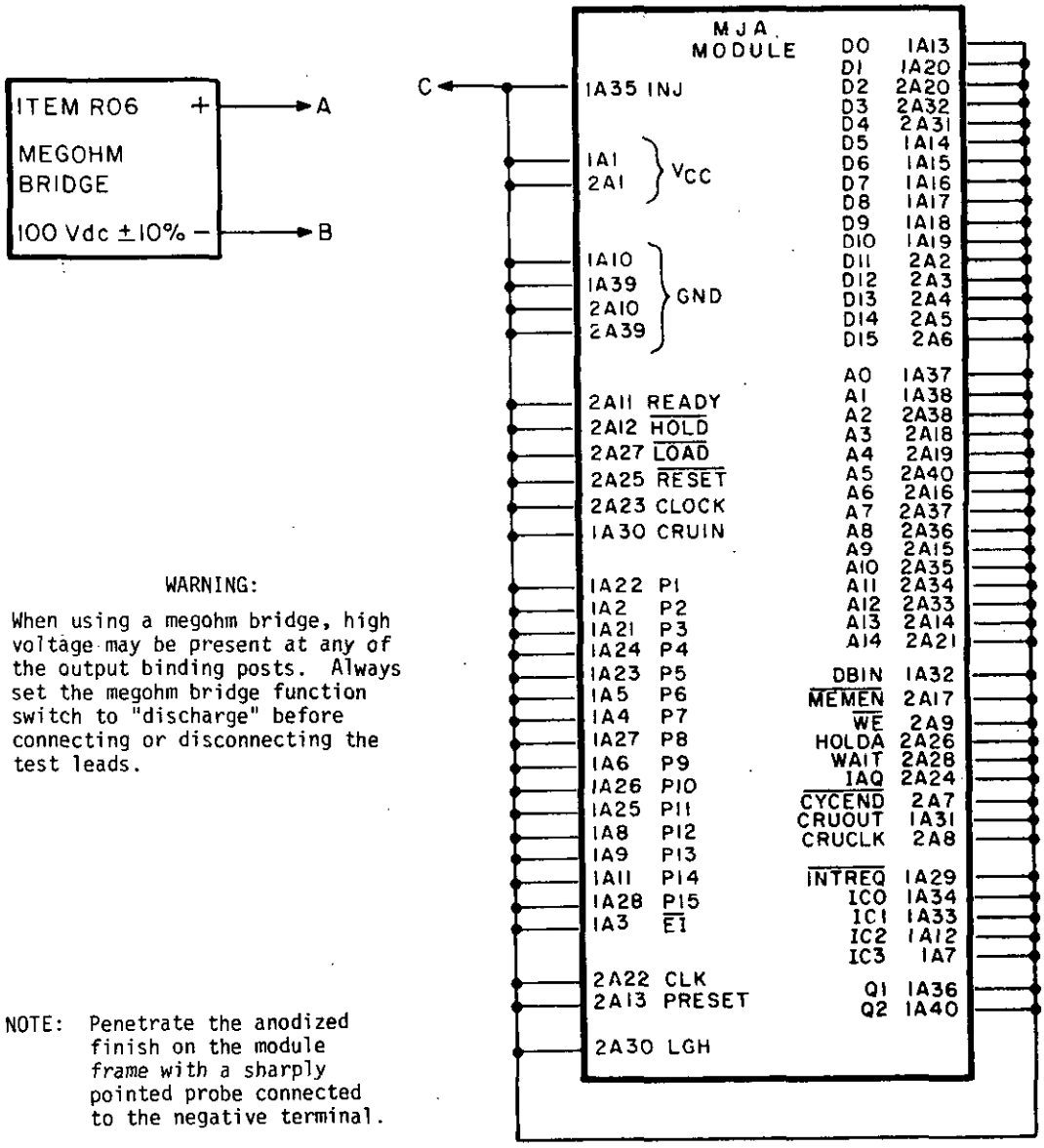
TABLE III. Continuity test.

Test condition	Parameter	Connect		Initial limits 25°+5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	Continuity	1A10	2A10	0.0	0.1	Ω	8
B2	↓	1A10	1A39	↓	↓	↓	↓
B3		1A10	2A39				
B4		1A1	2A1				

TABLE IV. Power supply stress check test.

Test condition	Parameter	Connect		Initial limits 25°+5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C	Power supply stress check	1A30,2A11,2A12, 2A23,2A25,2A27	1A2,1A3,1A4, 1A5,1A6,1A8, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A27, 1A28,2A13,2A22, 2A30	Module must pass all remaining tests			9

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 7. Isolation test circuit for table II.

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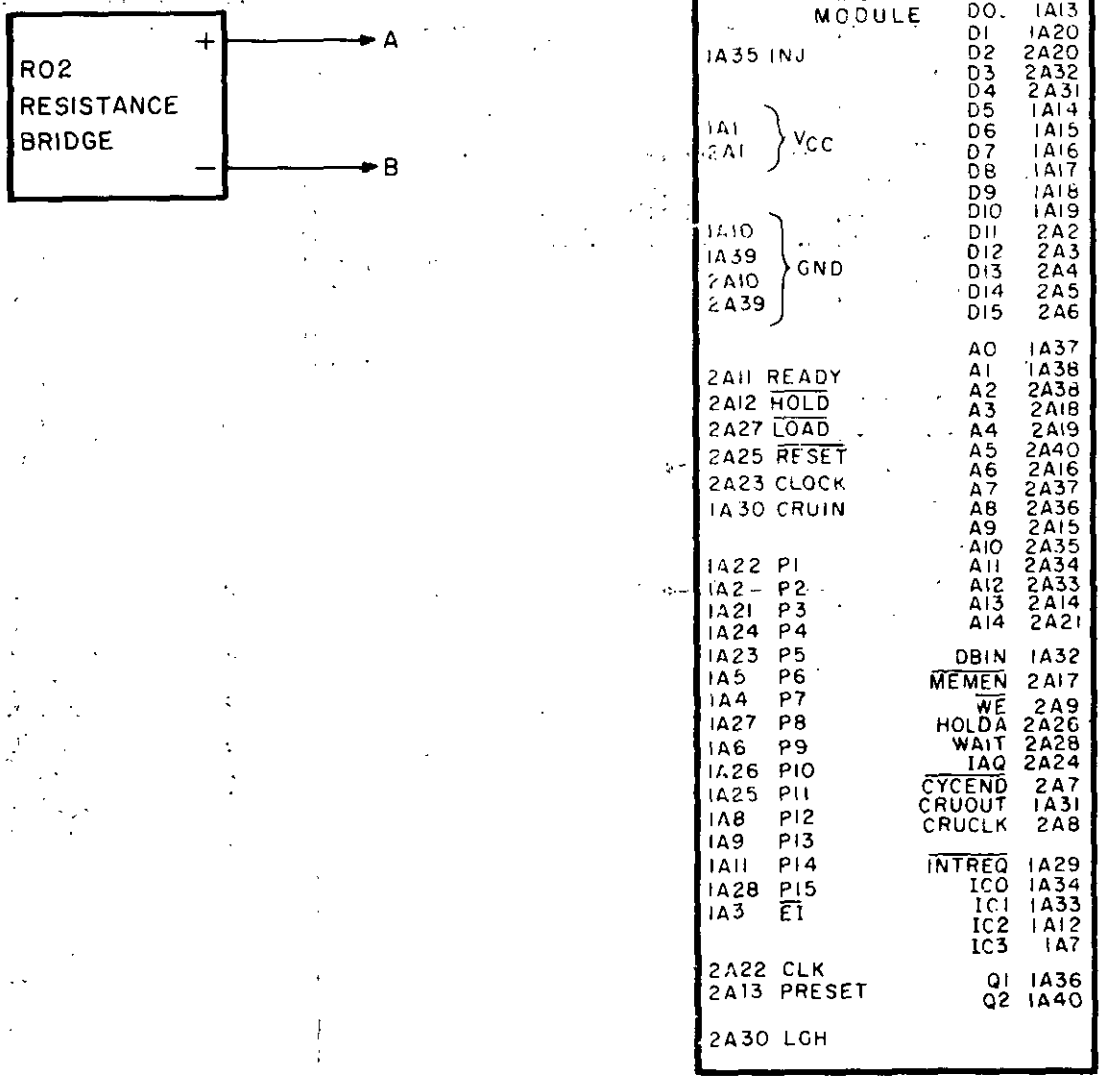
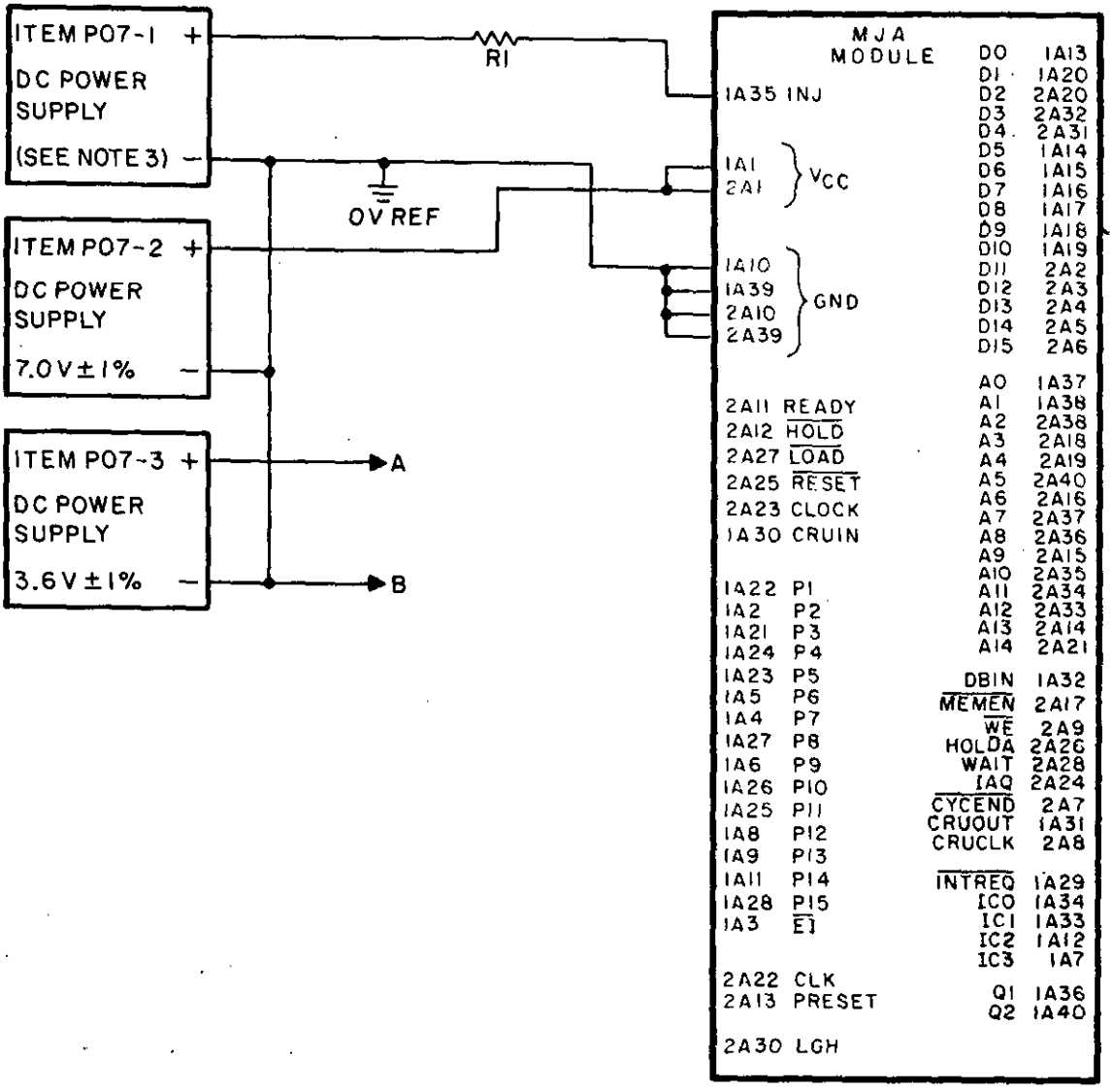


FIGURE 8. Continuity test circuit for table III.

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NOTES:
1. R1 = 4Ω ±1%, 2 W.
2. Hold this test condition for a minimum of 5 seconds and a maximum of 10 seconds.
3. Adjust item P07-1 voltage to limit pin 1A35 current to 750 mA ±1%.

FIGURE 9. Power supply stress check test circuit for table IV.

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TABLE V. Input breakdown current test.

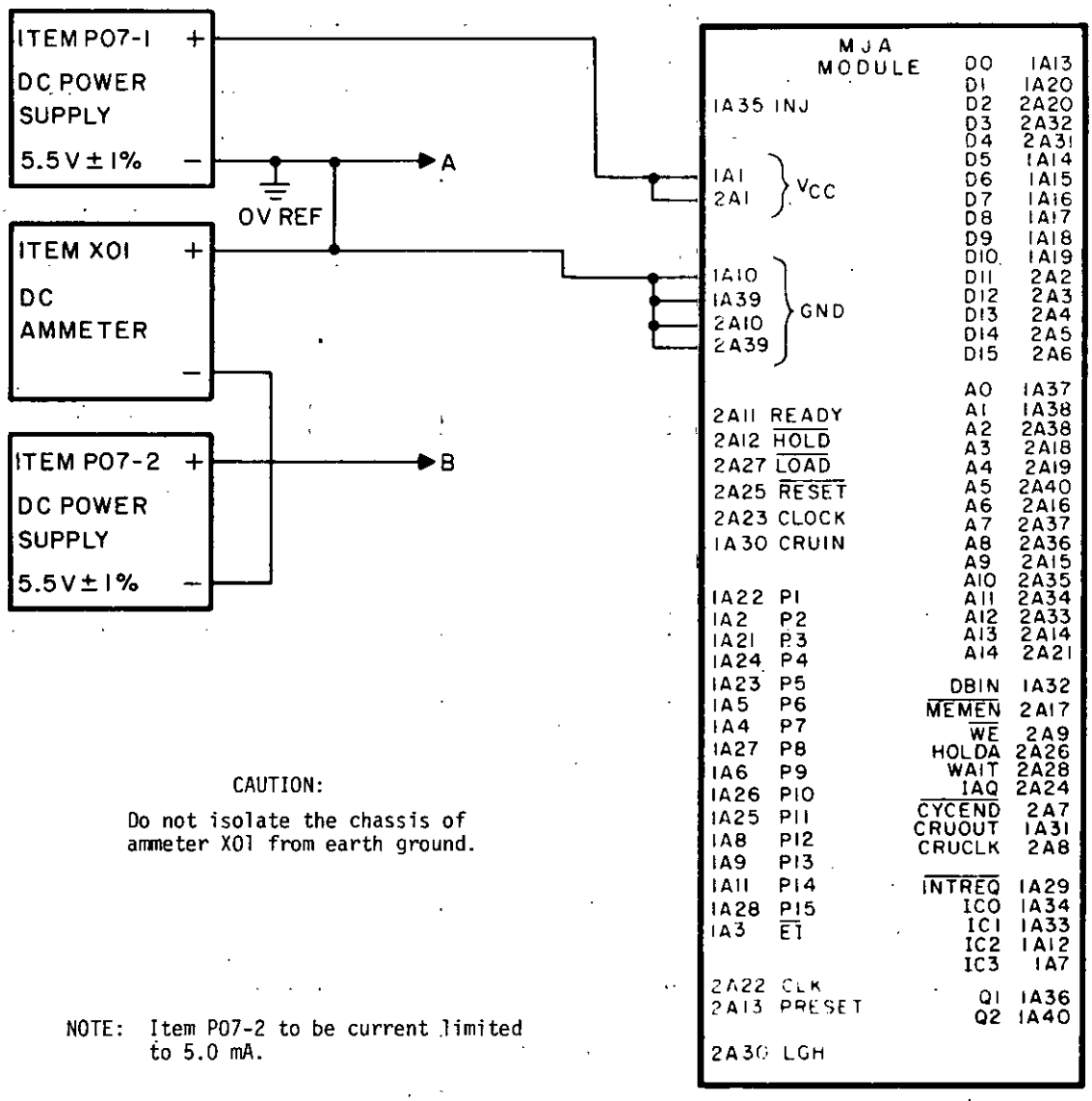
Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
D1	P1	Input breakdown current	1A2,1A3,1A4,1A5,1A6,1A8, 1A9,1A11,1A21,1A23,1A24, 1A25,1A26,1A27,1A28	1A22	0.0	1.0	mA	10
D2	P2		1A3,1A4,1A5,1A6,1A8,1A9, 1A11,1A21,1A22,1A23, 1A24,1A25,1A26,1A27, 1A28	1A2				
D3	P3		1A2,1A3,1A4,1A5,1A6,1A8, 1A9,1A11,1A22,1A23,1A24, 1A25,1A26,1A27,1A28	1A21				
D4	P4		1A2,1A3,1A4,1A5,1A6,1A8, 1A9,1A11,1A21,1A22,1A23, 1A25,1A26,1A27,1A28	1A24				
D5	P5		1A2,1A3,1A4,1A5,1A6,1A8, 1A9,1A11,1A21,1A22,1A24, 1A25,1A26,1A27,1A28	1A23				
D6	P6		1A2,1A3,1A4,1A6,1A8,1A9, 1A11,1A21,1A22,1A23, 1A24,1A25,1A26,1A27,1A28	1A5				
D7	P7		1A2,1A3,1A5,1A6,1A8,1A9, 1A11,1A21,1A22,1A23, 1A24,1A25,1A26,1A27,1A28	1A4				
D8	P8		1A2,1A3,1A4,1A5,1A6,1A8, 1A9,1A11,1A21,1A22,1A23, 1A24,1A25,1A26,1A28	1A27				
D9	P9		1A2,1A3,1A4,1A5,1A8,1A9, 1A11,1A21,1A22,1A23, 1A24,1A25,1A26,1A27,1A28	1A6				

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D10	P10 Input breakdown current	1A2,1A3,1A4,1A5,1A6,1A8, 1A9,1A11,1A21,1A22,1A23, 1A24,1A25,1A27,1A28	1A26	0.0	1.0	mA	10
D11	P11	1A2,1A3,1A4,1A5,1A6,1A8, 1A9,1A11,1A21,1A22,1A23, 1A24,1A26,1A27,1A28	1A25				
D12	P12	1A2,1A3,1A4,1A5,1A6,1A9, 1A11,1A21,1A22,1A23, 1A24,1A25,1A26,1A27,1A28	1A8				
D13	P13	1A2,1A3,1A4,1A5,1A6,1A8, 1A11,1A21,1A22,1A23, 1A24,1A25,1A26,1A27,1A28	1A9				
D14	P14	1A2,1A3,1A4,1A5,1A6,1A8, 1A9,1A21,1A22,1A23, 1A24,1A25,1A26,1A27, 1A28	1A11				
D15	P15	1A2,1A3,1A4,1A5,1A6,1A8, 1A9,1A11,1A21,1A22, 1A23,1A24,1A25,1A26, 1A27	1A28				
D16	$\overline{EI}$	1A2,1A4,1A5,1A6,1A8,1A9, 1A11,1A21,1A22,1A23, 1A24,1A25,1A26,1A27,1A28	1A3				
D17	CLK	2A30	2A22		0.1		
D18	PRESET	Open	2A13		0.2		

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTE: Item P07-2 to be current limited to 5.0 mA.

FIGURE 10. Input breakdown current test circuit for table V.

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TABLE VI. Output short circuit current test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E1	$\overline{\text{INTREQ}}$ Output short circuit current (interrupt section)	1A3	1A29	-20	-55	mA	11
E2	IC0	1A3	1A34	-20	-100		
E3	IC1	1A3	1A33	-20	-55		
E4	IC2	1A3	1A12	-20	-55		
E5	IC3	1A3	1A7	-20	-55		
E6	Q1 Output short circuit current (clock shaping section)	2A13	1A36	-20	-100		
E7	Q2 Output short circuit current (clock shaping section)	2A13	1A40	-20	-100		
E8	D0 Output short circuit current (microprocessor)	Open	1A13	-9.1	-12.4		
E9	D1		1A20				
E10	D2		2A20				
E11	D3		2A32				
E12	D4		2A31				
E13	D5		1A14				
E14	D6		1A15				
E15	D7		1A16				
E16	D8		1A17				

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TABLE VI. Output short circuit current test - Continued.

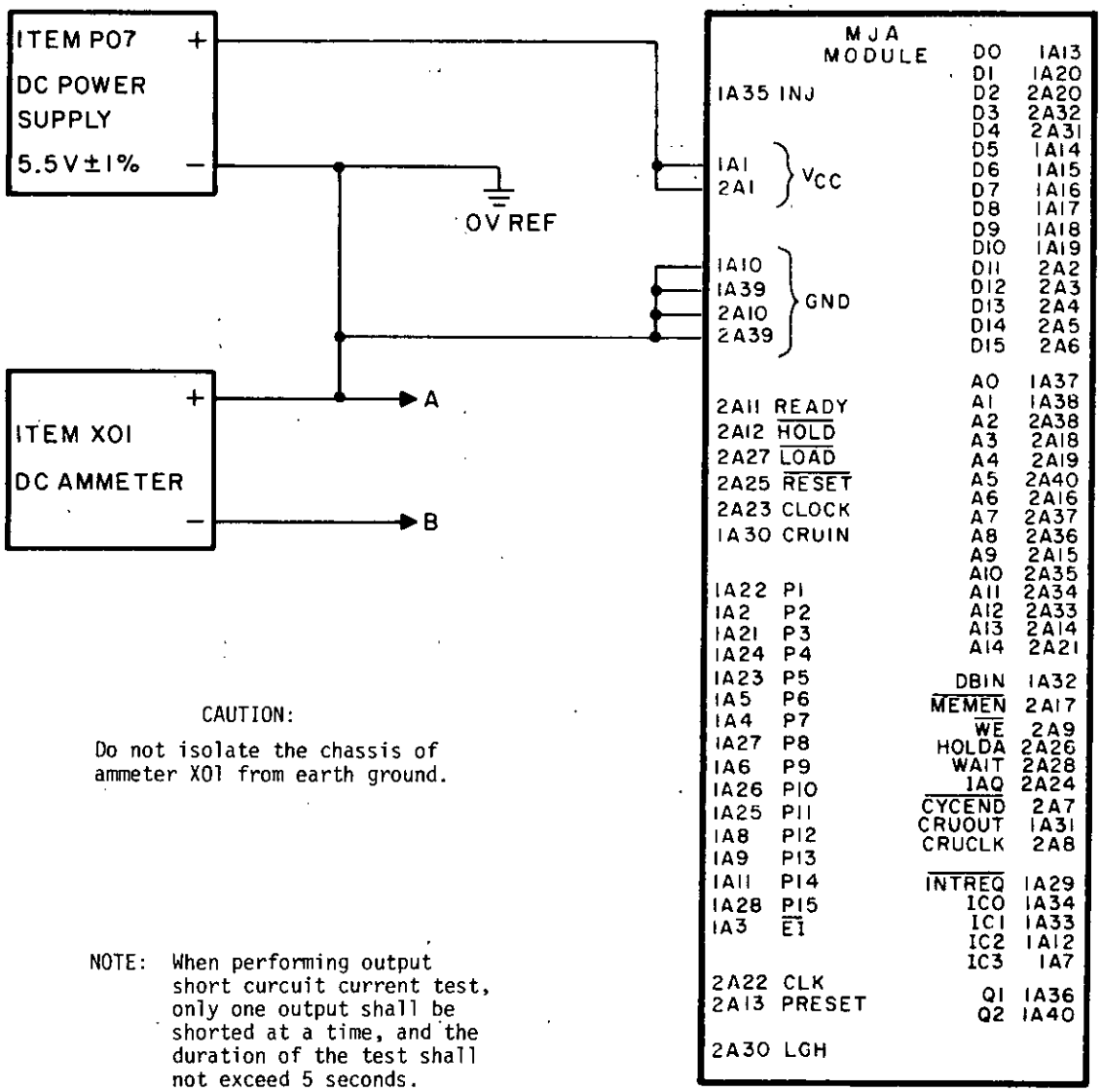
Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
E17	D9	Output short circuit current (microprocessor)	Open	1A18	-9.1	-12.4	mA	11
E18	D10			1A19				
E19	D11			2A2				
E20	D12			2A3				
E21	D13			2A4				
E22	D14			2A5				
E23	D15			2A6				
E24	A0			1A37				
E25	A1			1A38				
E26	A2			2A38				
E27	A3			2A18				
E28	A4			2A19				
E29	A5			2A40				
E30	A6			2A16				
E31	A7			2A37				
E32	A8			2A36				
E33	A9			2A15				

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TABLE VI. Output short circuit current test - Continued.

Test condition	Parameter	Connect		Initial limits 25°+5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E34	A10 Output short circuit current (microprocessor)	Open	2A35	-9.1	-12.4	mA	11
E35	A11		2A34				
E36	A12		2A33				
E37	A13		2A14				
E38	A14		2A21				
E39	DBIN		1A32				
E40	$\overline{\text{MEMEN}}$		2A17				
E41	$\overline{\text{WE}}$		2A9				
E42	HOLDA		2A26				
E43	WAIT		2A28				
E44	IAQ		2A24				
E45	$\overline{\text{CYCEND}}$		2A7				
E46	CRUOUT		1A31				
E47	CRUCLK		2A8				

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CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

NOTE: When performing output short curcuit current test, only one output shall be shorted at a time, and the duration of the test shall not exceed 5 seconds.

FIGURE 11. Output short circuit test circuit for table VI.

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TABLE VII. Input clamp diode voltage test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F1	P1 Input clamp diode voltage	Open	1A22	Open	Open	-0.2	-1.5	V	12
F2	P2		1A2						
F3	P3		1A21						
F4	P4		1A24						
F5	P5		1A23						
F6	P6		1A5						
F7	P7		1A4						
F8	P8		1A27						
F9	P9		1A6						
F10	P10		1A26						
F11	P11		1A25						
F12	P12		1A8						
F13	P13		1A9						
F14	P14		1A11						
F15	P15		1A28						
F16	$\overline{EI}$		1A3						
F17	CLK		2A22						
F18	PRESET		2A13						
F19	READY		2A11						
F20	$\overline{HOLD}$		2A12						

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TABLE VII. Input clamp diode voltage test - Continued.

Test condition	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
F21	$\overline{\text{LOAD}}$	Input clamp diode voltage	Open	2A27	Open	Open	-0.2	-1.5	V	12
F22	$\overline{\text{RESET}}$			2A25						
F23	CLOCK			2A23						
F24	CRUIN			1A30						
F25	D0		1A3	1A13	2A23	2A25				12,13
F26	D1			1A20						
F27	D2			2A20						
F28	D3			2A32						
F29	D4			2A31						
F30	D5			1A14						
F31	D6			1A15						
F32	D7			1A16						
F33	D8			1A17						
F34	D9			1A18						
F35	D10			1A19						
F36	D11			2A2						
F37	D12			2A3						
F38	D13			2A4						
F39	D14			2A5						
F40	D15			2A6						

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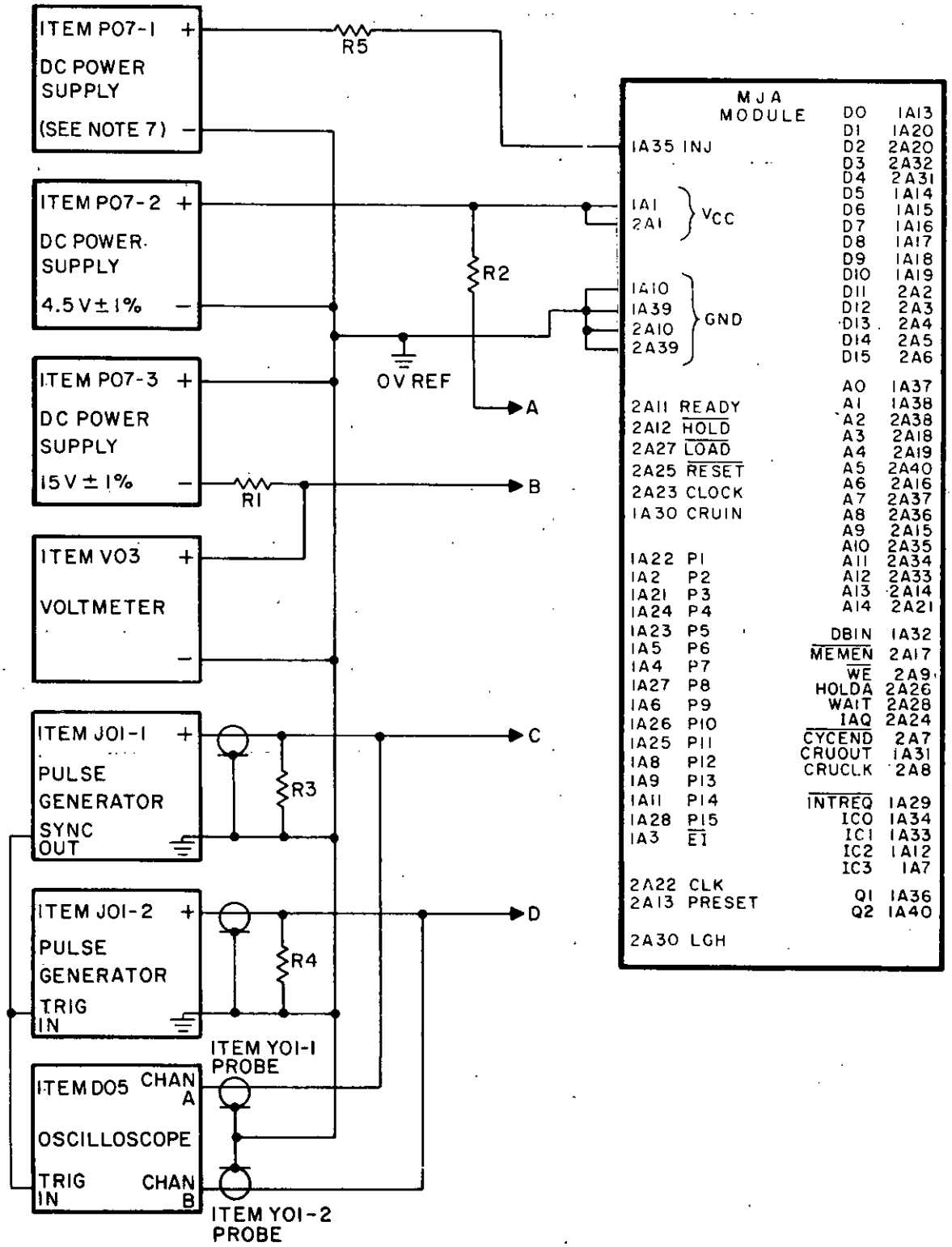


FIGURE 12. Input clamp diode voltage test circuit for table VII.

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NOTES:
1. R1 = 1.13 kΩ ±1%, 1/4 W for test conditions F1-F16 and F19-F40.
2. R1 = 750Ω ±1%, 1/4 W for test conditions F17 and F18.
3. R2 = 1.0 kΩ ±5%, 1/4 W.
4. R3, R4 = 51Ω ±5%, 1/2 W.
5. R5 = 4Ω ±1%, 2W.
6. For test conditions F25-F40 use the CPU preconditioning flow chart on figure 13 to set the CPU DATA lines into the input mode.
7. Adjust item P07-1 voltage to limit pin 1A35 current to 450 mA ±1%.

FIGURE 12. Input clamp diode voltage test circuit for table VII - Continued.

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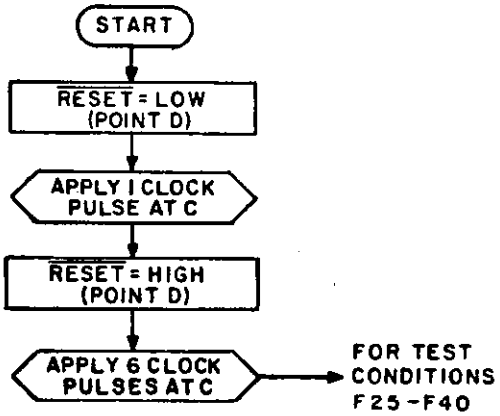
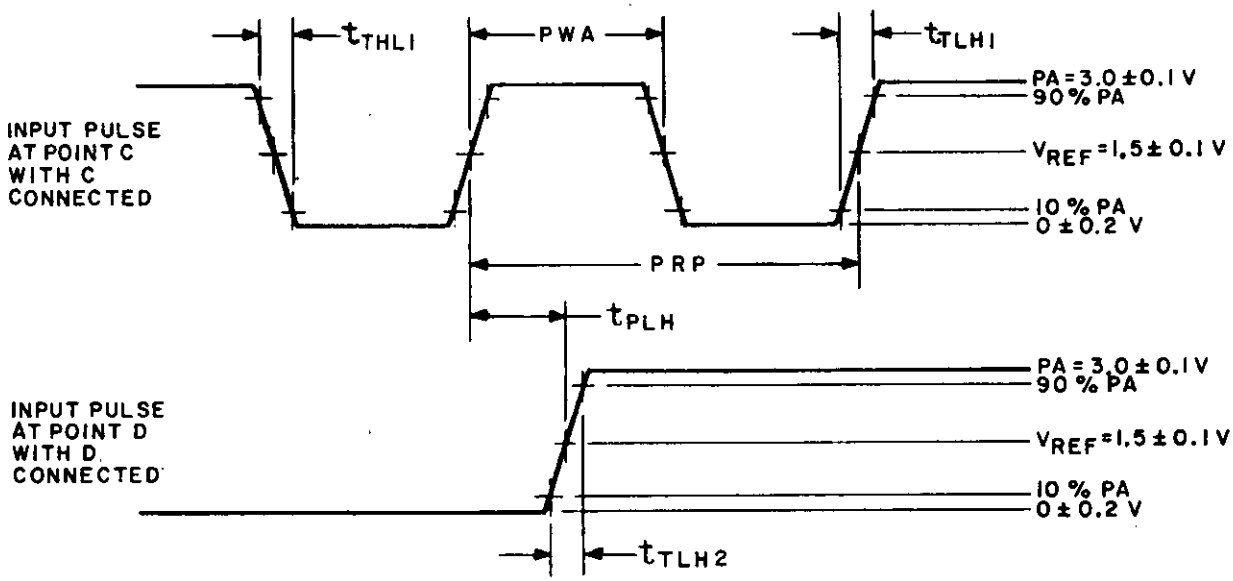


FIGURE 13. CPU preconditioning flow chart and waveforms (for test conditions F25-F40) for table VII.

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NOTES:
1. Input pulse parameters:

   a. $t_{TLH1} = t_{THL1} = t_{TLH2} = 20 \pm 2$ ns.
      PWA = 250 ns minimum.
      PRP = 2 x PWA minimum.
      t_{PLH} = 1/2 x PWA ±10%.

   b. The CPU can be operated in a static mode and thus PWA has no maximum limit and PRP can be much more than two times PWA.

2. Only one pass through the flow chart is required for test conditions F25-F40.

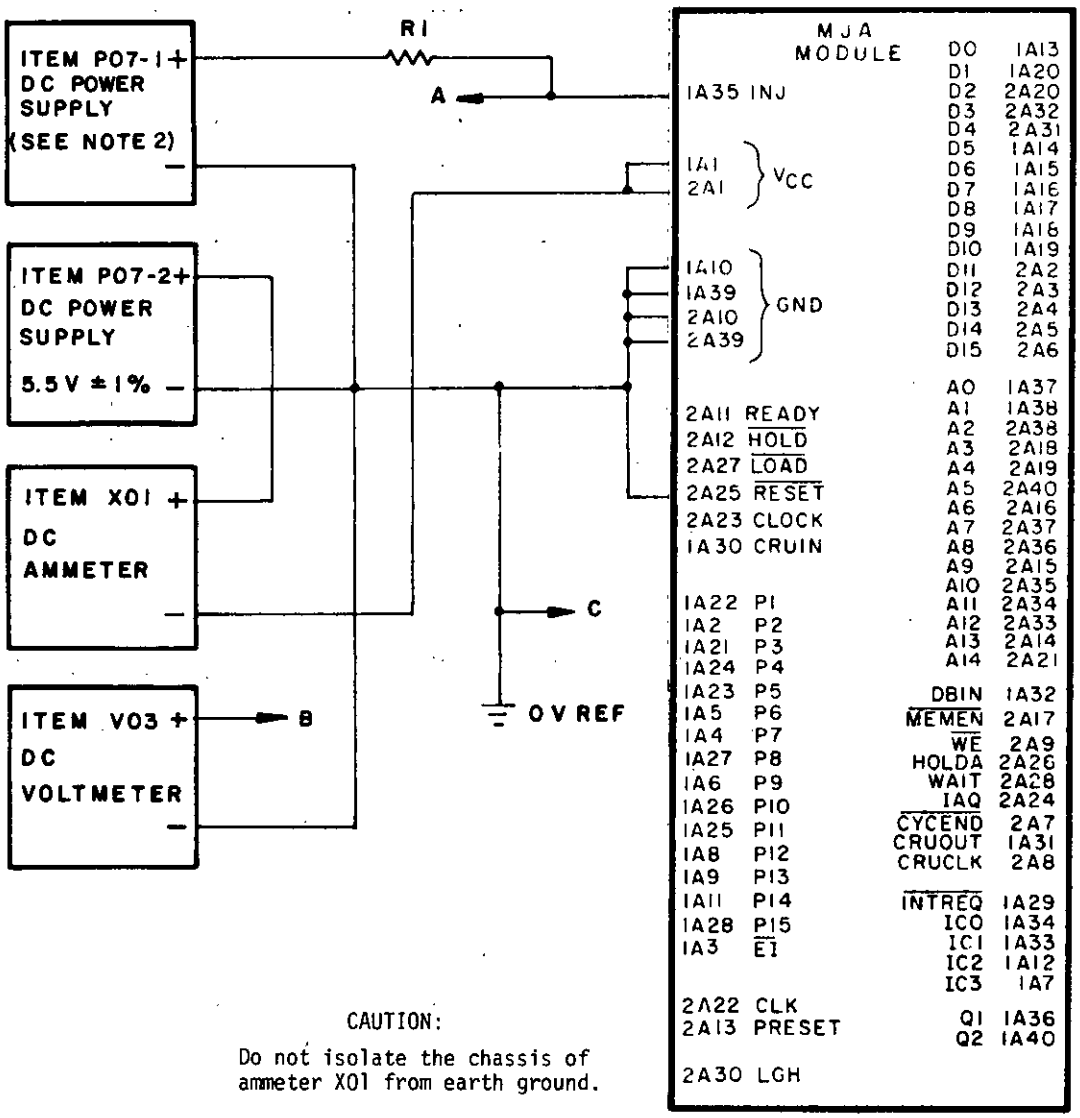
FIGURE 13. CPU preconditioning flow chart and waveforms (for test conditions F25-F40) for table VII - Continued.

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TABLE VIII. Injector supply voltage and power supply current test.

Test condition	Parameter	Connect			Initial limits 25°+5°C ambient		Unit	Test figure.
		A to pin	B to pin	C to pin	Min	Max		
G1	Injector supply voltage	B	A	Open	0.71	1.32	V	14
G2	Injector supply voltage	B	A	Open	0.74	1.37	V	14
G3	V_{CC} power supply current	Open	Open	2A30,1A3	20	175	mA	14

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. R1 = 4Ω ±1%, 2 W.
2. Adjust item P07-1 voltage to limit pin 1A35 current to 450 mA ±1% for G1 and 550 mA ±1% for G2. Set to 500 mA for G3.

FIGURE 14. Injector supply voltage and power supply current test circuit for table VIII.

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TABLE IX. Input current (low-level and high-level) test.

Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
H1	P1	Input current, high level	1A22	Open	0	80	µA	15
H2	P2		1A2					
H3	P3		1A21					
H4	P4		1A24					
H5	P5		1A23					
H6	P6		1A5					
H7	P7		1A4			40		
H8	P8		1A27			80		
H9	P9		1A6					
H10	P10		1A26					
H11	P11		1A25					
H12	P12		1A8					
H13	P13		1A9					
H14	P14		1A11					
H15	P15		1A28			40		
H16	$\overline{EI}$		1A3			80		
H17	CLK		2A22			40		
H18	PRESET		2A13			80		

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TABLE IX. Input current (low-level and high-level) test - Continued.

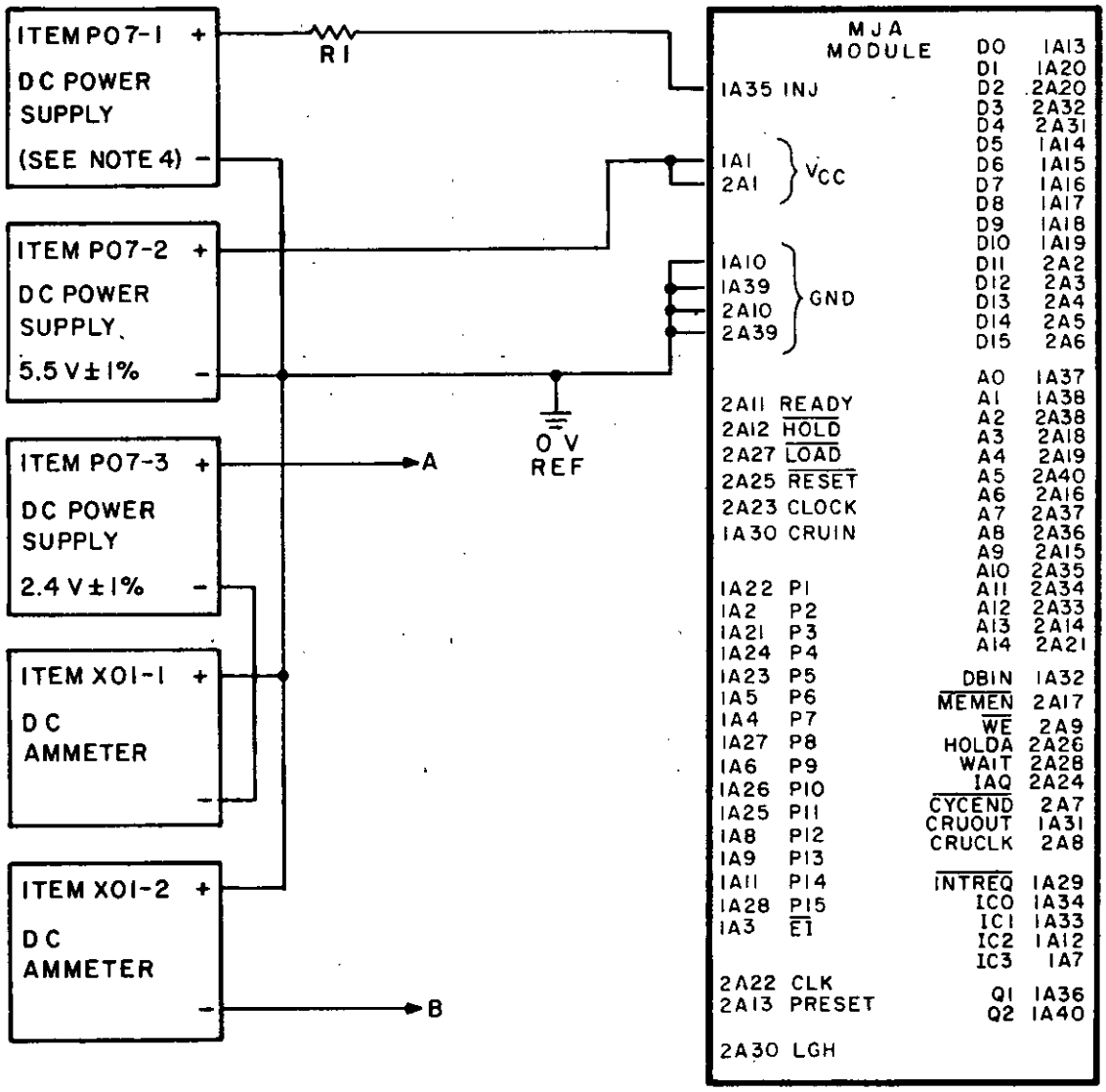
Test condition	Parameter		Connect A to pin	Connect B to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
H19	LGH	Input current, high level	2A30	Open	-2.6	-3.4	mA	15
H20	READY		2A11		0	300	µA	
H21	$\overline{\text{HOLD}}$		2A12					
H22	$\overline{\text{LOAD}}$		2A27					
H23	$\overline{\text{RESET}}$		2A25					
H24	CLOCK		2A23			1000		
H25	CRUIN		1A30			300		
J1	P1	Input current, low level	Open	1A22	-0.2	-3.50	mA	
J2	P2			1A2				
J3	P3			1A21				
J4	P4			1A24				
J5	P5			1A23				
J6	P6			1A5				
J7	P7			1A4	-0.1	-1.75		
J8	P8			1A27	-0.2	-3.50		
J9	P9			1A6				
J10	P10			1A26				
J11	P11			1A25				

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TABLE IX. Input current (low-level and high-level) test - Continued.

Test condition	Parameter		Connect A to pin	Connect B to pin	Initial limits 25°+5°C ambient Min	Initial limits 25°+5°C ambient Max	Unit	Test figure
J12	P12	Input current, low level	Open	1A8	-0.2	-3.50	mA	15
J13	P13			1A9	-0.2	-3.50		
J14	P14			1A11	-0.2	-3.50		
J15	P15			1A28	-0.1	-1.75		
J16	$\overline{EI}$			1A3	-0.2	-3.50		
J17	CLK			2A22	-0.1	-0.88		
J18	PRESET			2A13	-0.1	-1.75		
J19	LGH			2A30	-5.0	-10.8		
J20	READY			2A11	0	-10	µA	
J21	$\overline{HOLD}$			2A12				
J22	$\overline{LOAD}$			2A27				
J23	$\overline{RESET}$			2A25				
J24	CLOCK			2A23		-20		
J25	CRUIN			1A30		-10		

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CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

NOTES:
1. R1 = 4Ω ±1%, 2 W.
2. Since the ammeters are not used simultaneously, only one dc ammeter XO1 is required.
3. With point A disconnected, ammeter XO1-1 should indicate zero current on the 1 μA range.
4. Adjust item PO7-1 voltage to limit pin 1A35 current to 550 mA ±1%.

FIGURE 15. Input current (low-level and high-level) test circuit for table IX.

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TABLE X. Output voltage (high-level) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
K1	DBIN	1A32	5.22	5.55	V	16,17
K2	$\overline{\text{MEMEN}}$	2A17				
K3	$\overline{\text{WE}}$	2A9				
K4	HOLDA	2A26				
K5	WAIT	2A28				
K6	IAQ	2A24				
K7	$\overline{\text{CYCEND}}$	2A7				
K8	CRUOUT	1A31				16,18
K9	CRUCLK	2A8				16,18
K10	A0	1A37				16,17
K11	A1	1A38				
K12	A2	2A38				
K13	A3	2A18				
K14	A4	2A19				
K15	A5	2A40				
K16	A6	2A16				
K17	A7	2A37				
K18	A8	2A36				
K19	A9	2A15				
K20	A10	2A35				

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TABLE X. Output voltage (high-level) test - Continued.

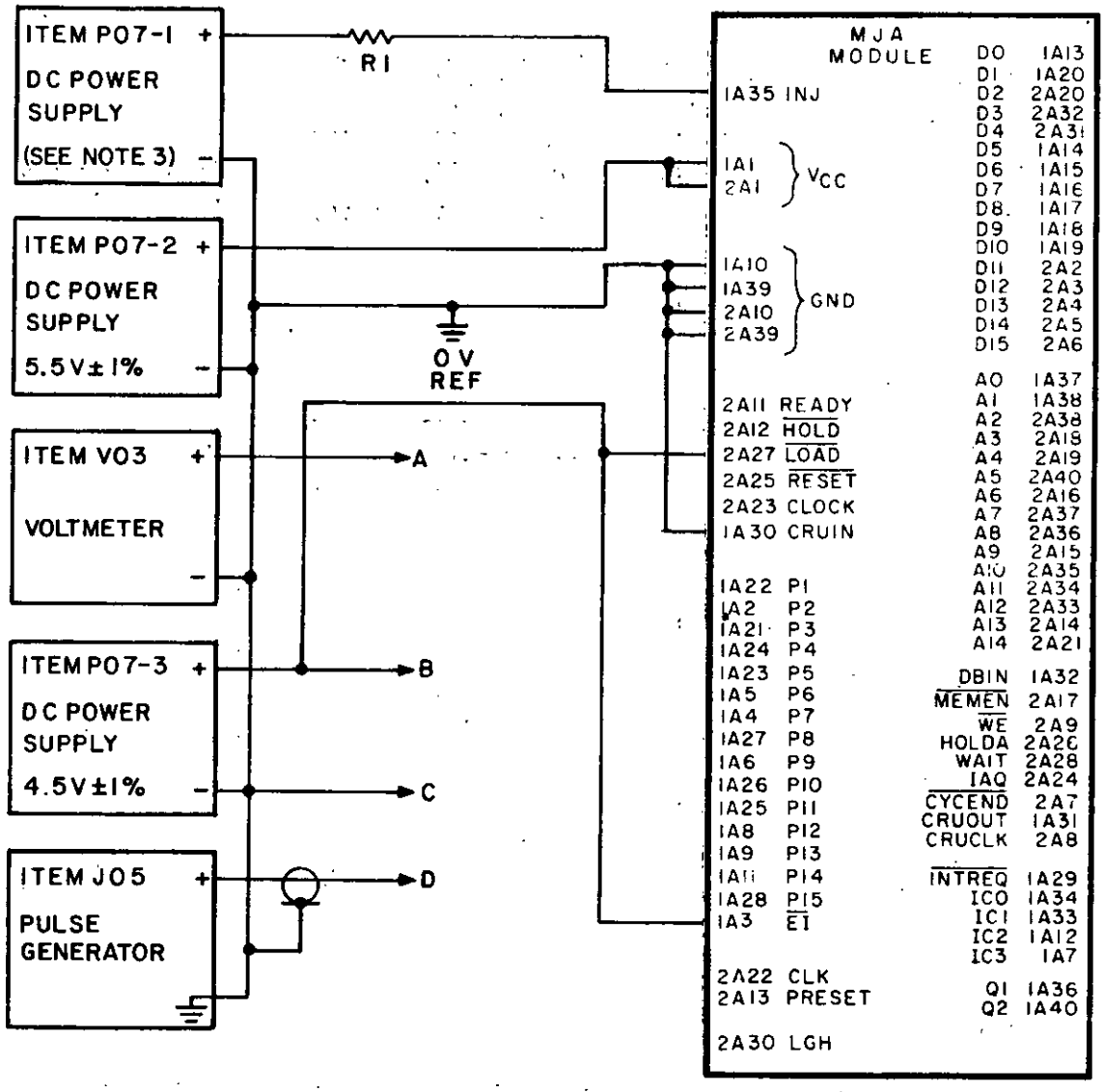
Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
K21	A11	Output voltage, high level	2A34	5.22	5.55	V	16,17
K22	A12		2A33				
K23	A13		2A14				
K24	A14		2A21				
K25	D0		1A13	5.05			
K26	D1		1A20				
K27	D2		2A20				
K28	D3		2A32				
K29	D4		2A31				
K30	D5		1A14				
K31	D6		1A15				
K32	D7		1A16				
K33	D8		1A17				
K34	D9		1A18				
K35	D10		1A19				
K36	D11		2A2				
K37	D12		2A3				
K38	D13		2A4				
K39	D14		2A5				
K40	D15		2A6				

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TABLE X. Output voltage (high-level) test - Continued.

Test condition	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
K41	$\overline{\text{INTREQ}}$	Output voltage, high level	1A3	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A27,1A28	Open	Open	1A29	2.4	4.5	V	19
K42	ICO						1A34				
K43	IC1						1A33				
K44	IC2						1A12				
K45	IC3						1A7				
K46	Q1		2A13	Open			1A36				
K47	Q2		2A13	Open			1A40				

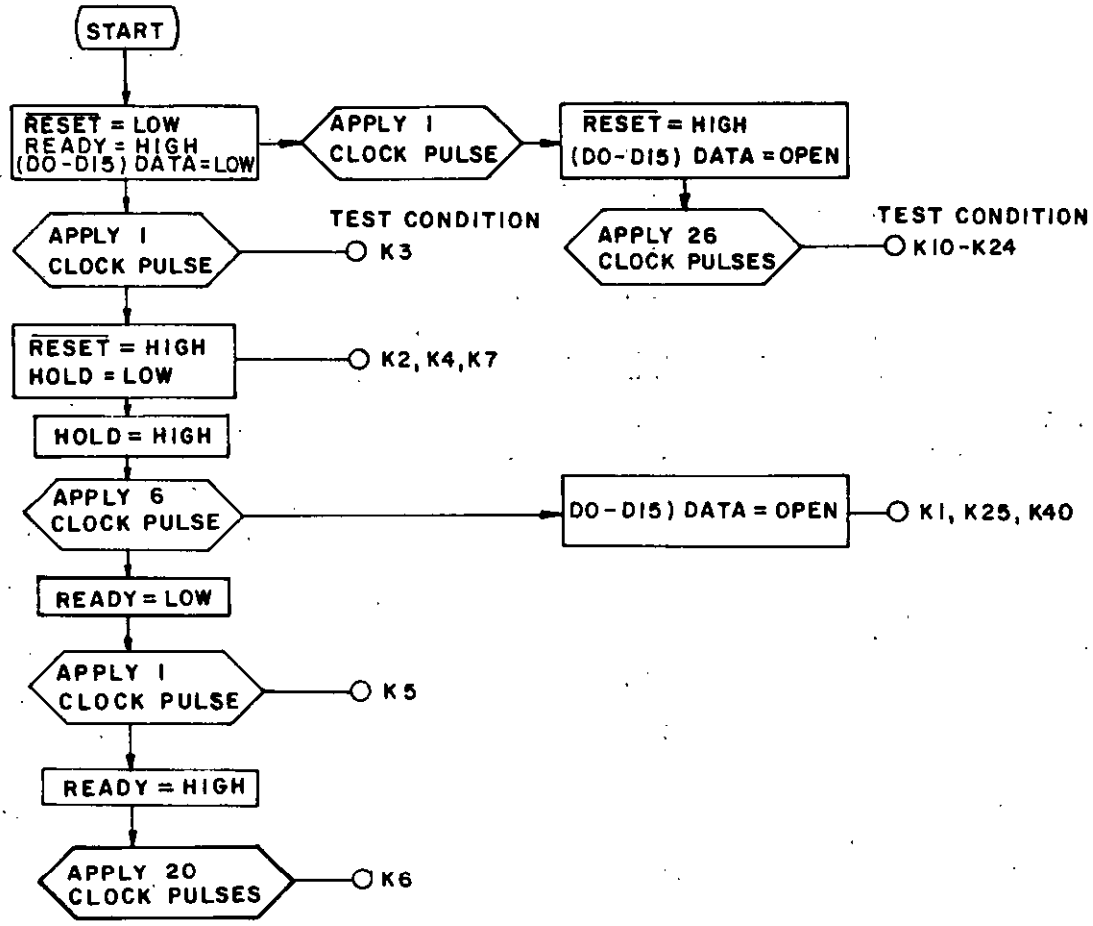
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NOTES:
1. R1 = 4Ω ±1%, 2 W.
2. Use the preconditioning pattern of figures 17 and 18.
3. Adjust item P07-1 voltage to limit pin 1A35 current to 550 mA ±1%.

FIGURE 16. Output voltage (high-level) test circuit for test conditions K1-K40 in table X.

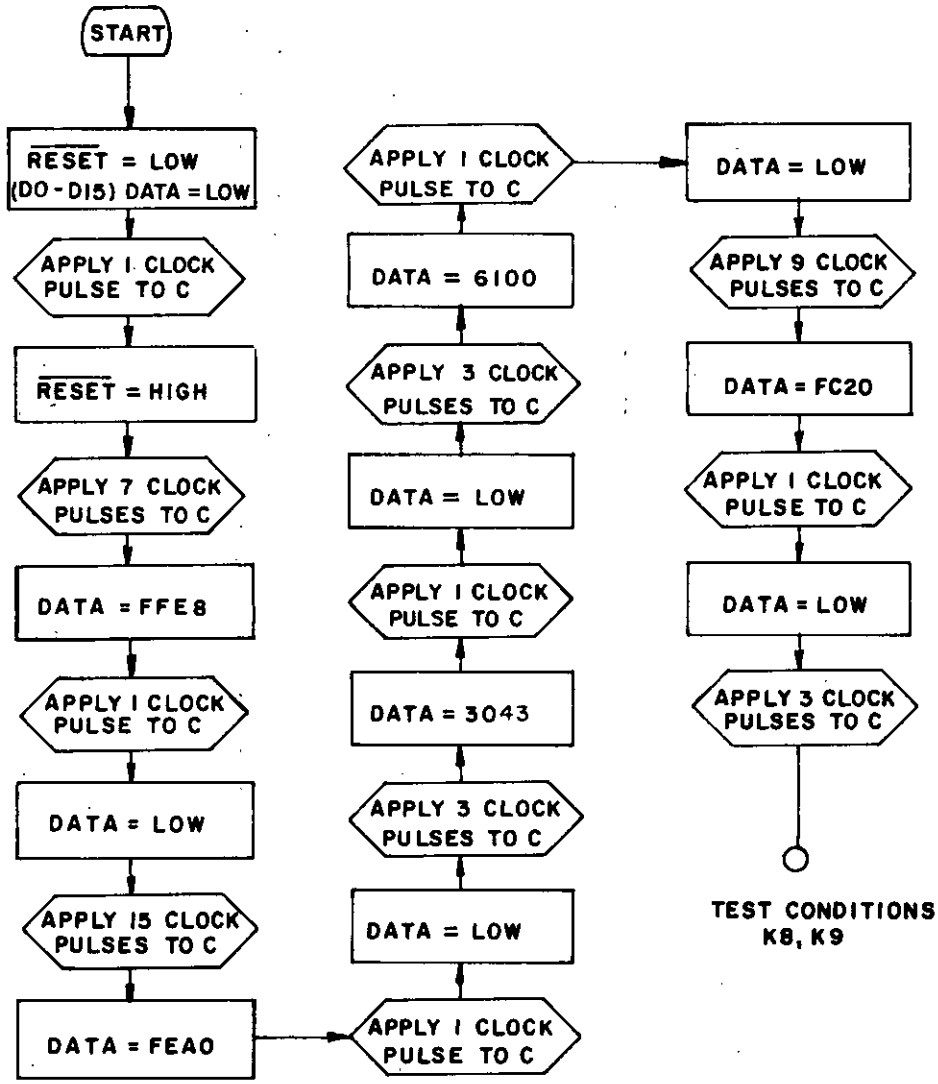
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NOTES:
1. Low = Logic low level = point C voltage.
2. High = Logic high level = point B voltage.
3. CLOCK, $\overline{\text{RESET}}$ and DO-D15 are pins 2A23, 2A25, 1A13-1A20, 2A2-2A6, 2A20, 2A31, 2A32, respectively although the DO-D15 pins are not in numerical order (DO, D1, D2 etc.).
4. Refer to figure 21 for the timing waveforms and input voltages.
5. CLOCK is at point C and $\overline{\text{RESET}}$ is at point D.

FIGURE 17. CPU preconditioning flow chart for test conditions K1-K7 and K10-K40 in table X.

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NOTES:
1. Low = Logic low level = point C.
2. High = Logic high level = point B.
3. CLOCK and $\overline{\text{RESET}}$ are at pins 2A13 and 2A15 resepctively.
4. DO-D15 are pins 1A13, 1A20, 2A20, 2A32, 2A31, 1A14-1A19, 2A2-2A6 respectively.
5. In the data entry point when data = FFE8, the FFE8 is a hexadecimal representation of 16 data bits. That is FFE8 corresponds to 1111111111101000 where the left most bit is the most significant bit. The most significant bit of the CPU is DO (pin 2A13). Then in this example DO = high and D15 = low.
6. Refer to figure 21 for the timing waveforms and input voltages.
7. CLOCK is at point C and $\overline{\text{RESET}}$ is at point D.

FIGURE 18. CPU preconditioning flow chart for test conditions K8 and K9 in table X.

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TABLE XI. Output voltage (low-level) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Connect: E to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
L1	DBIN Output voltage, low level	1A13,1A14,1A15, 1A16,1A17,1A18, 1A19, 1A20,2A2, 2A3,2A4,2A5,2A6, 2A20,2A31,2A32, 1A30	2A11, 2A12, 2A27, 1A3	2A23	2A25	1A32	0.0	0.4	V	19,20
L2	$\overline{\text{MEMEN}}$					2A17				
L3	$\overline{\text{WE}}$					2A9				
L4	HOLDA					2A26				
L5	WAIT					2A28				
L6	IAQ					2A24				
L7	$\overline{\text{CYCEND}}$					2A7				
L8	CRUOUT					1A31				
L9	CRUCLK		2A3, 2A11, 2A12, 2A27			2A8				
L10	A0	1A19,1A30,2A20, 2A31,2A32	2A11, 2A12, 2A27, 1A3			1A37				
L11	A1					1A38				
L12	A2					2A38				
L13	A3					2A18				

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TABLE XI. Output voltage (low-level) test - Continued.

Test condition		Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
L14	A4	Output voltage, low level	1A19, 1A30, 2A20, 2A31, 2A32	2A11, 2A12, 2A27, 1A3	2A23	2A25	2A19	0.0	0.4	V	19,20
L15	A5		1A19, 1A30, 2A20, 2A31, 2A32				2A40				
L16	A6		1A20,1A30,2A2, 2A3,2A4,2A5,2A6, 2A20,2A31,2A32				2A16				
L17	A7		1A20,1A30,2A20, 2A31,2A32				2A37				
L18	A8						2A36				
L19	A9						2A15				
L20	A10						2A35				
L21	A11						2A34				
L22	A12						2A33				
L23	A13						2A14				
L24	A14						2A21				
L25	D0						1A13				

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TABLE XI. Output voltage (low-level) test - Continued.

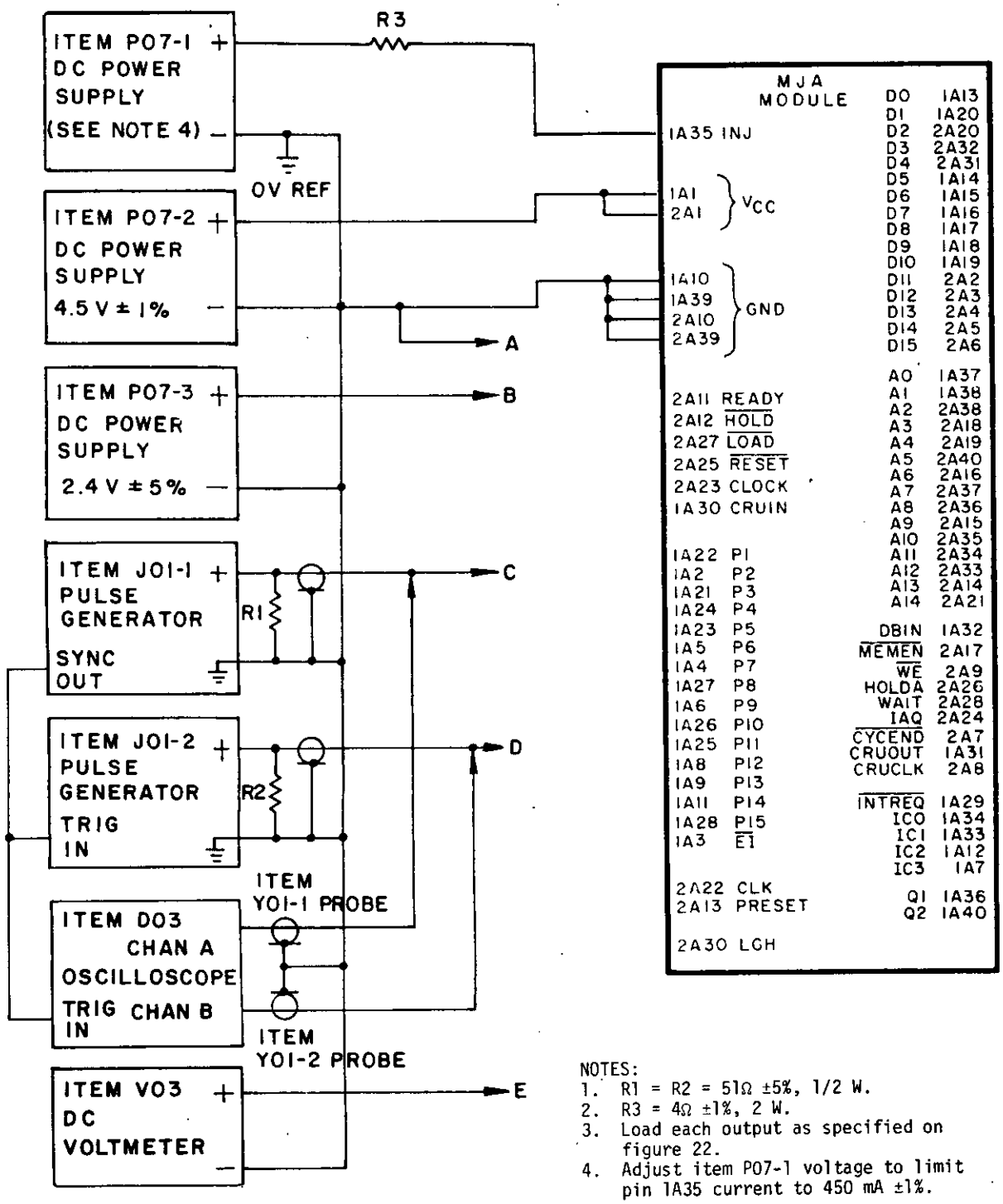
Test condition	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Initial limits 25°+5°C ambient Min	Initial limits 25°+5°C ambient Max	Unit	Test figure
L26	D1	Output voltage, low level	1A20,1A30,2A20, 2A31,2A32	2A11, 2A12, 2A27, 1A3	2A23	2A25	1A20	0.0	0.4	V	19,20
L27	D2						2A20				
L28	D3						2A32				
L29	D4						2A31				
L30	D5						1A14				
L31	D6						1A15				
L32	D7						1A16				
L33	D8						1A17				
L34	D9						1A18				
L35	D10						1A19				
L36	D11						2A2				
L37	D12						2A3				
L38	D13						2A4				
L39	D14						2A5				
L40	D15						2A6				

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TABLE XI. Output voltage (low-level) test - Continued.

Test condition	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
L41	$\overline{\text{INTREQ}}$	Output voltage, low level	1A3, 2A30	Open	Open	Open	1A29	0.0	0.4	V	19,20
L42	IC0	↓	↓	↓	↓	↓	1A34	↓	↓	↓	↓
L43	IC1						1A33				
L44	IC2						1A12				
L45	IC3						1A7				
L46	Q1		2A30				1A36				
L47	Q2		2A30				1A40				

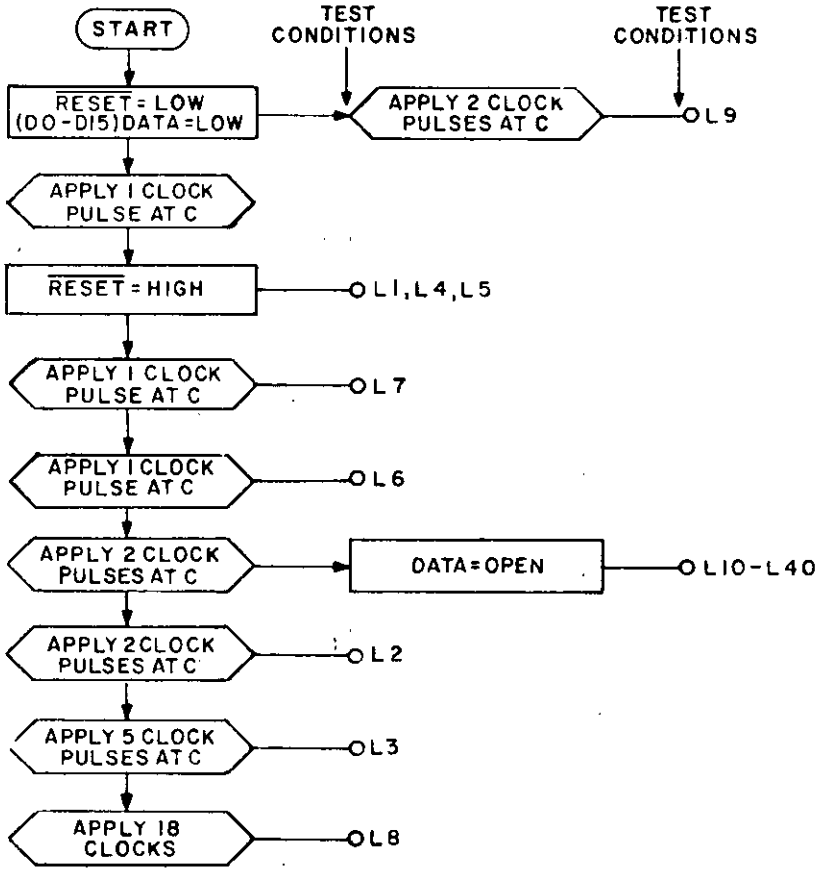
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. R3 = 4Ω ±1%, 2 W.
3. Load each output as specified on figure 22.
4. Adjust item PO7-1 voltage to limit pin 1A35 current to 450 mA ±1%.

FIGURE 19. Output voltage (high-level and low-level) test circuit for test conditions K41-K47 in table X and for test conditions L1-L40 in table XI.

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NOTES:
1. Low = Logic low level = point A voltage.
2. High = Logic high level = point B voltage.
3. CLOCK and $\overline{\text{RESET}}$ are pins 2A23 and 2A25 respectively.
4. D0-D15 are pins 1A13, 1A20, 2A20, 2A32, 2A31, 1A14-1A19, 2A2-2A6 respectively.
5. Open = No connection.
6. CLOCK is at point C and $\overline{\text{RESET}}$ is at point D.
7. Refer to figure 21 for timing waveforms and input voltages.

FIGURE 20. CPU preconditioning flow chart for test conditions L1-L40 in table XI.

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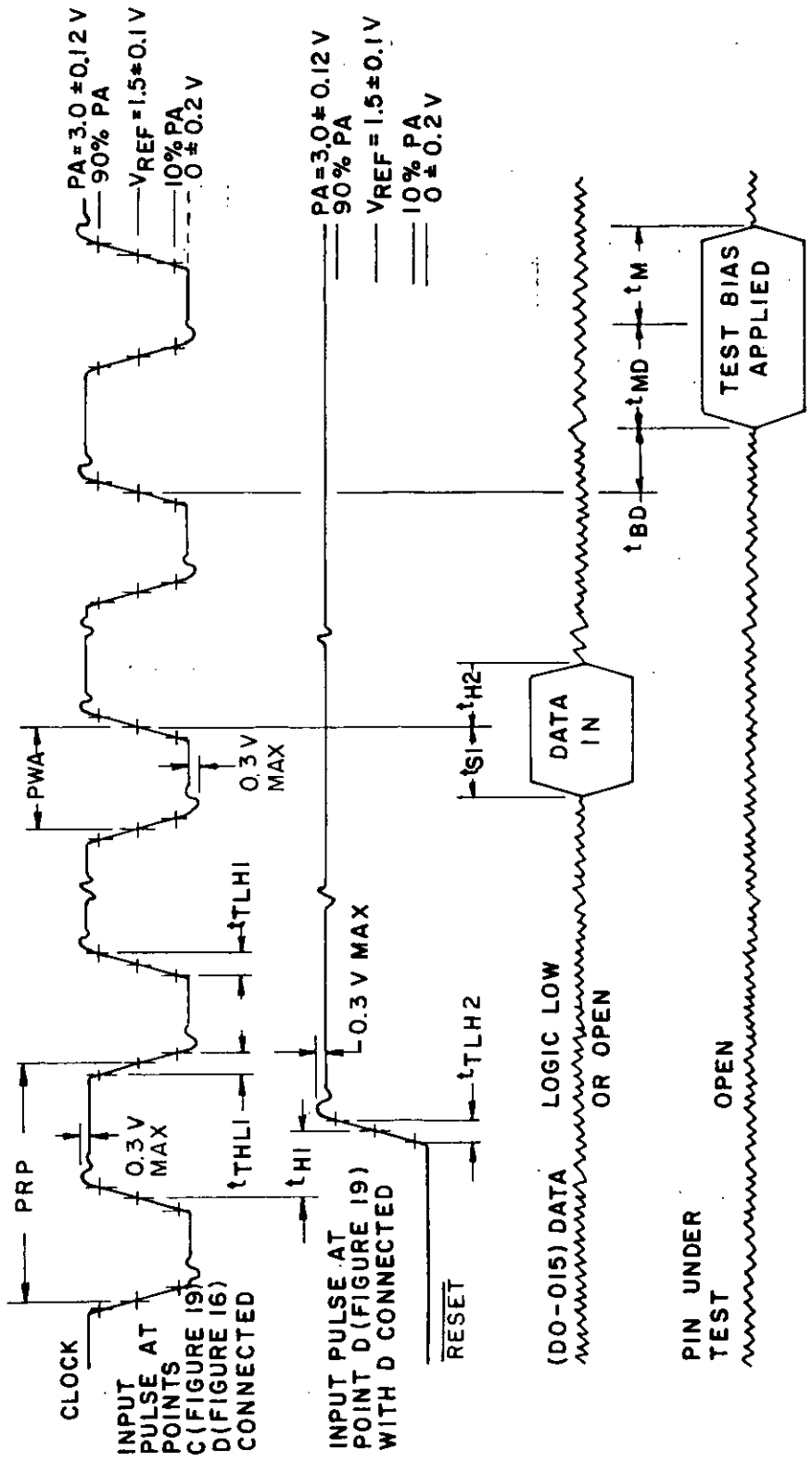


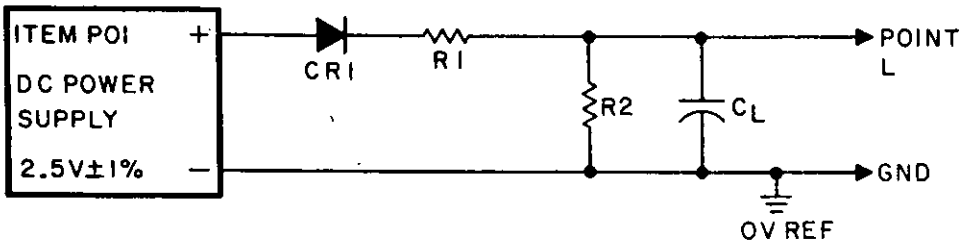
FIGURE 21. Output voltage (high-level and low-level) test circuit waveforms for tables X and XI.

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NOTES:
1. Input pulse parameters:
   PWA = 500 ns minimum (see note 2).
   PRP = 3 x PWA minimum.
   $t_{LH1} = t_{LH2} = t_{HL1}$ = 20 ±2 ns.
   $t_{H1} = t_{H2}$ = 1/2 x PWA ±10%.
   t_{S1} = 1/2 x PWA ±10%.
   t_{BD} = Bias delay time = PWA ±10%.
   t_{MD} = Measurement delay time = PWA ±10%.
   t_{M} = Measurement time = PWA ±10%.
2. The CPU is a static device and therefore can be clocked at a slower rate.
3. For test conditions K3 and L9 on figure 21, the RESET input (point D) does not go high. Therefore disregard the t_{H1} hold time requirement during these test conditions.

FIGURE 21. Output voltage (high-level and low-level) test circuit waveforms for tables X and XI - Continued.

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NOTES:
1. CR1 = 1N4150 or equivalent.
2. C_L = 50 pF ±5% (for pins 1A7, 1A12, 1A29, 1A33, 1A34, 1A36, 1A40).
   C_L = 150 pF ±5% (for all other pins).
3. R2 = Select value for each output load as specified in the following table.
4. R1 = Select value for a current flow of I_{OL} when point L is held at 0.4 V. I_{OL} is found in the following table.

Outputs 3/	R2 1/	I_{OL} 2/
Q1	6.34 kΩ	3.20 mA
Q2	6.34 kΩ	3.60 mA
IC0	17.8 kΩ	4.0 mA
$\overline{\text{INTREQ}}$, IC1, IC2, IC3	4.32 kΩ	16.0 mA
CPU outputs	---	10.7 mA

1/ All resistors are metal film, ±1%, 1/4 W.
2/ I_{OL} tolerance, ±1%.
3/ CPU outputs are A0-A14, $\overline{\text{MEMEN}}$, $\overline{\text{WE}}$, HOLDA, WAIT, IAQ, $\overline{\text{CYCEND}}$, CRUOUT, CRUCLK, D0-D15.

FIGURE 22. Logic load circuit.

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TABLE XII. Static functional (priority interrupt and clock shaping) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M1	Static functional test, priority interrupt	1A3	1A2,1A4,1A5,1A6,1A8, 1A9,1A11,1A21,1A22,1A23, 1A24,1A25,1A26,1A27, 1A28	Open	1/	Verify the logic states in table XIII			23,24, 25
M2		1A2,1A4,1A5,1A6,1A8,1A9, 1A11,1A21,1A22,1A23,1A24, 1A25,1A26,1A27,1A28	1A3						
M3		1A2,1A4,1A5,1A6,1A8,1A9, 1A11,1A21,1A23,1A24,1A25, 1A26,1A27,1A28	1A3,1A22						
M4		1A4,1A5,1A6,1A8,1A9,1A11, 1A21,1A22,1A23,1A24,1A25, 1A26,1A27,1A28	1A2,1A3						
M5		1A4,1A5,1A6,1A8,1A9,1A11, 1A22,1A23,1A24,1A25,1A26, 1A27,1A28,1A2	1A3,1A21						
M6		1A2,1A4,1A5,1A6,1A8,1A9, 1A11,1A21,1A22,1A23,1A25, 1A26,1A27,1A28	1A3,1A24						
M7		1A2,1A4,1A5,1A6,1A8,1A9, 1A11,1A21,1A22,1A24,1A25, 1A26,1A27,1A28	1A3,1A23						
M8		1A4,1A6,1A8,1A9,1A11, 1A25,1A26,1A27,1A28	1A2,1A3,1A5,1A21,1A22, 1A23,1A24						
M9		1A2,1A5,1A6,1A8,1A9,1A11, 1A21,1A22,1A23,1A24,1A25, 1A26,1A27,1A28	1A3,1A4						
M10		1A2,1A4,1A5,1A6,1A8,1A9, 1A11,1A21,1A22,1A23,1A24, 1A25,1A26,1A28	1A3,1A27						
M11		1A2,1A4,1A5,1A8,1A9,1A11, 1A21,1A22,1A23,1A24,1A25, 1A26,1A27,1A28	1A3,1A6						
M12		1A2,1A4,1A5,1A6,1A8,1A9, 1A11,1A21,1A22,1A23,1A24, 1A25,1A27,1A28	1A3,1A26						

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TABLE XII. Static functional (priority interrupt and clock shaping) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M13	Static functional test, priority interrupt	1A2,1A4,1A5,1A6,1A8,1A9, 1A11,1A21,1A22,1A23,1A24, 1A26,1A27,1A28	1A3,1A25	Open	1/	Verify the logic states in table XIII			23,24, 25
M14		1A2,1A4,1A5,1A6,1A9,1A11, 1A21,1A22,1A23,1A24,1A25, 1A26,1A27,1A28	1A3,1A8						
M15		1A2,1A4,1A5,1A6,1A8,1A11, 1A21,1A22,1A23,1A24,1A25, 1A26,1A27,1A28	1A3,1A9						
M16		1A2,1A4,1A5,1A6,1A8,1A9, 1A21,1A22,1A23,1A24,1A25, 1A26,1A27,1A28	1A3,1A11						
M17		1A2,1A4,1A5,1A6,1A8,1A9, 1A11,1A21,1A22,1A23,1A24, 1A25,1A26,1A27	1A3,1A28						
M18	Static functional test, clock shaping	Open	2A13	2A22					
M19		Open	2A13						
M20		2A13	2A30						
M21		2A13	2A30						
M22		Open	2A13						
M23		2A13	Open						
M24									
M25									
M26									
M27									
M28									

1/ Use item V03 (point D) to verify the output logic states.

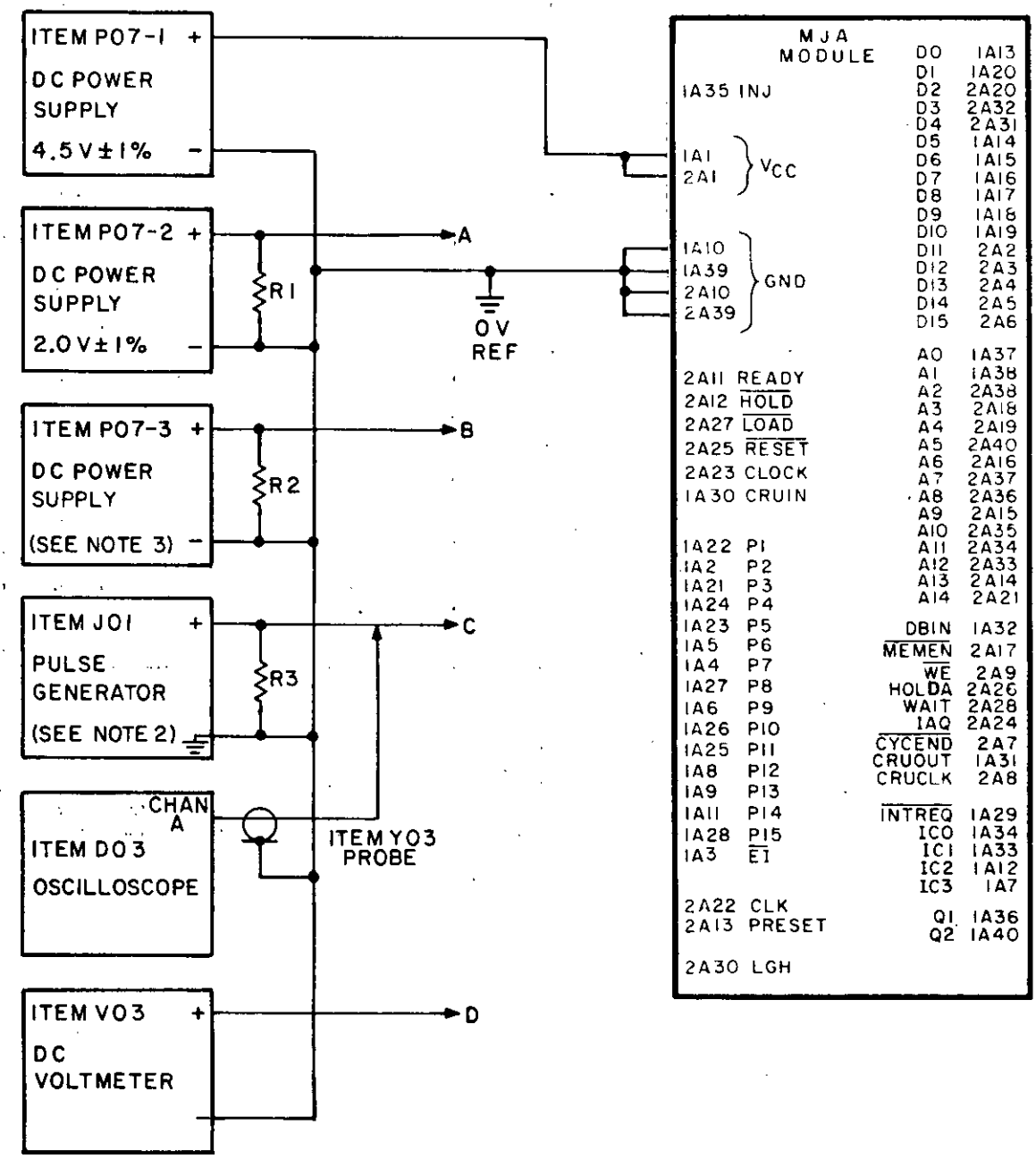
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TABLE XIII. Output conditions for the static functional test in table XII.

Test condition	Output logic states						
	$\overline{\text{INTREQ}}$	IC0	IC1	IC2	IC3	Q1	Q2
M1	H	L	H	H	H	X	X
M2	H	H	H	H	H	X	X
M3	L	L	L	L	H	X	X
M4	L	L	L	H	L	X	X
M5	L	L	L	H	H	X	X
M6	L	L	H	L	L	X	X
M7	L	L	H	L	H	X	X
M8	L	L	H	H	L	X	X
M9	L	L	H	H	H	X	X
M10	L	H	L	L	L	X	X
M11	L	H	L	L	H	X	X
M12	L	H	L	H	L	X	X
M13	L	H	L	H	H	X	X
M14	L	H	H	L	L	X	X
M15	L	H	H	L	H	X	X
M16	L	H	H	H	L	X	X
M17	L	H	H	H	H	X	X
M18	X	X	X	X	X	H	H
M19	X	X	X	X	X	H	H
M20	X	X	X	X	X	L	L
M21	X	X	X	X	X	L	L
M22	X	X	X	X	X	H	H
M23	X	X	X	X	X	L	H
M24	X	X	X	X	X	H	L
M25	X	X	X	X	X	H	H
M26	X	X	X	X	X	L	H
M27	X	X	X	X	X	H	L
M28	X	X	X	X	X	H	H

NOTES:
1. X = Don't care.
2. H = High level voltage = 2.4 to 4.5 V.
3. L = Low level voltage = 0.0 to 0.4 V.
4. For test conditions M1-M28 verification of the output logic states in notes 2 and 3 are obtained by connecting item VO3 (point D, figure 23) to each output pin in succession after input conditions have been accomplished.

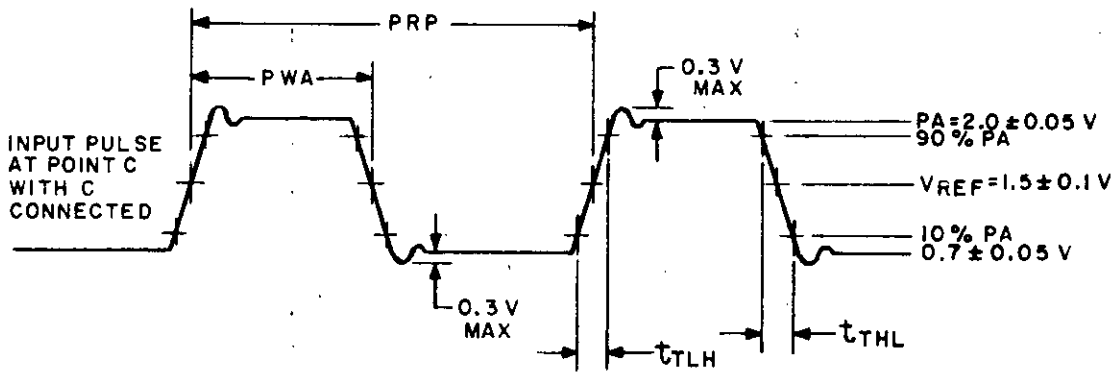
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NOTES:
1. R1-R3 = 51Ω ±5%, 1/2 W.
2. Use the flow chart on figure 25 to obtain the required clock pulse numbers and input setups for test conditions M18-M28 in table XII.
3. Item P07-3 voltage:
   V = 0.8 V ±1% for test conditions M1-M17.
   V = 0.7 V ±1% for test conditions M18-M28.

FIGURE 23. Static functional (priority interrupt and clock shaping) test circuit for table XII.

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NOTES:
1. Clock input pulse parameters:
   PWA = 500 ±100 ns.
   PRP = 1,000 ns minimum.
   $t_{TLH} = t_{THL}$ = 50 ±25 ns.
2. Clock threshold level setup: Make circuit connections per table XII. Place item J01 in single pulse mode. Use item D03 and adjust dc offset of item J01 for a low level of 0.7 ±0.05 V (clock low level threshold). Adjust item J01 to obtain the required PWA and set PRP to allow reading of all output states and input setup times in tests M18-M28. Use item D03 and adjust item J01 amplitude for a high level of 2.0 ±0.05 V (clock high level threshold).
3. The clock is the CLK input at pin 2A22.

FIGURE 24. Static functional (priority interrupt and clock shaping) test circuit waveform for table XII.

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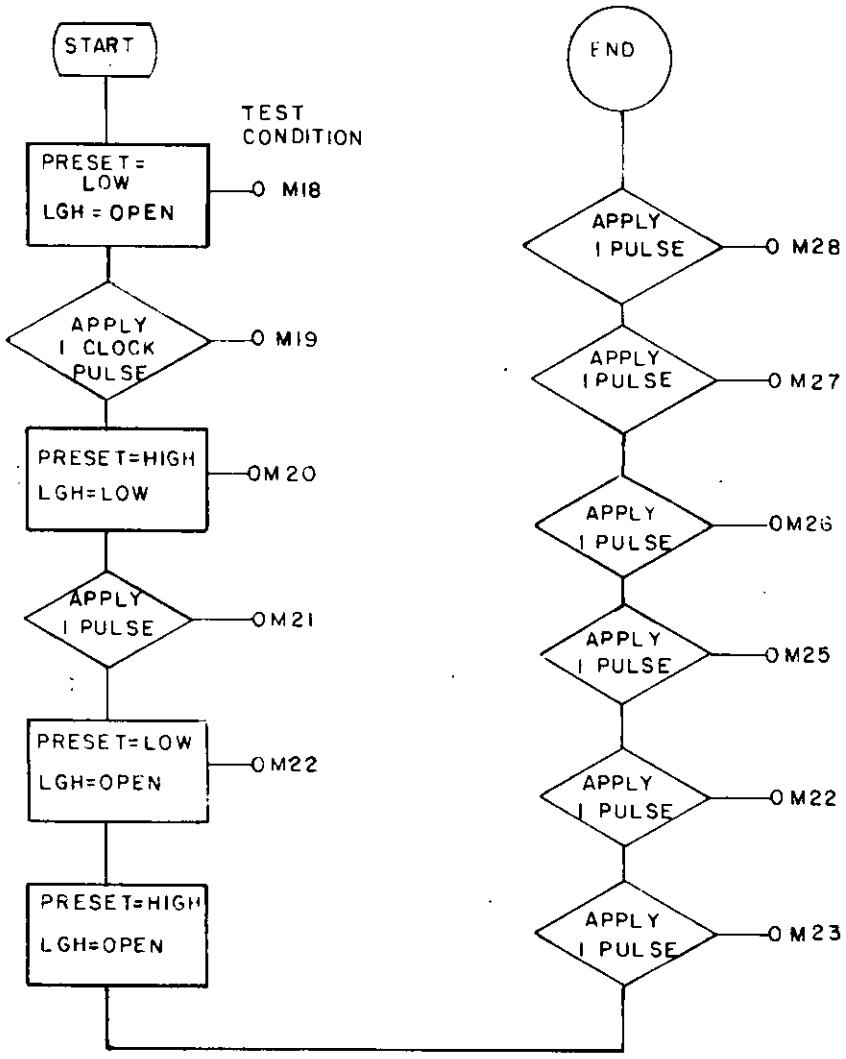


FIGURE 25. Flow chart for the static functional (clock shaping) test in table XII.

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TABLE XIV. Dynamic functional (minimum and maximum injector current) test.

Test condition	Parameter	Input table	Waveform figure	Test figure
N1	Dynamic functional test, minimum injector current	XV	27	26
N2	Dynamic functional test, maximum injector current	XV	27	26

NOTE: Apply the vectors (see 6.3) for each test condition using the input setup and hold times, power supply voltage and injector current as specified in table XV and figure 26. Verify the output levels are $V_{OL} \leq 0.6$ V and $V_{OH} \geq 2.0$ V at a point in time not less than the maximum propagation delay time shown in tables XVIII and XIX.

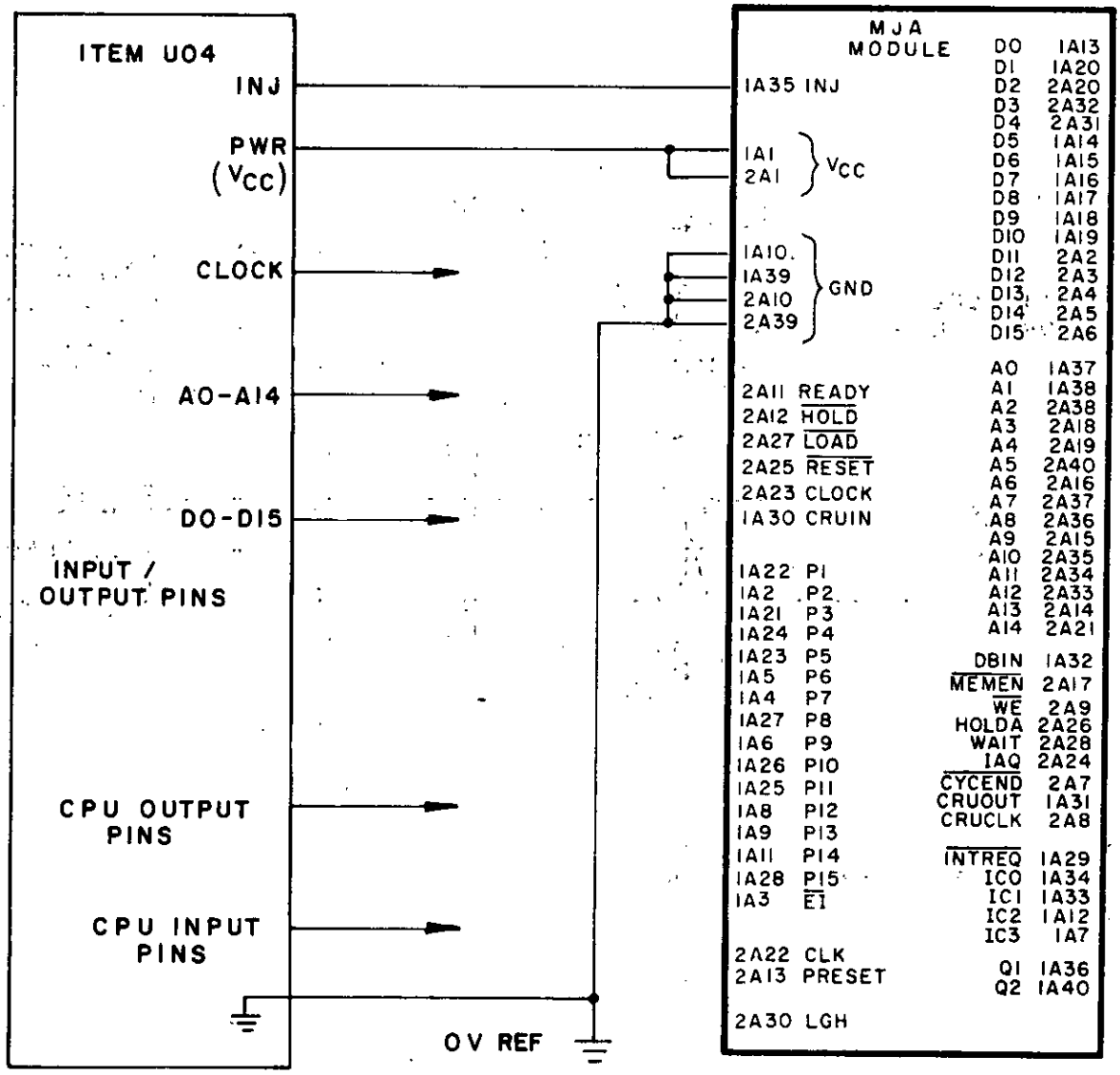
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TABLE XV. CPU input conditions for tables XIV and XX.

Parameter	Pin(s)	Input conditions	Unit	Figure
INJ	1A35	1/,2/	mA	26,27
V_{CC}	1A1,2A1	3/	V	
PRP	2A23	222 ±1%	ns	
PWA	2A23	111 ±1%		
t_{TLH1}, t_{TLH2}, t_{THL1}, t_{THL2}	All CPU inputs	15 ±5		
t_S	1A13,1A14,1A15,1A16,1A17,1A18,1A19, 2A2,2A3,2A4,2A5,2A6,2A20,2A31,2A32	65 +2, -0		
t_S	2A11	70 +2, -0		
t_S	2A12	110 +2, -0		
t_S	1A30	50 +2, -0		
t_S	1A29	25 +2, -0		
t_S	1A34,1A33,1A12,1A7	25 +2, -0		
t_H	1A13,1A14,1A15,1A16,1A17,1A18,1A19, 2A2,2A3,2A4,2A5,2A6,2A20,2A31,2A32	40 +2, -0		
t_H	2A11	50 +2, -0		
t_H	2A12	25 +2, -0		
t_H	1A30	40 +2, -0		
t_H	1A29	55 +2, -0		
t_H	1A34,1A33,1A12,1A7	55 +2, -0		
V_{IH}	All	2.0 +0.1,-0.0	V	
V_{IL}	All	0.6 +0.0,-0.1	V	

1/ INJ = 450 mA ±1% for test conditions N1, T1.
2/ INJ = 550 mA ±1% for test conditions N2, T2.
3/ V_{CC} = 4.5 V ±1% for test conditions N1, N2,T1,T2.

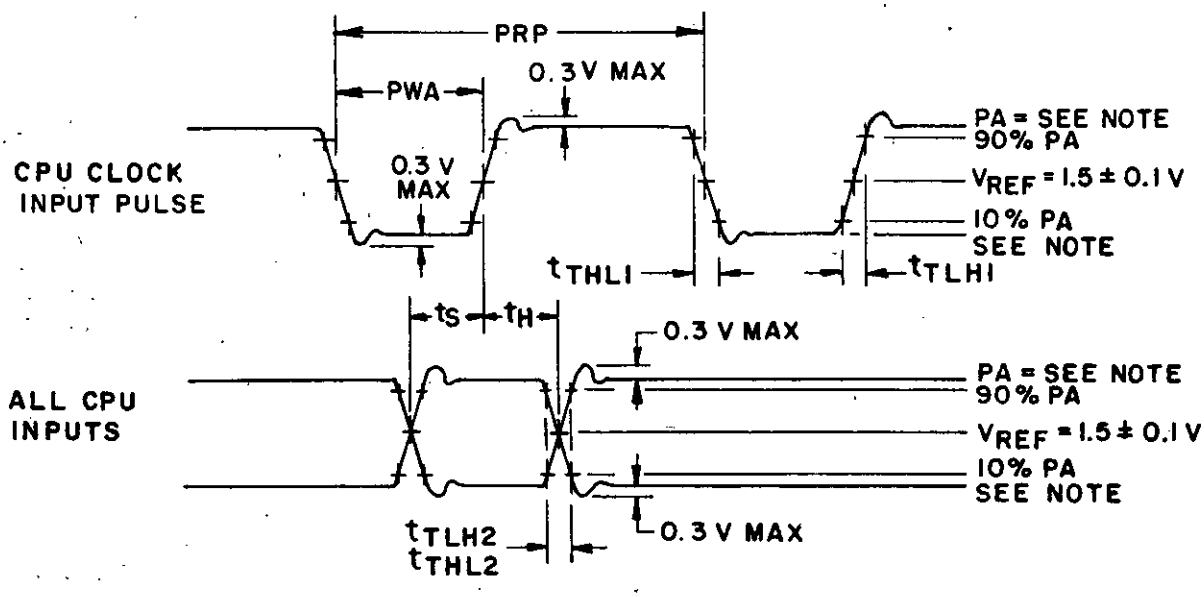
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NOTES:
1. Load each output as specified on figure 22.
2. Item U04 must be capable of sinking 20 mA when connected to any data buss pin (DO-DI5).
3. INTREQ, ICO-IC3 inputs are connected to the priority interrupt outputs and therefore item U04 must be capable of sinking 100 mA when attached to either of these inputs and forcing a logic low level of 0.4 V. The inputs shall never be held low in a static condition for more than 0.5 second.

FIGURE 26. Dynamic functional (minimum and maximum injector current) test and propagation delay time (t_{PLH} and t_{PHL}) check test circuit for tables XIV and XX.

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NOTE: See table XV for input parameters.

FIGURE 27. Dynamic functional (minimum and maximum injector current) test circuit waveforms for table XIV.

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TABLE XVI. Transition time (t_{THL}) test.

Test condition	Parameter	Pin	Flow chart figure	Waveform figure	Initial limits 25° ±5°C Min	Initial limits 25° ±5°C Max	Unit	Test figure
P1	DBIN Transition time (t_{THL}), C_L = 150 pF	1A32	29	37,38	5	40	ns	28
P2	$\overline{MEMEN}$	2A17	30	37,38,41				
P3	$\overline{WE}$	2A9	30	37,38,40				
P4	HOLDA	2A26	29	37,38,39				
P5	WAIT	2A28	31	37,38,42				
P6	IAQ	2A24	32	37,38,41				
P7	$\overline{CYCEND}$	2A7	32	37,38,41				
P8	CRUOUT	1A31	33	37,38,40,43				
P9	CRUCLK	2A8	34	37,38,40,43				
P10	A0	1A37	35	37,38,41,43				
P11	A1	1A38						
P12	A2	2A38						
P13	A3	2A18						
P14	A4	2A19						
P15	A5	2A40						
P16	A6	2A16						
P17	A7	2A37						
P18	A8	2A36						
P19	A9	2A15						

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TABLE XVI. Transition time (t_{THL}) test - Continued.

Test condition	Parameter		Pin	Flow chart figure	Waveform figure	Initial limits 25° ±5°C		Unit	Test figure
						Min	Max		
P20	A10	Transition time (t_{THL}), C_L = 150 pF	2A35	35	37,38,41,43	5	40	ns	28
P21	A11		2A34						
P22	A12		2A33						
P23	A13		2A14						
P24	A14		2A21						
P25	D0		1A13	36	37,38,40,43				
P26	D1		1A20						
P27	D2		2A20						
P28	D3		2A32						
P29	D4		2A31						
P30	D5		1A14						
P31	D6		1A15						
P32	D7		1A16						
P33	D8		1A17						
P34	D9		1A18						
P35	D10		1A19						
P36	D11		2A2						
P37	D12		2A3						
P38	D13		2A4						
P39	D14		2A5						
P40	D15		2A6						

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TABLE XVI. Transition time (t_{THL}) test - Continued.

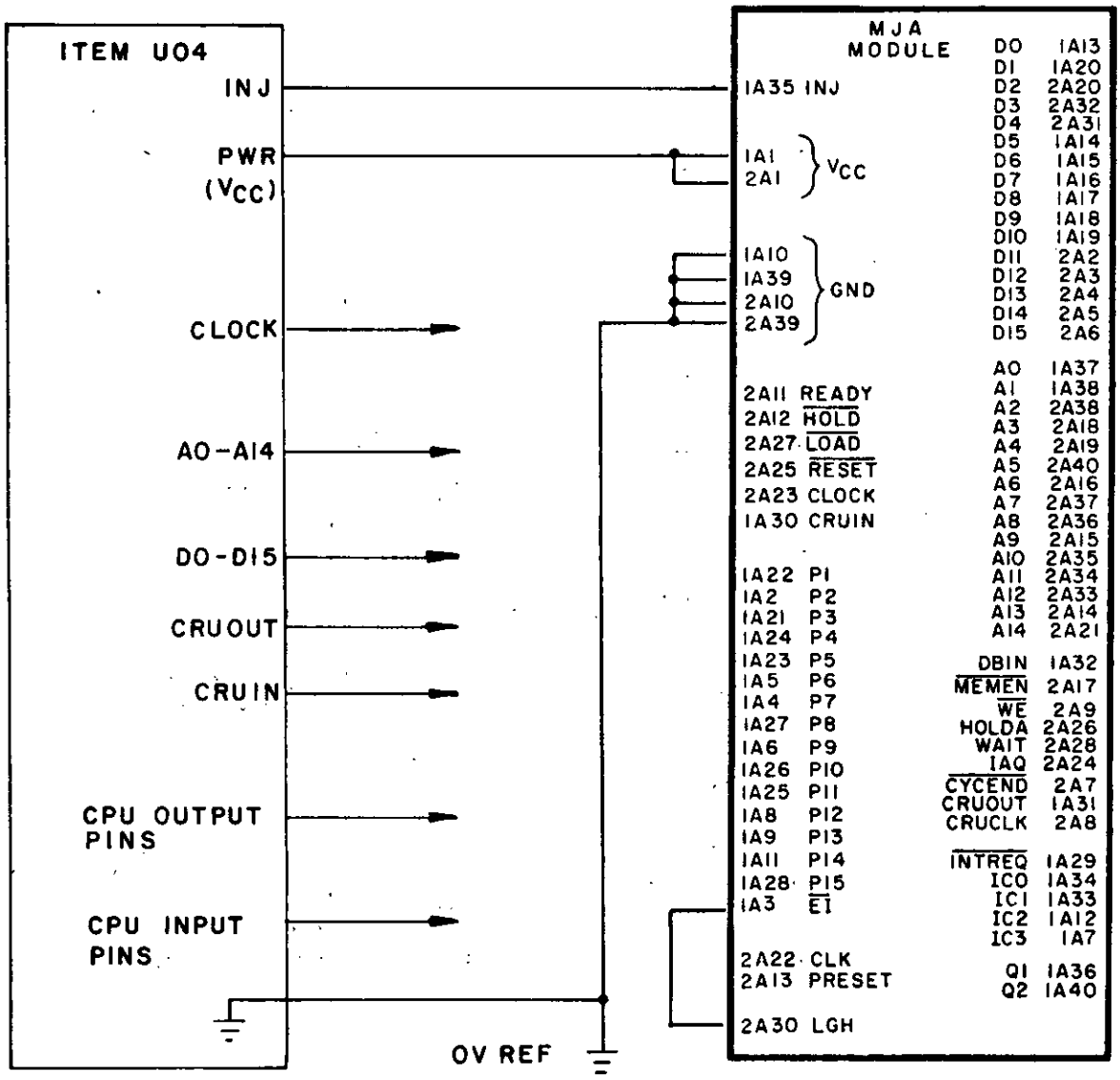
Test condition	Parameter		Connect: A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Initial limits 25°±5°C ambient: Min	Max	Unit	Test figure
P41	$\overline{INTREQ}$	Transition time (t_{THL}), C_L = 50 pF	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A27,1A28	1A3	2A30	Open	C	1A29	4	50	ns	44,45
P42	ICO							1A34		20		
P43	IC1							1A33		50		
P44	IC2							1A12		50		
P45	IC3							1A7		50		
P46	Q1		2A13	Open	2A22			1A36		20		44,46
P47	Q2		2A13	Open	2A22			1A40		20		44,46

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TABLE XVII. Transition time (t_{TLH}) test.

Test condition	Parameter		Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Connect: E to pin	Connect: F to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
Q1	$\overline{\text{INTREQ}}$	Transition time (t_{TLH}), C_L = 50 pF	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A27,1A28	1A3	2A30	Open	C	1A24	10	80	ns	44,45
Q2	IC0							1A34	3	45		
Q3	IC1							1A33	10	80		
Q4	IC2							1A12	10	80		
Q5	IC3							1A7	10	80		
Q6	Q1		2A13	Open	2A22			1A36	3	45		44,46
Q7	Q2		2A13	Open	2A22			1A40	3	45		44,46

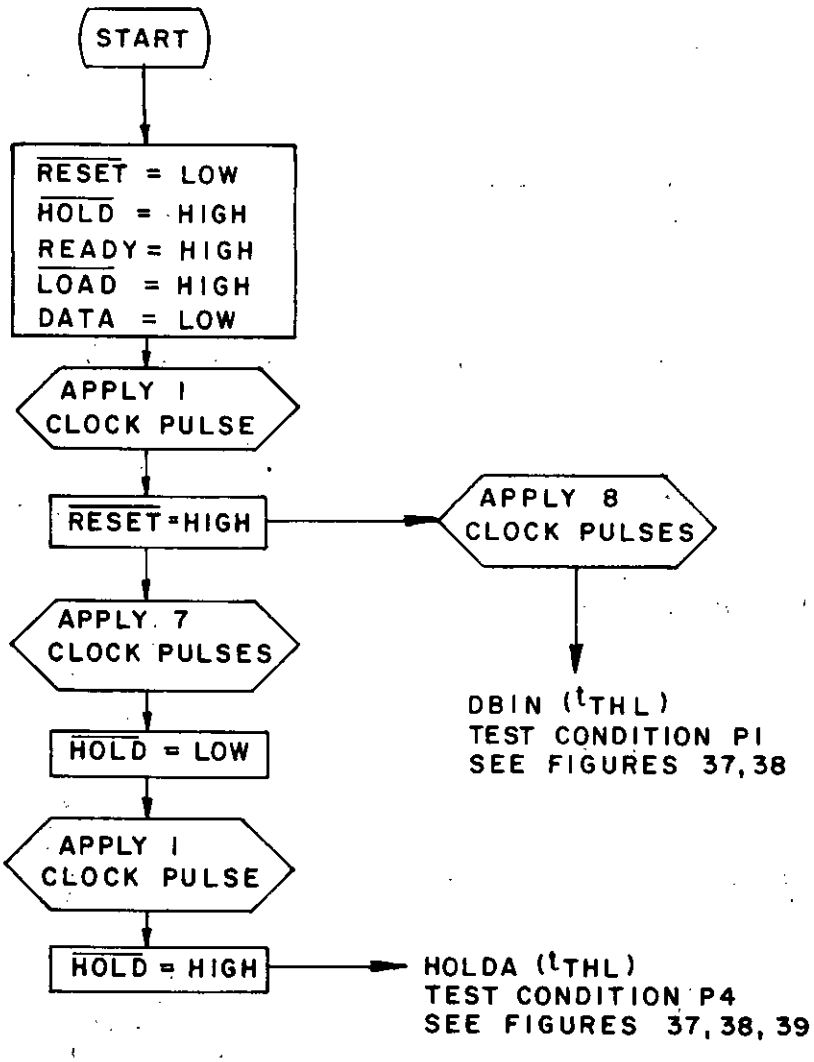
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NOTES:
1. INJ = 450 mA ±1%.
2. V_{CC} = 5.0 V ±1%.
3. All CPU inputs not specified in the flow chart shall be left open.
4. Load each output under test as specified on figure 22.

FIGURE 28. Transition time (t_{THL}) test circuit for table XVI.

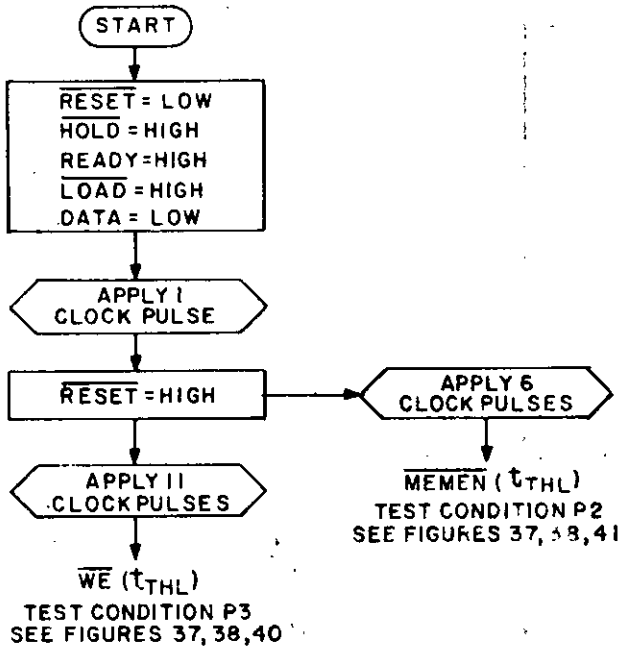
Source: https://assist.dla.mil -- Downloaded: 20260-20T17:35Z
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NOTES:
1. Data = D0-D15 pins.
2. D0 = MSB.
3. D15 = LSB.
4. Low level = 0.0 ±0.2 V.
5. High level = 3.0 ±0.3 V.

FIGURE 29. Transition time (t_{THL}) test preconditioning flow chart for table XVI.

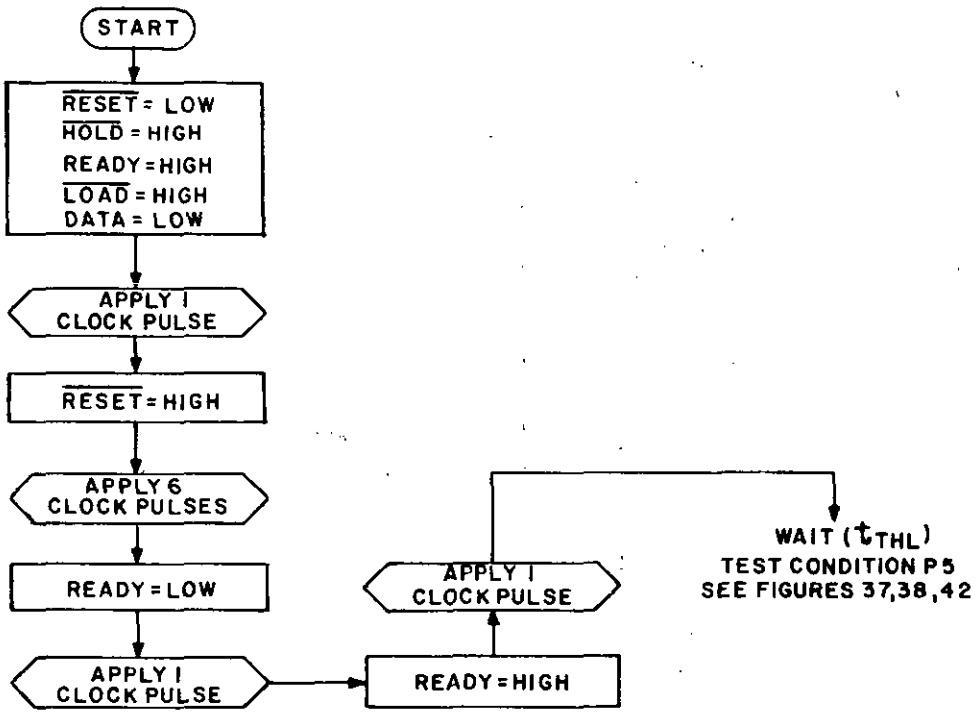
Source: https://assist.dla.mil -- Downloaded: 202610-20T17:35Z
Check the source to verify that this is the current version before use.



NOTES:
1. Data = D0 - D15 pins.
2. Low level = 0.0 ±0.2 V.
3. High level = 3.0 ±0.3 V.

FIGURE 30. Transition time (t_{THL}) test preconditioning flow chart for table XVI.

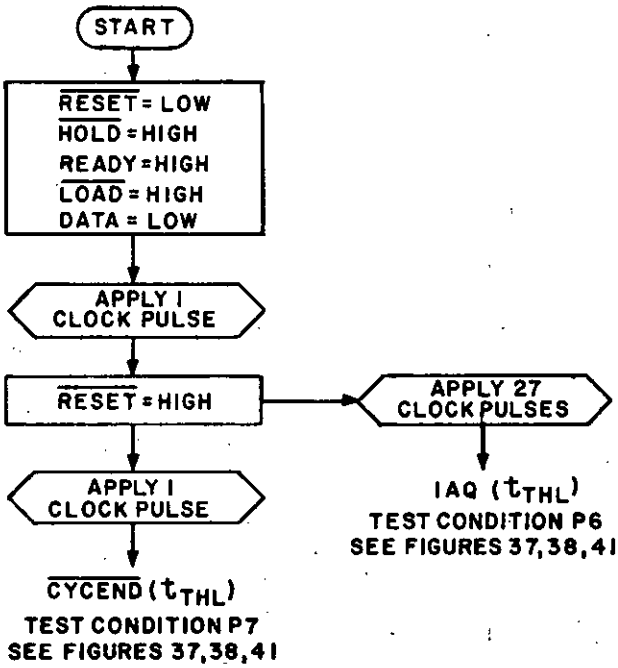
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.



NOTES:
1. Data = D0 - D15 pins.
2. Low level = 0.0 ±0.2 V.
3. High level = 3.0 ±0.3 V.

FIGURE 31. Transition time (t_{THL}) test preconditioning flow chart for table XVI.

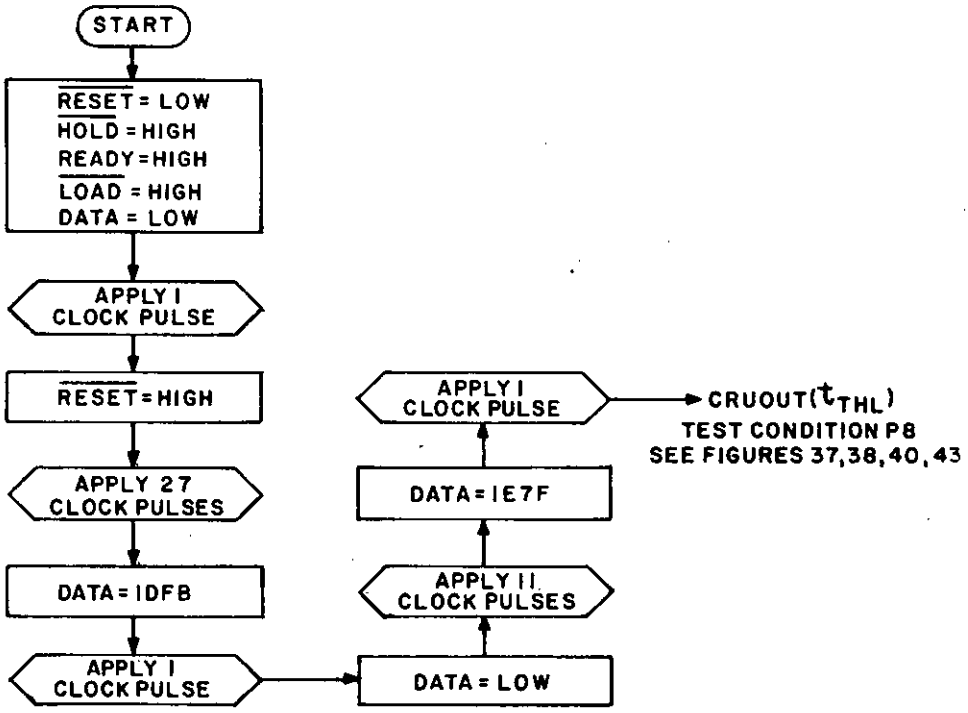
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.



NOTES:
1. Data = D0-D15 pins.
2. D0 = MSB.
3. D15 = LSB.
4. Low level = 0.0±0.2 V.
5. High level = 3.0 ±0.3 V.

FIGURE 32. Transition time (t_{THL}) test preconditioning flow chart for table XVI.

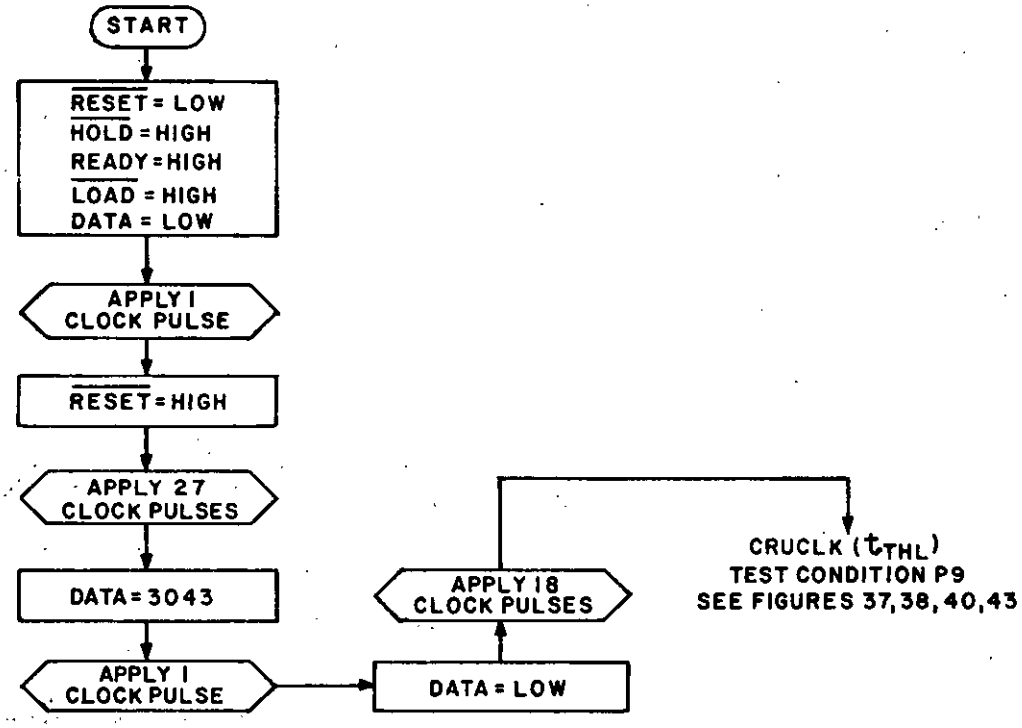
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.



NOTES:
1. Data = D0-D15 pins.
2. D0 = MSB.
3. D15 = LSB.
4. Low level = 0.0 ±0.2 V.
5. High level = 3.0 ±0.3 V.

FIGURE 33. Transition time (t_{THL}) test preconditioning flow chart for table XVI.

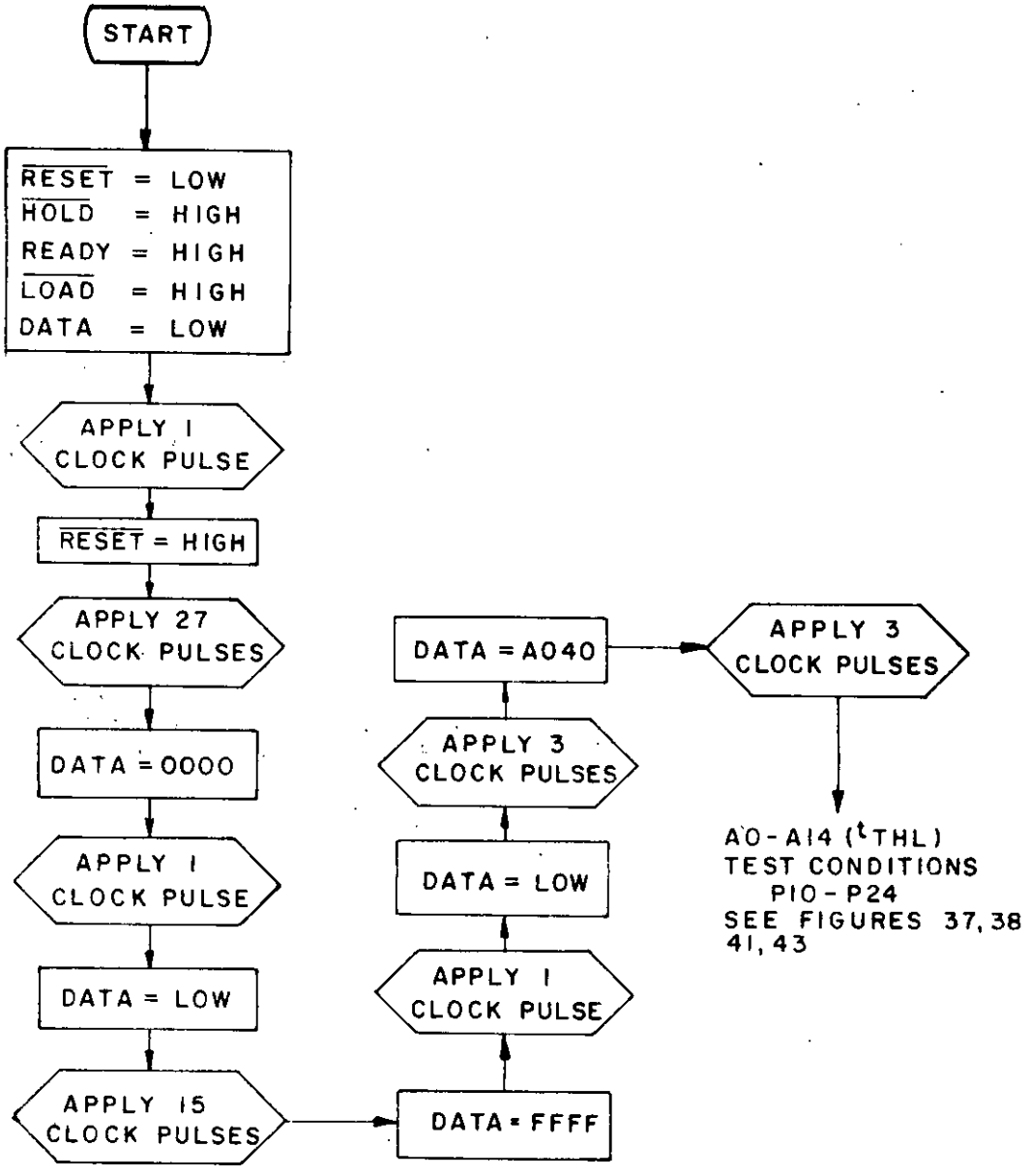
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.



NOTES:
1. Data = D0-D15 pins.
2. D0 = MSB.
3. D15 = LSB.
4. High level = 3.0 ±0.3 V.
5. Low level = 0.0 ±0.2 V.

FIGURE 34. Transition time (t_{THL}) test preconditioning flow chart for table XVI.

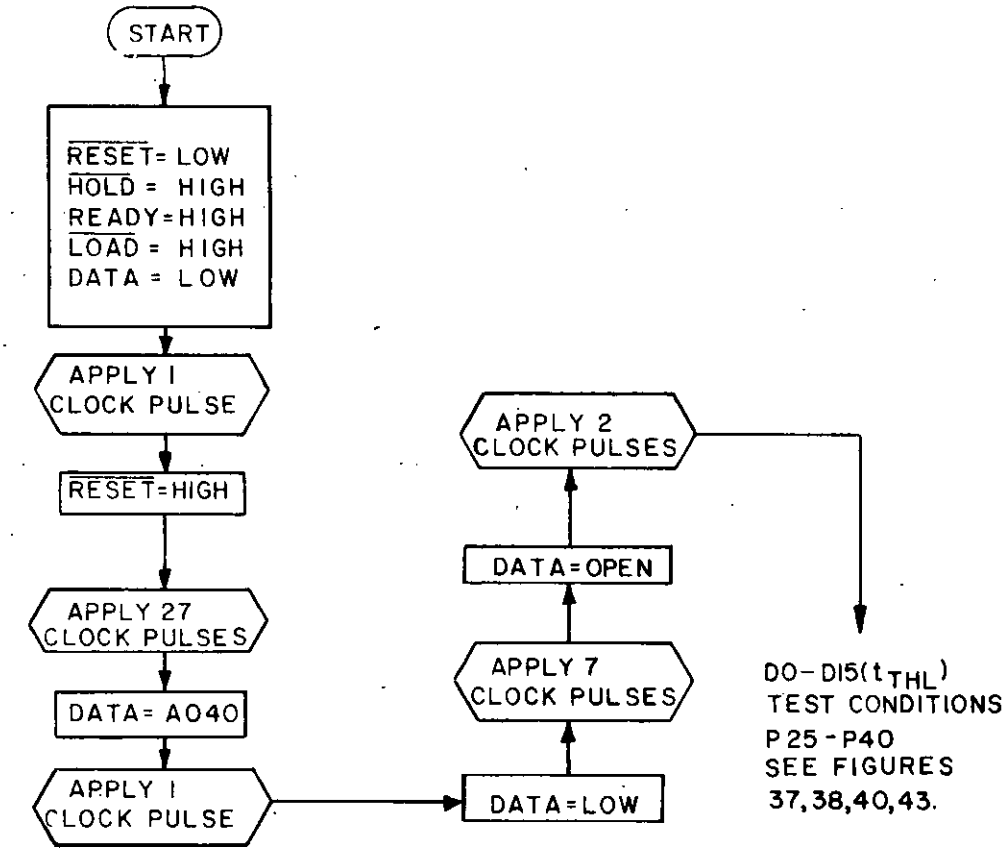
Source: https://assist.dla.mil -- Downloaded: 20260-20T17:35Z
Check the source to verify that this is the current version before use.



NOTES:
1. Data = D0 - D15 pins.
2. D0 = MSB.
3. D15 = LSB.
4. High level = 3.0 ±0.3 V.
5. Low level = 0.0 ±0.2 V.

FIGURE 35. Transition time (t_{THL}) test preconditioning flow chart for table XVI.

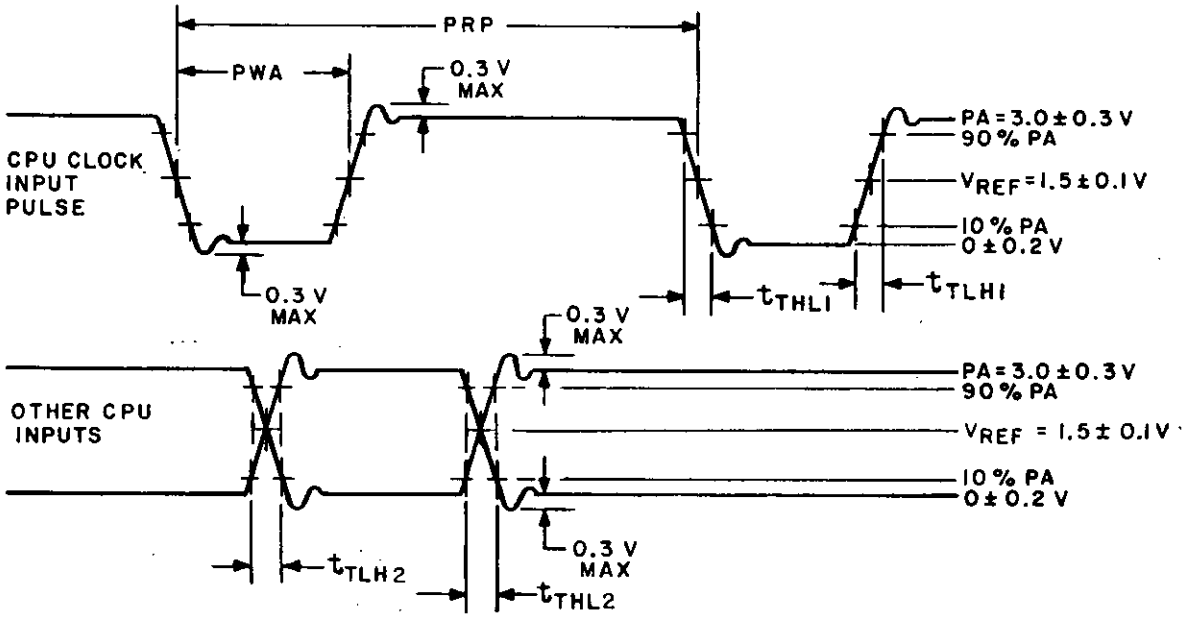
Source: https://assist.dla.mil -- Downloaded: 202610-20T17:35Z
Check the source to verify that this is the current version before use.



NOTES:
1. Data = DO-D15 pins.
2. DO = MSB.
3. D15 = LSB.
4. Open = No connection.
5. High level = 3.0 ±0.3 V.
6. Low level = 0.0 ±0.2 V.

FIGURE 36. Transition time (t_{THL}) test preconditioning flow chart for table XVI.

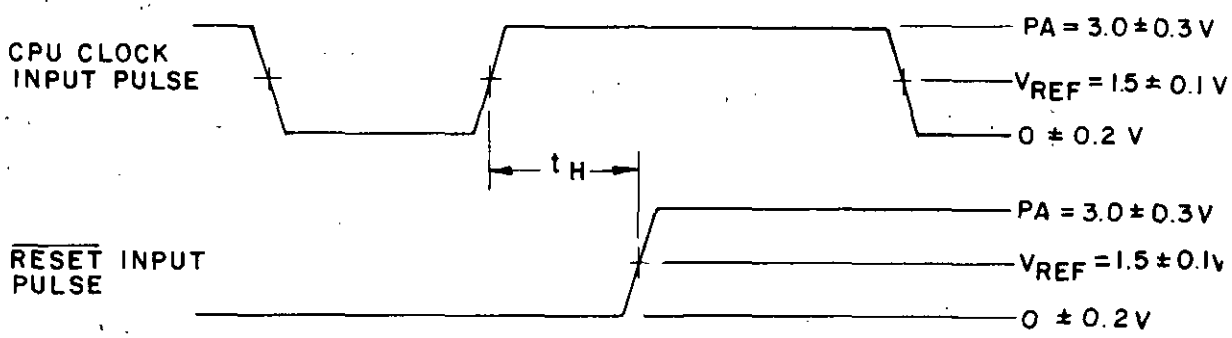
Source: https://assist.dla.mil -- Downloaded: 20260-20T17:35Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters:
   PRP = 1.0 ±0.2 µs.
   PWA = 0.3 ±0.06 µs.
   $t_{TLH1} = t_{TLH2} = t_{THL1} = t_{THL2} = 20 \pm 2$ ns.
2. Other CPU inputs consist of D0-D15, $\overline{\text{RESET}}$, $\overline{\text{HOLD}}$, and READY.

FIGURE 37. Transition time (t_{THL}) test circuit waveforms for table XVI.

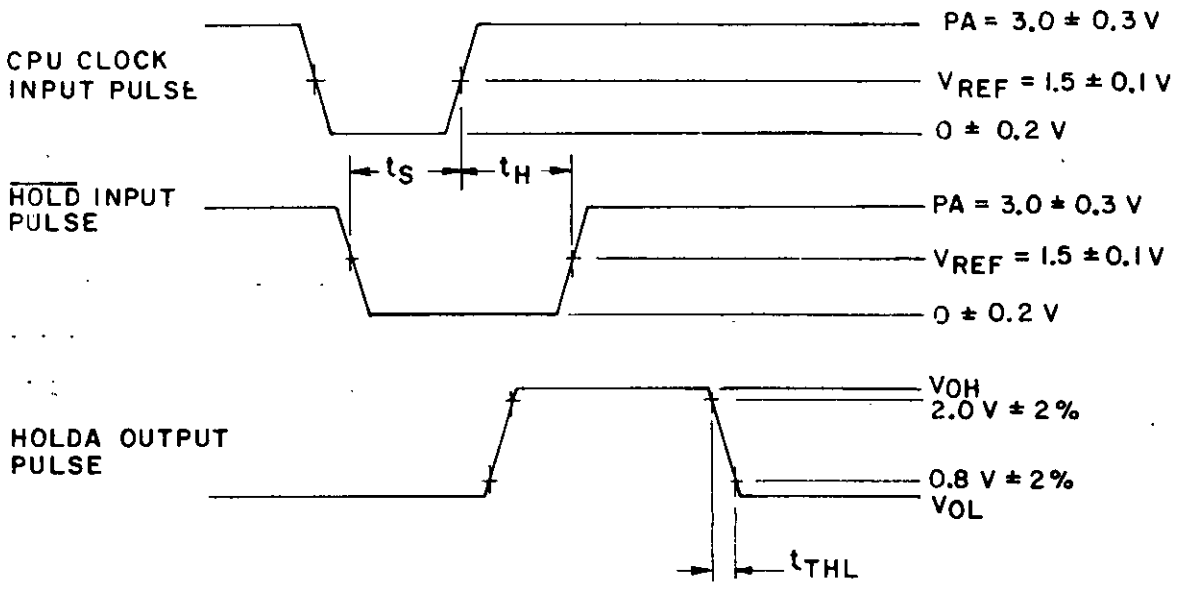
Source: https://assist.dla.mil -- Downloaded: 202610-20T17:35Z
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NOTES:
1. t_H = 100 +5, -0 ns.
2. Input pulse parameters are found on figure 37.

FIGURE 38. Transition time (t_{THL}) test circuit preconditioning waveforms for table XVI.

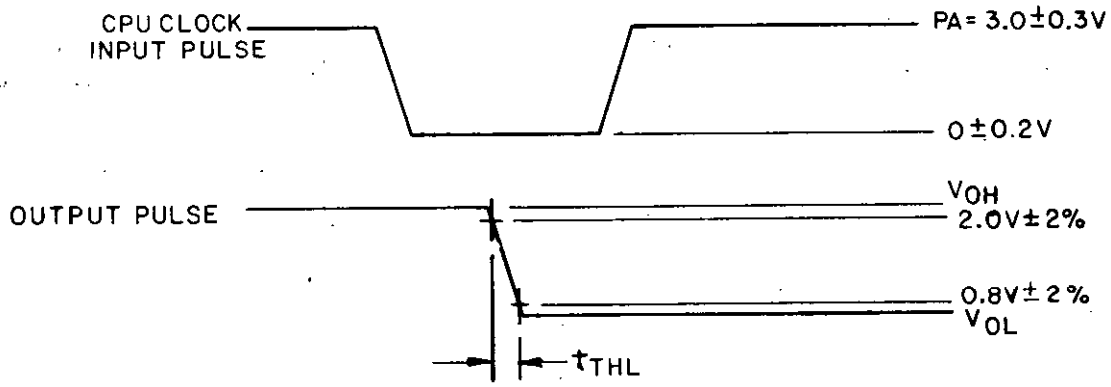
Source: https://assist.dla.mil -- Downloaded: 20260-20T17:35Z
Check the source to verify that this is the current version before use.



NOTES:
1. $t_S = t_H = 100$ ns ±10%.
2. $V_{OH} = 2.4$ V minimum.
3. $V_{OL} = 0.4$ V maximum.
4. Input pulse parameters are found on figure 37.

FIGURE 39. Transition time (t_{THL}) test circuit preconditioning waveforms for table XVI.

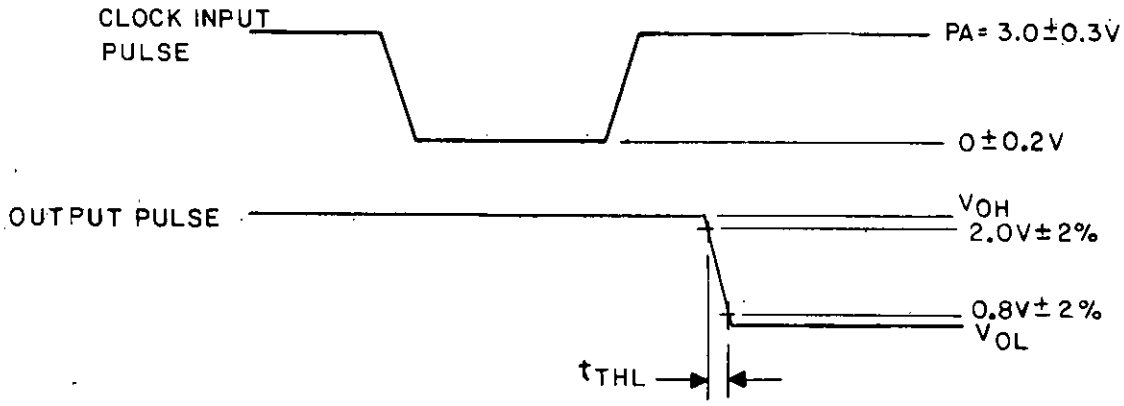
Source: https://assist.dla.mil -- Downloaded: 202610-20T17:35Z
Check the source to verify that this is the current version before use.



NOTES:
1. V_{OH} = 2.4 V minimum.
2. V_{OL} = 0.4 V maximum.
3. Input pulse parameters are found on figure 37.

FIGURE 40. Transition time (t_{THL}) test circuit preconditioning waveforms for table XVI.

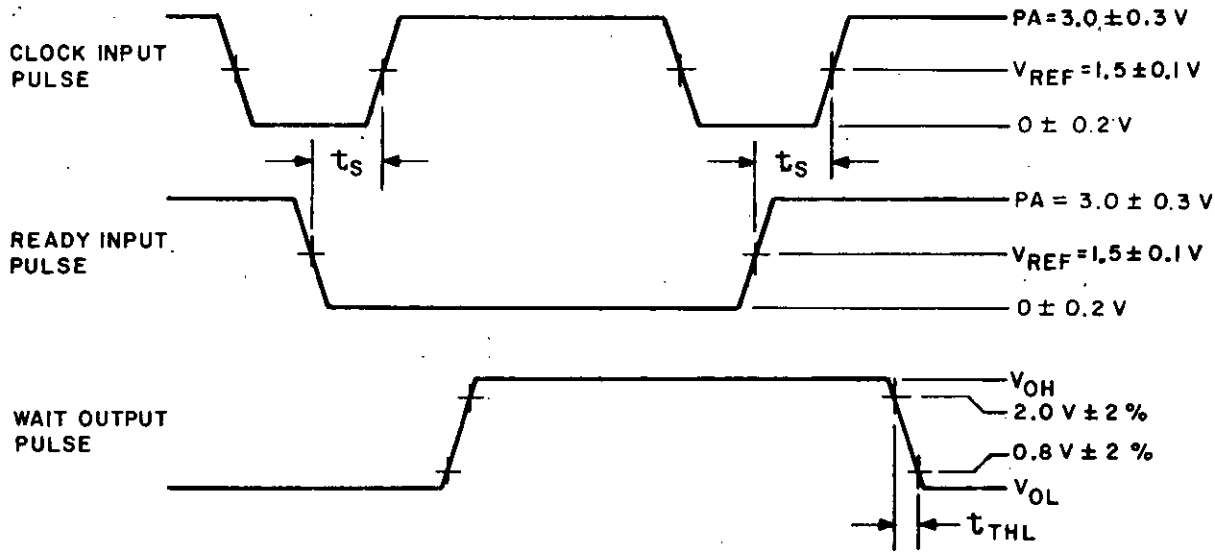
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.



NOTES:
1. V_{OH} = 2.4 V minimum.
2. V_{OL} = 0.4 V maximum.
3. Input pulse parameters are found on figure 37.

FIGURE 41. Transition time (t_{THL}) test circuit preconditioning waveforms for table XVI.

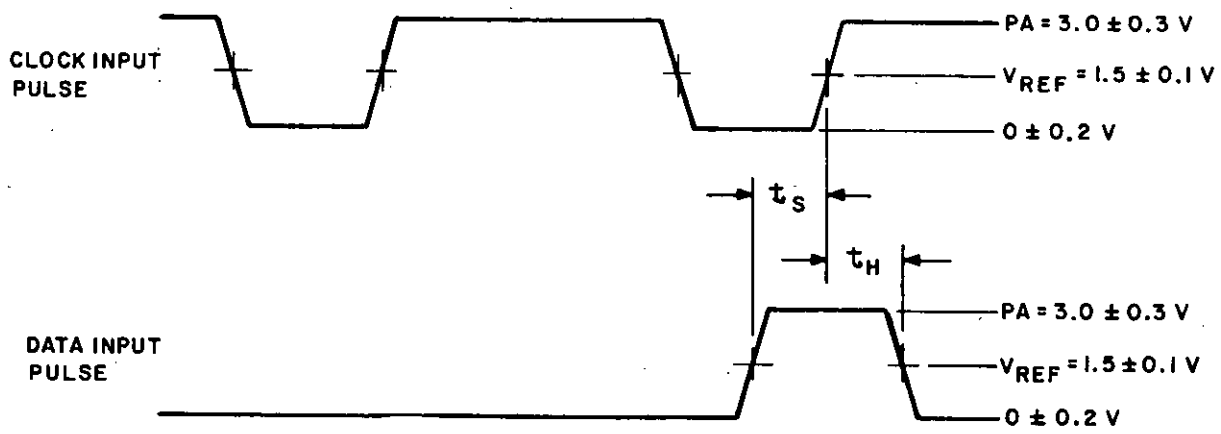
Source: https://assist.dla.mil -- Downloaded: 202610-20T17:35Z
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NOTES:
1. t_S = 100 ±20 ns.
2. Input pulse parameters are found on figure 37.

FIGURE 42. Transition time (t_{THL}) test circuit preconditioning waveforms for table XVI.

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NOTE: $t_S = t_H = 100 \pm 20$ ns.

FIGURE 43. Transition time (t_{THL}) test circuit preconditioning waveforms for table XVI.

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Check the source to verify that this is the current version before use.

TABLE XVIII. Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect: A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Initial limits 25°+5°C ambient: Min	Max	Unit	Test figure
R1	$\overline{EI}$ to $\overline{INTREQ}$ Propagation delay time (t_{PLH}), C_L = 50 pF	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A28	1A27	1A3	Open	C	1A29	10	58	ns	44,49
R2	$\overline{EI}$ to $\overline{INTREQ}$	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A27,1A28	2A30	1A3					43		
R3	LGH to $\overline{INTREQ}$	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A27,1A28	1A3	2A30					61		
R4	P1 to $\overline{INTREQ}$	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A23,1A24, 1A25,1A26,1A27, 1A28		1A22							
R5	P2 to $\overline{INTREQ}$	1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A27,1A28		1A2							
R6	P3 to $\overline{INTREQ}$	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A22,1A23,1A24, 1A25,1A26,1A27, 1A28		1A21							
R7	P4 to $\overline{INTREQ}$	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A25,1A26,1A27, 1A28		1A24							
R8	P5 to $\overline{INTREQ}$	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A24, 1A25,1A26,1A27, 1A28		1A23							
R9	P6 to $\overline{INTREQ}$	1A2,1A4,1A6,1A8, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A27, 1A28		1A5							

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TABLE XVIII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Connect F to pin	Initial limits 25°+5°C ambient Min	Initial limits 25°+5°C ambient Max	Unit	Test figure
R10	P7 to $\overline{\text{INTREQ}}$ Propagation delay time (t_{PLH}), C_L = 50 pF	1A2,1A5,1A6,1A8, 1A9,1A11,1A21, 1A22,1A23,1A24 1A25,1A26,1A27, 1A28	1A3	1A4	Open	C	1A29	10	61	ns	44,49
R11	P8 to $\overline{\text{INTREQ}}$	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A28		1A27							
R12	P9 to $\overline{\text{INTREQ}}$	1A2,1A4,1A5,1A8, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A27, 1A28		1A6							
R13	P10 to $\overline{\text{INTREQ}}$	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A27, 1A28		1A26							
R14	P11 to $\overline{\text{INTREQ}}$	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A26,1A27, 1A28		1A25							
R15	P12 to $\overline{\text{INTREQ}}$	1A2,1A4,1A5,1A6, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A27, 1A28		1A8							
R16	P13 to $\overline{\text{INTREQ}}$	1A2,1A4,1A5,1A6, 1A8,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A27, 1A28		1A9							
R17	P14 to $\overline{\text{INTREQ}}$	1A2,1A4,1A5,1A6, 1A8,1A9,1A21, 1A22,1A23,1A24, 1A25,1A26,1A27, 1A28		1A11							
R18	P15 to $\overline{\text{INTREQ}}$	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A27		1A28							

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TABLE XVIII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Connect F to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
R19	$\overline{E1}$ to ICO Propagation delay time (t_{PLH}), C_L = 50 pF	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A27,1A28	Open	1A3	Open	C	1A34	10	56	ns	44,48
R20	LGH to ICO	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A27,1A28	1A3	2A30					51		44,47
R21	P1 to ICO	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A23,1A24, 1A25,1A26,1A27, 1A28		1A22							
R22	P2 to ICO	1A4,1A5,1A6,1A8, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A27 1A28		1A2							
R23	P3 to ICO	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A22,1A23,1A24, 1A25,1A26,1A27, 1A28		1A21							
R24	P4 to ICO	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A25,1A26,1A27, 1A28		1A24							
R25	P5 to ICO	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A24, 1A25,1A26,1A27, 1A28		1A23							
R26	P6 to ICO	1A2,1A4,1A6,1A8, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A27, 1A28		1A5							
R27	P7 to ICO	1A2,1A5,1A6,1A8, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A27, 1A28		1A4							

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TABLE XVIII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Connect F to pin	Initial limits 25°+5°C ambient Min	Initial limits 25°+5°C ambient Max	Unit	Test figure
R28	$\overline{EI}$ to IC1 Propagation delay time (t_{PLH}), C_L = 50 pF	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A28	1A27	1A3	Open	C	1A33	10	61	ns	44,49
R29	$\overline{EI}$ to IC1	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A27,1A28	2A30	1A3					46		
R30	LGH to IC1	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A27,1A28	1A3	2A30							
R31	PI to IC1	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A23,1A24, 1A25,1A26,1A27, 1A28		1A22							
R32	P2 to IC1	1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A27,1A28		1A2							
R33	P3 to IC1	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A22,1A23,1A24, 1A25,1A26,1A27, 1A28		1A21							
R34	P4 to IC1	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A25,1A26,1A28	1A3,1A27	1A24					56		44,48
R35	P5 to IC1	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A24, 1A25,1A26,1A28	1A3,1A27	1A23					56		44,,48
R36	P6 to IC1	1A2,1A4,1A6,1A8, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A28	1A3,1A27	1A5					56		44,48

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TABLE XVIII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Connect F to pin	Initial limits 25°+5°C ambient Min	Initial limits 25°+5°C ambient Max	Unit	Test figure
R37	P7 to IC1 Propagation delay time (t_{PLH}), C_L = 50 pF	1A2,1A5,1A6,1A8, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A28	1A3,1A27	1A4	Open	C	1A33	10	56	ns	44,48
R38	P8 to IC1	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A28	1A3	1A27					46		44,49
R39	P9 to IC1	1A2,1A4,1A5,1A8, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A27, 1A28		1A6							
R40	P10 to IC1	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A27, 1A28		1A26							
R41	P11 to IC1	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A26,1A27, 1A28		1A25							
R42	$\overline{EI}$ to IC2	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A28	1A27	1A3			1A12		61		
R43	$\overline{EI}$ to IC2	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A27,1A28	2A30	1A3					46		
R44	LGH to IC2	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A27,1A28	1A3	2A30					46		
R45	P1 to IC2	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A23,1A24, 1A25,1A26,1A27, 1A28	1A3	1A22					46		
R46	P2 to IC2	1A4,1A5,1A6,1A8, 1A9,1A11,1A21, 1A22,1A23,1A25, 1A26,1A27,1A28	1A3,1A24	1A2					50		44,48

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TABLE XVIII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
R47	P2 to IC2 Propagation delay time (t_{PLH}), C_L = 50 pF	1A4,1A5,1A6,1A8, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A28	1A3,1A27	1A2	Open	C	1A12	10	56	ns	44,48
R48	P3 to IC2	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A22,1A23,1A25, 1A26,1A27,1A28	1A3,1A24	1A21					50		
R49	P3 to IC2	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A28	1A3,1A27	1A21					56		
R50	P4 to IC2	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A25,1A26,1A27, 1A28	1A3	1A24					46		44,49
R51	P5 to IC2	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A24, 1A25,1A26,1A27, 1A28	1A3	1A23					46		44,49
R52	P6 to IC2	1A2,1A4,1A6,1A8, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A28	1A3,1A27	1A5					56		44,48
R53	P7 to IC2	1A2,1A5,1A6,1A8, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A28	1A3,1A27	1A4					56		44,48
R54	P8 to IC2	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A28	1A3	1A27					46		44,49
R55	P9 to IC2	1A2,1A4,1A5,1A8, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A27, 1A28	1A3	1A6					46		44,49

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TABLE XVIII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect: A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Initial limits 25°+5°C ambient: Min	Max	Unit	Test figure
R56	P10 to IC2 Propagation delay time (t_{PLH}), C_L = 50 pF	1A2,1A4,1A5, 1A6,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A27, 1A28	1A3,1A8	1A26	Open	C	1A12	10	50	ns	44,48
R57	P11 to IC2	1A2,1A4,1A5,1A6, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A26,1A27,1A28	1A3,1A8	1A25					50		44,48
R58	P12 to IC2	1A2,1A4,1A5,1A6, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A27, 1A28	1A3	1A8					46		44,49
R59	P13 to IC2	1A2,1A4,1A5,1A6, 1A8,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A27, 1A28	1A3	1A9					46		
R60	$\overline{EI}$ to IC3	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A28	1A27	1A3			1A7		61		
R61	$\overline{EI}$ to IC3	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A27,1A28	2A30	1A3					46		
R62	LGH to IC3	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A27,1A28	1A3	2A30					46		
R63	P1 to IC3	1A4,1A5,1A6,1A8, 1A9,1A11,1A21, 1A23,1A24,1A25, 1A26,1A27,1A28	1A2,1A3	1A22					50		44,48
R64	P1 to IC3	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A23,1A24, 1A25,1A26,1A28	1A3,1A27	1A22					56		44,48

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TABLE XVIII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect: A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Initial limits 25°+5°C ambient: Min	Max	Unit	Test figure
R65	P2 to IC3 Propagation delay time (t_{PLH}), C_L = 50 pF	1A4,1A5,1A6,1A8, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A27, 1A28	1A3	1A2	Open	C	1A7	10	46	ns	44,49
R66	P3 to IC3	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A22,1A23, 1A25,1A26,1A27, 1A28	1A3,1A24	1A21					50		44,48
R67	P3 to IC3	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A22,1A23,1A24, 1A25,1A26,1A28	1A3,1A27	1A21					56		44,48
R68	P4 to IC3	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A25,1A26,1A27, 1A28	1A3	1A24					46		44,49
R69	P5 to IC3	1A2,1A4,1A6,1A8, 1A9,1A11,1A21, 1A22,1A24,1A25, 1A26,1A27,1A28	1A3,1A5	1A23					50		44,48
R70	P5 to IC3	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A24, 1A25,1A26,1A28	1A3,1A27	1A23					56		44,48
R71	P6 to IC3	1A2,1A4,1A6,1A8, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A27, 1A28	1A3	1A5					46		44,49
R72	P7 to IC3	1A2,1A5,1A6,1A8, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A28	1A3,1A27	1A4					56		44,48
R73	P8 to IC3	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A28	1A3	1A27					46		44,49

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TABLE XVIII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Connect F to pin	Initial limits 25°+5°C ambient Min	Initial limits 25°+5°C ambient Max	Unit	Test figure
R74	P9 to IC3 Propagation delay time (t_{PLH}), C_L = 50 pF	1A2,1A4,1A5, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A27, 1A28	1A3,1A26	1A6	Open	C	1A7	10	50	ns	44,48
R75	P10 to IC3	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A27, 1A28	1A3	1A26					46		44,49
R76	P11 to IC3	1A2,1A4,1A5,1A6, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A26,1A27,1A28	1A3,1A8	1A25					50		44,48
R77	P12 to IC3	1A2,1A4,1A5,1A6, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A27, 1A28	1A3	1A8					46		44,49
R78	P13 to IC3	1A2,1A4,1A5,1A6, 1A8,1A21,1A22, 1A23,1A24,1A25, 1A26,1A27,1A28	1A3,1A11	1A9					50		44,48
R79	P14 to IC3	1A2,1A4,1A5,1A6, 1A8,1A9,1A21, 1A22,1A23,1A24, 1A25,1A26,1A27, 1A28	1A3	1A11					46		44,48
R80	CLK to Q1	2A13	Open	2A22			1A36	5	29		44,46
R81	CLK to Q2	2A13		2A22			1A40				44,46
R82	PRESET to Q1	Open		2A13	2A30		1A36				44,50
R83	PRESET to Q2	Open		2A13	2A30		1A40				44,50

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TABLE XIX. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Connect: E to pin	Connect: F to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
S1	$\overline{EI}$ to $\overline{INTREQ}$ Propagation delay time (t_{PHL}), C_L = 50 pF	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A28	1A27	1A3	Open	C	1A29	10	68	ns	44,49
S2	$\overline{EI}$ to $\overline{INTREQ}$	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A27,1A28	2A30	1A3					38		
S3	LGH to $\overline{INTREQ}$	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A27,1A28	1A3	2A30					48		
S4	P1 to $\overline{INTREQ}$	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A23,1A24, 1A25,1A26,1A27, 1A28		1A22							
S5	P2 to $\overline{INTREQ}$	1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A27,1A28		1A2							
S6	P3 to $\overline{INTREQ}$	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A22,1A23;1A24, 1A25,1A26,1A27, 1A28		1A21							
S7	P4 to $\overline{INTREQ}$	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A25,1A26,1A27, 1A28		1A24							
S8	P5 to $\overline{INTREQ}$	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A24, 1A25,1A26,1A27, 1A28		1A23							
S9	P6 to $\overline{INTREQ}$	1A2,1A4,1A6,1A8, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A27, 1A28		1A5							

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TABLE XIX. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Connect F to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
S10	P7 to $\overline{\text{INTREQ}}$ Propagation delay time (t_{PHL}), C_L = 50 pF	1A2,1A5,1A6,1A8, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A27, 1A28	1A3	1A4	Open	C	1A29	10	48	ns	44,49
S11	P8 to $\overline{\text{INTREQ}}$	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A28		1A27							
S12	P9 to $\overline{\text{INTREQ}}$	1A2,1A4,1A5,1A8, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A27, 1A28		1A6							
S13	P10 to $\overline{\text{INTREQ}}$	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A27, 1A28		1A26							
S14	P11 to $\overline{\text{INTREQ}}$	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A26,1A27, 1A28		1A25							
S15	P12 to $\overline{\text{INTREQ}}$	1A2,1A4,1A5,1A6, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A27, 1A28		1A8							
S16	P13 to $\overline{\text{INTREQ}}$	1A2,1A4,1A5,1A6, 1A8,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A27, 1A28		1A9							
S17	P14 to $\overline{\text{INTREQ}}$	1A2,1A4,1A5,1A6, 1A8,1A9,1A21, 1A22,1A23,1A24, 1A25,1A26,1A27, 1A28		1A11							
S18	P15 to $\overline{\text{INTREQ}}$	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A27		1A28							

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TABLE XIX. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Connect F to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
S19	$\overline{E1}$ to ICO Propagation delay time (t_{PHL}), C_L = 50 pF	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A27,1A28	Open	1A3	Open	C	1A34	10	41	ns	44,48
S20	LGH to ICO	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A27,1A28	1A3	2A30					36		44,47
S21	P1 to ICO	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A23,1A24, 1A25,1A26,1A27, 1A28		1A22							
S22	P2 to ICO	1A4,1A5,1A6,1A8, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A27, 1A28		1A2							
S23	P3 to ICO	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A22,1A23,1A24, 1A25,1A26,1A27, 1A28		1A21							
S24	P4 to ICO	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A25,1A26,1A27, 1A28		1A24							
S25	P5 to ICO	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A24, 1A25,1A26,1A27, 1A28		1A23							
S26	P6 to ICO	1A2,1A4,1A6,1A8, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A27, 1A28		1A5							
S27	P7 to ICO	1A2,1A5,1A6,1A8, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A27, 1A28		1A4							

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TABLE XIX. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect: A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Initial limits 25°+5°C ambient: Min	Max	Unit	Test figure
S28	$\overline{E1}$ to IC1 Propagation delay time (t_{PHL}), C_L = 50 pF	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A28	1A27	1A3	Open	C	1A33	10	68	ns	44,49
S29	$\overline{E1}$ to IC1	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A27,1A28	2A30	1A3					38		
S30	LGH to IC1	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A27,1A28	1A3	2A30					37		
S31	P1 to IC1	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A23,1A24, 1A25,1A26,1A27, 1A28		1A22							
S32	P2 to IC1	1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A27,1A28		1A2							
S33	P3 to IC1	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A22,1A23,1A24, 1A25,1A26,1A27, 1A28		1A21							
S34	P4 to IC1	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A25,1A26,1A28	1A3,1A27	1A24					63		44,48
S35	P5 to IC1	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A24, 1A25,1A26,1A28	1A3,1A27	1A23					63		44,48
S36	P6 to IC1	1A2,1A4,1A6,1A8, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A28	1A3,1A27	1A5					63		44,48

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TABLE XIX. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Connect F to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
S37	P7 to IC1 Propagation delay time (t_{PHL}), C_L = 50 pF	1A2,1A5,1A6,1A8, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A28	1A3,1A27	1A4	Open	C	1A33	10	63	ns	44,48
S38	P8 to IC1	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A28	1A3	1A27					37		44,49
S39	P9 to IC1	1A2,1A4,1A5,1A8, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A27, 1A28		1A6							
S40	P10 to IC1	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A27, 1A28		1A26							
S41	P11 to IC1	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A26,1A27, 1A28		1A25							
S42	$\overline{EI}$ to IC2	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A28	1A27	1A3			1A12		68		
S43	$\overline{EI}$ to IC2	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A27,1A28	2A30	1A3					38		
S44	LGH to IC2	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A27,1A28	1A3	2A30					37		
S45	P1 to IC2	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A23,1A24, 1A25,1A26,1A27, 1A28	1A3	1A22					37		
S46	P2 to IC2	1A4,1A5,1A6,1A8, 1A9,1A11,1A21, 1A22,1A23,1A25, 1A26,1A27,1A28	1A3,1A24	1A2					42		44,48

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TABLE XIX. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Connect F to pin	Initial limits 25°+5°C ambient Min	Initial limits 25°+5°C ambient Max	Unit	Test figure
S47	P2 to IC2 Propagation delay time (t_{PHL}), C_L = 50 pF	1A4,1A5,1A6,1A8, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A28	1A3,1A27	1A2	Open	C	1A12	10	63	ns	44,48
S48	P3 to IC2	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A22,1A23,1A25, 1A26,1A27,1A28	1A3,1A24	1A21					42		
S49	P3 to IC2	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A22,1A23,1A24, 1A25,1A26,1A28	1A3,1A27	1A21					63		
S50	P4 to IC2	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A25,1A26,1A27, 1A28	1A3	1A24					37		44,49
S51	P5 to IC2	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A24, 1A25,1A26,1A27, 1A28	1A3	1A23					37		44,49
S52	P6 to IC2	1A2,1A4,1A6,1A8, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A28	1A3,1A27	1A5					63		44,48
S53	P7 to IC2	1A2,1A5,1A6,1A8, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A28	1A3,1A27	1A4					63		44,48
S54	P8 to IC2	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A28	1A3	1A27					37		44,49
S55	P9 to IC2	1A2,1A4,1A5,1A8, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A27, 1A28	1A3	1A6					37		44,49

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TABLE XIX. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Connect F to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
S56	P10 to IC2 Propagation delay time (t_{PHL}), C_L = 50 pF	1A2,1A4,1A5, 1A6,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A27, 1A28	1A3,1A8	1A26	Open	C	1A12	10	42	ns	44,49
S57	P11 to IC2	1A2,1A4,1A5,1A6, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A26,1A27,1A28	1A3,1A8	1A25					42		44,48
S58	P12 to IC2	1A2,1A4,1A5,1A6, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A27, 1A28	1A3	1A8					37		44,49
S59	P13 to IC2	1A2,1A4,1A5,1A6, 1A8,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A27, 1A28	1A3	1A9					37		
S60	$\overline{E1}$ to IC3	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A28	1A27	1A3			1A7		68		
S61	$\overline{EI}$ to IC3	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A27,1A28	2A30	1A3					38		
S62	LGH to IC3	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A27,1A28	1A3	2A30					37		
S63	P1 to IC3	1A4,1A5,1A6,1A8, 1A9,1A11,1A21, 1A23,1A24,1A25, 1A26,1A27,1A28	1A2,1A3	1A22					42		44,48
S64	P1 to IC3	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A23,1A24, 1A25,1A26,1A28	1A3,1A27	1A22					63		44,48

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TABLE XIX. Propagation delay time (t_{PHL}) test - Continued.

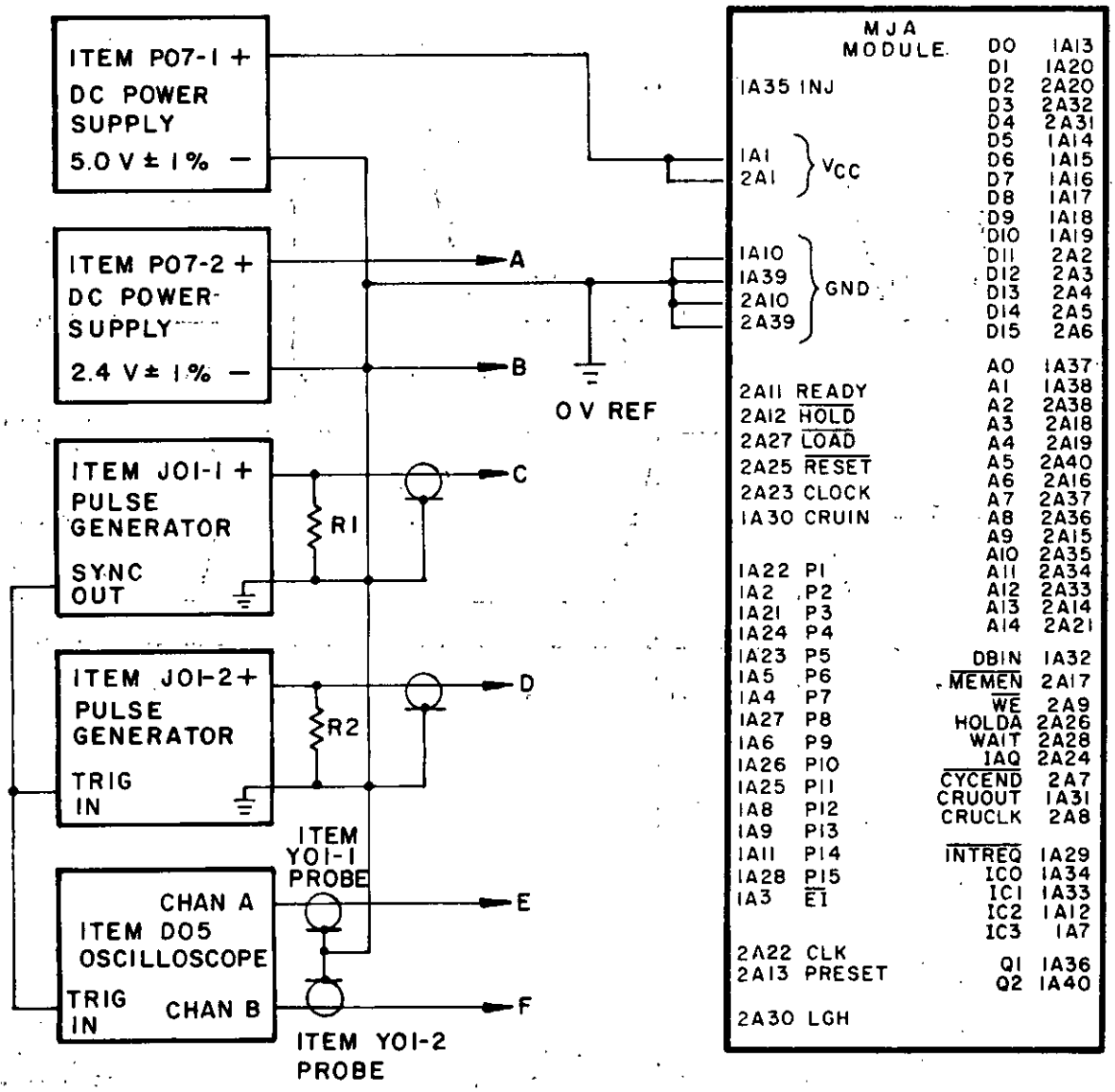
Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Connect F to pin	Initial limits 25°+5°C ambient Min	Initial limits 25°+5°C ambient Max	Unit	Test figure
S65	P2 to IC3 Propagation delay time (t_{PHL}), C_L = 50 pF	1A4,1A5,1A6,1A8, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A27, 1A28	1A3	1A2	Open	C	1A7	10	37	ns	44,49
S66	P3 to IC3	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A22,1A23,1A25, 1A26,1A27,1A28	1A3,1A24	1A21					42		44,48
S67	P3 to IC3	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A22,1A23,1A24, 1A25,1A26,1A28	1A3,1A27	1A21					63		44,48
S68	P4 to IC3	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A25,1A26,1A27, 1A28	1A3	1A24					37		44,49
S69	P5 to IC3	1A2,1A4,1A6,1A8, 1A9,1A11,1A21, 1A22,1A24,1A25, 1A26,1A27,1A28	1A3,1A5	1A23					42		44,48
S70	P5 to IC3	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A24, 1A25,1A26,1A28	1A3,1A27	1A23					63		44,49
S71	P6 to IC3	1A2,1A4,1A6,1A8, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A27, 1A28	1A3	1A5					37		
S72	P7 to IC3	1A2,1A5,1A6,1A8, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A28	1A3,1A27	1A4					63		
S73	P8 to IC3	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A26, 1A28	1A3	1A27					37		

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TABLE XIX. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Connect F to pin	Initial limits 25°+5°C ambient Min	Initial limits 25°+5°C ambient Max	Unit	Test figure
S74	P9 to IC3 Propagation delay time (t_{PHL}), C_L = 50 pF	1A2,1A4,1A5, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A27, 1A28	1A3,1A26	1A6	Open	C	1A7	10	42	ns	44,48
S75	P10 to IC3	1A2,1A4,1A5,1A6, 1A8,1A9,1A11, 1A21,1A22,1A23, 1A24,1A25,1A27, 1A28	1A3	1A26					37		44,49
S76	P11 to IC3	1A2,1A4,1A5,1A6, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A26,1A27,1A28	1A3,1A8	1A25					42		44,48
S77	P12 to IC3	1A2,1A4,1A5,1A6, 1A9,1A11,1A21, 1A22,1A23,1A24, 1A25,1A26,1A27, 1A28	1A3	1A8					37		44,49
S78	P13 to IC3	1A2,1A4,1A5,1A6, 1A8,1A21,1A22, 1A23,1A24,1A25, 1A26,1A27,1A28	1A3,1A11	1A9					42		44,48
S79	P14 to IC3	1A2,1A4,1A5,1A6, 1A8,1A9,1A21, 1A22,1A23,1A24, 1A25,1A26,1A27, 1A28	1A3	1A11					37		44,49
S80	CLK to Q1	2A13	Open	2A22			1A36	5	44		44,46
S81	CLK to Q2	2A13		2A22			1A40				44,46
S82	CLEAR to Q1	Open		2A13	2A30	D	1A36				44,50
S83	CLEAR to Q2	Open		2A13	2A30	D	1A40				44,50

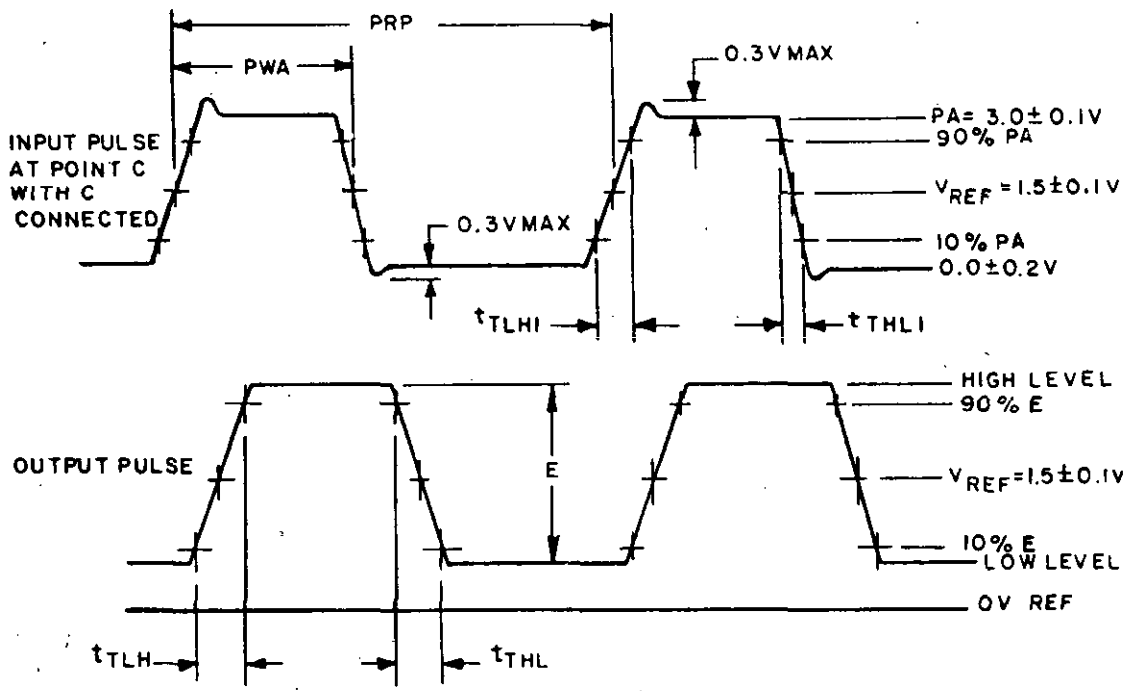
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load each output under test as specified on figure 22.

FIGURE 44. Transition time (t_{THL} and t_{TLH}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XVI, XVII, XVIII and XIX.

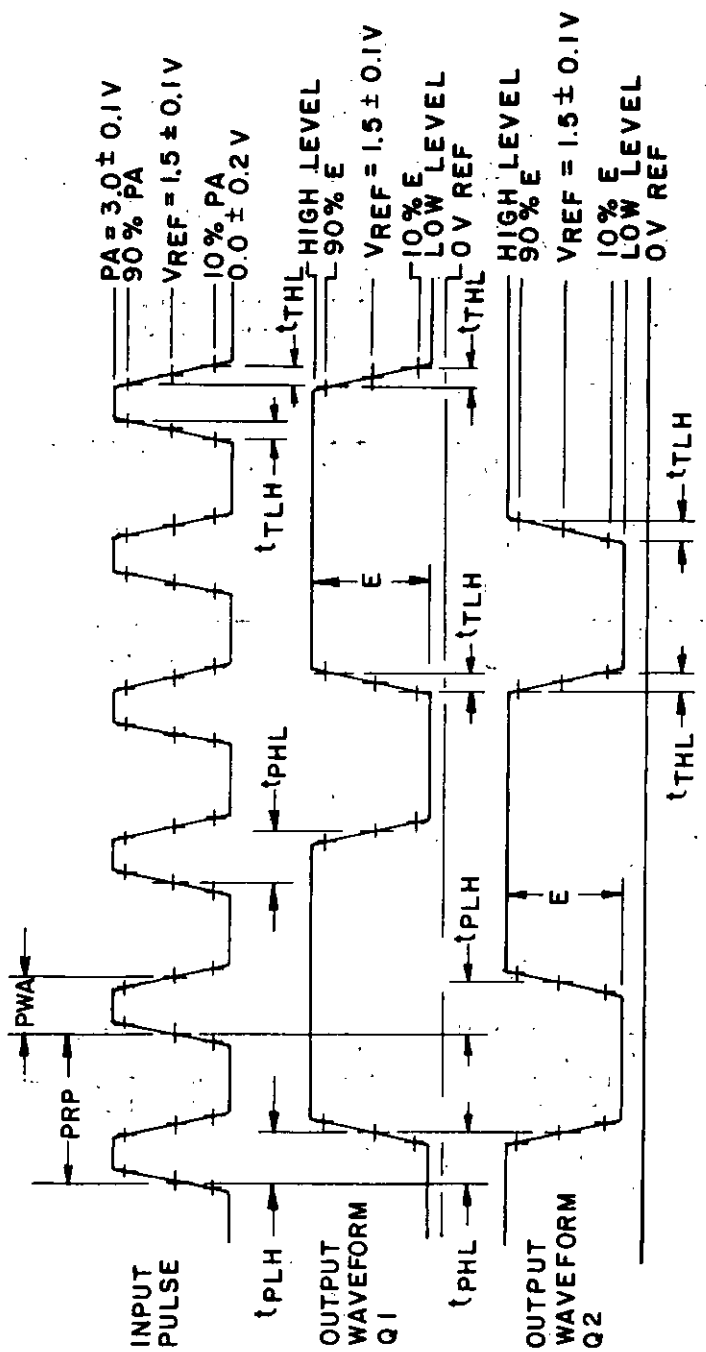
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NOTES:
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1}$ = 10 ±2 ns.
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. Before making the measurements specified in table XVI and XVII verify that the output pulse high level is ≥ 2.4 V and the low level is ≤ 0.4 V and ≤ 0.5 V for test conditions P42 and Q2 only.
3. The lead length of electrical conductors between resistor R1 and input pins; between probes Y01-1 and Y01-2 and output pins; and between load circuits and output pins, shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

FIGURE 45. Transition time (t_{THL} and t_{TLH}) test circuit waveforms for tables XVI and XVII.

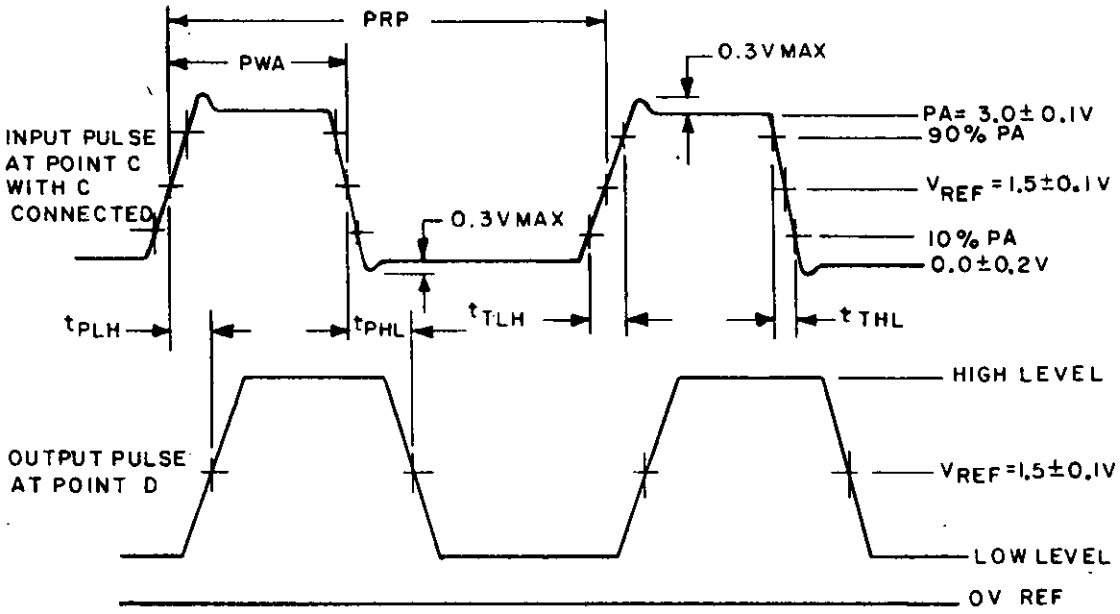
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NOTES:
1. Input pulse parameters:
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
   PWA = 400 ±40 ns.
   PRP = 1,000 ±100 ns.
2. Before making the measurements specified in tables XVI, XVII, XVIII and XIX, verify that the output pulse high level is ≥ 2.5 V and the low level is ≤ 0.5 V.
3. The lead length of electrical conductors between resistor R1 and input pins; between items YO1-1 and YO1-2 and output pins; and between load circuit and output pins, shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

FIGURE 46. Transition time (t_{THL} and t_{TLH}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XVI, XVII, XVIII and XIX.

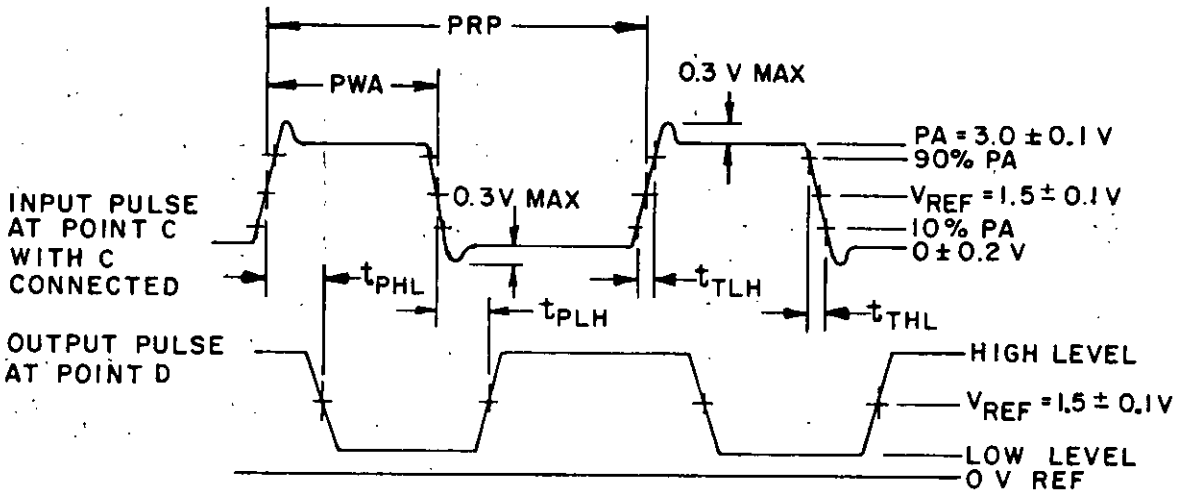
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NOTES:
1. Input pulse parameters:
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. Before making the measurements specified in tables XVIII and XIX, verify that the output pulse high level is $\geq$ 2.5 V and the low level is $\leq$ 0.5 V.
3. The lead length of electrical conductors between resistor R1 and input pins; between items Y01-1 and Y01-2 and output pins; and between load circuits and output pins, shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

FIGURE 47. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XVIII and XIX.

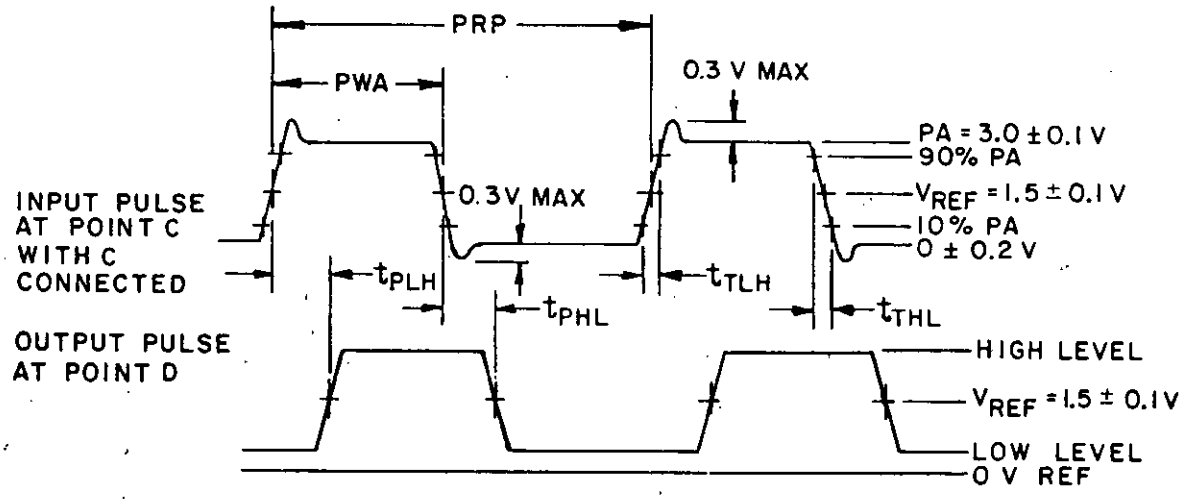
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NOTES:
1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. Before making the measurements specified in tables XVIII and XIX, verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$ 0.4 V.
3. The lead length of electrical conductors between resistor R1 and input pins; between probes YO1-1 and YO1-2 and output pins; and between load circuits and output pins, shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

FIGURE 48. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XVIII and XIX.

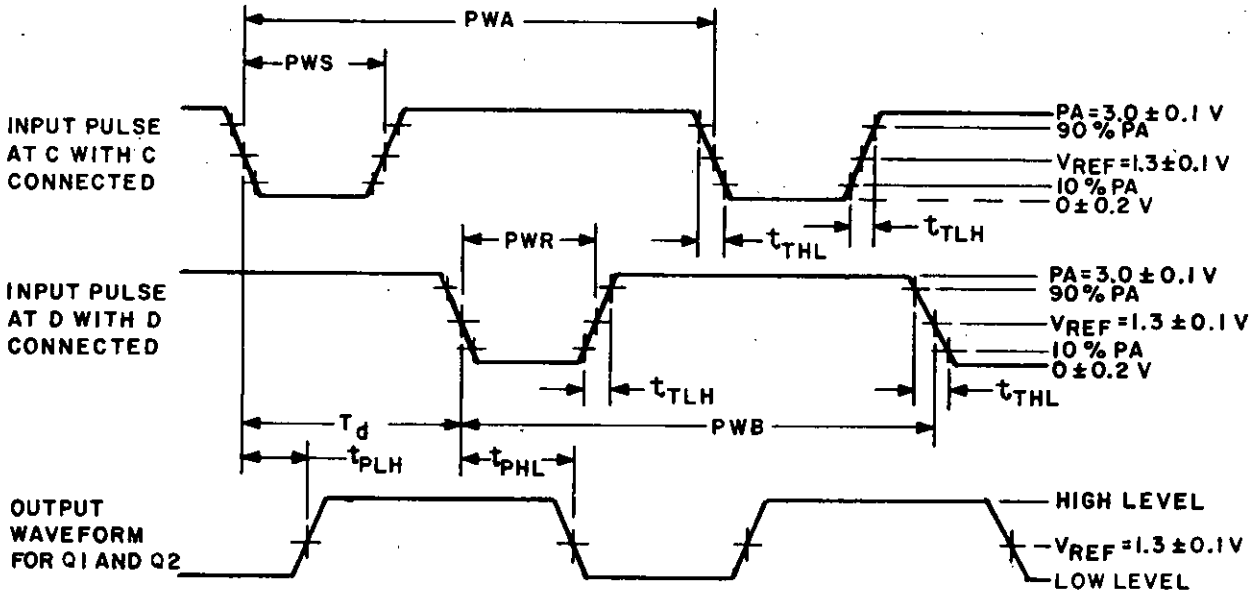
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NOTES:
1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. Before making the measurements specified in tables XVIII and XIX, verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$ 0.4 V.
3. The lead length of electrical conductors between resistor R1 and input pins; between probes Y01-1 and Y01-2 and output pins; and between load circuits and output pins, shall be less than 1 inch (25.4 mm).
4. All voltages are reference to ground.

FIGURE 49. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XVIII and XIX.

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NOTES:
1. Adjust items J01-1 and J01-2 to provide input pulse parameters as follows:
   PWS = PWR = 100 +2, -0 ns.
   PWA = PWB = 400 ±40 ns.
   T_d = 200 ±20 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. All voltage levels are referenced to ground.

FIGURE 50. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XVIII and XIX.

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TABLE XX. Dynamic functional (minimum and maximum injector current) test and propagation delay time (t_{PLH} and t_{PHL}) check.

Test condition	Test description	Input table	Output table	Waveform	Test figure
T1	Dynamic functional test, minimum injector current and propagation delay time check	XV	XXI	51	26
T2	Dynamic functional test, maximum injector current and propagation delay time check	XV	XXI	51	26

NOTE: Apply the vectors (see 6.3) with the input parameters of table XV and checking the output parameters of table XXI.

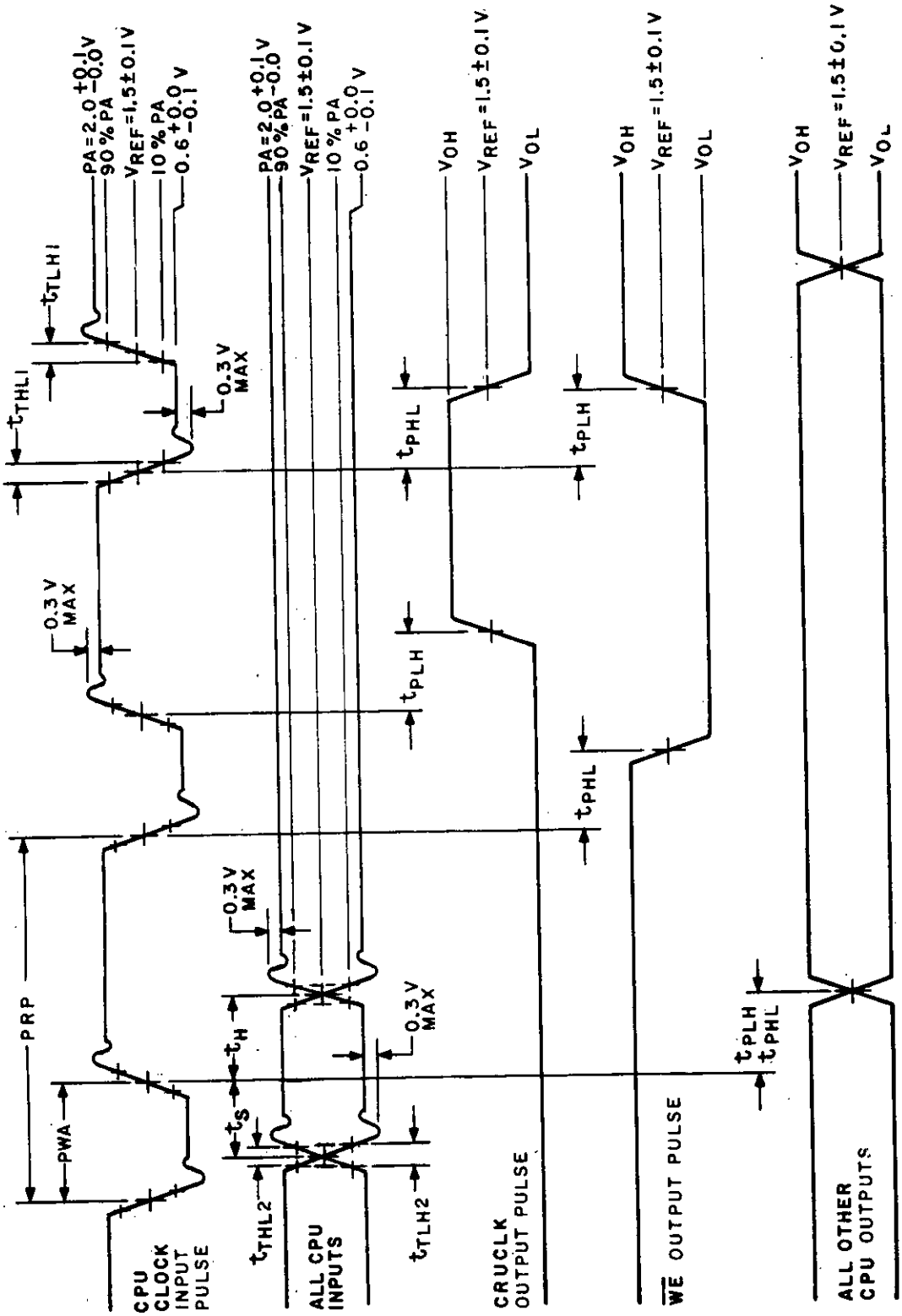
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TABLE XXI. CPU output conditions for table XX.

Parameter	Pin(s)	Output conditions		Unit	Test figure
		Min	Max		
A0-A14 Propagation delay time	1A37, 1A38, 2A14, 2A15, 2A16, 2A17, 2A18, 2A19, 2A21, 2A33, 2A34, 2A35, 2A36, 2A37, 2A38, 2A40	---	140	ns	26, 51
D0-D15	1A13, 1A14, 1A15, 1A16, 1A17, 1A18, 1A19, 1A20, 2A2, 2A3, 2A4, 2A5, 2A6, 2A20, 2A31, 2A32	---	155		
DBIN	1A32	---	160		
$\overline{\text{MEMEN}}$	2A17	---	140		
$\overline{\text{WE}}$	2A28	---	185		
HOLDA	2A26	---	290		
WAIT	2A28	---	130		
IAQ	2A24	---	130		
$\overline{\text{CYCEND}}$	2A7	---	145		
CRUOUT	1A31	---	185		
CRUCLK	2A8	---	155		
Output switch voltage		1.5	1.5	V	

NOTE: Check propagation delay times by starting output comparison at maximum propagation delay time limit and continuing for 5 ns.

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NOTES:
1. All input parameters are found in table XV.
2. All output parameters are found in table XXI.

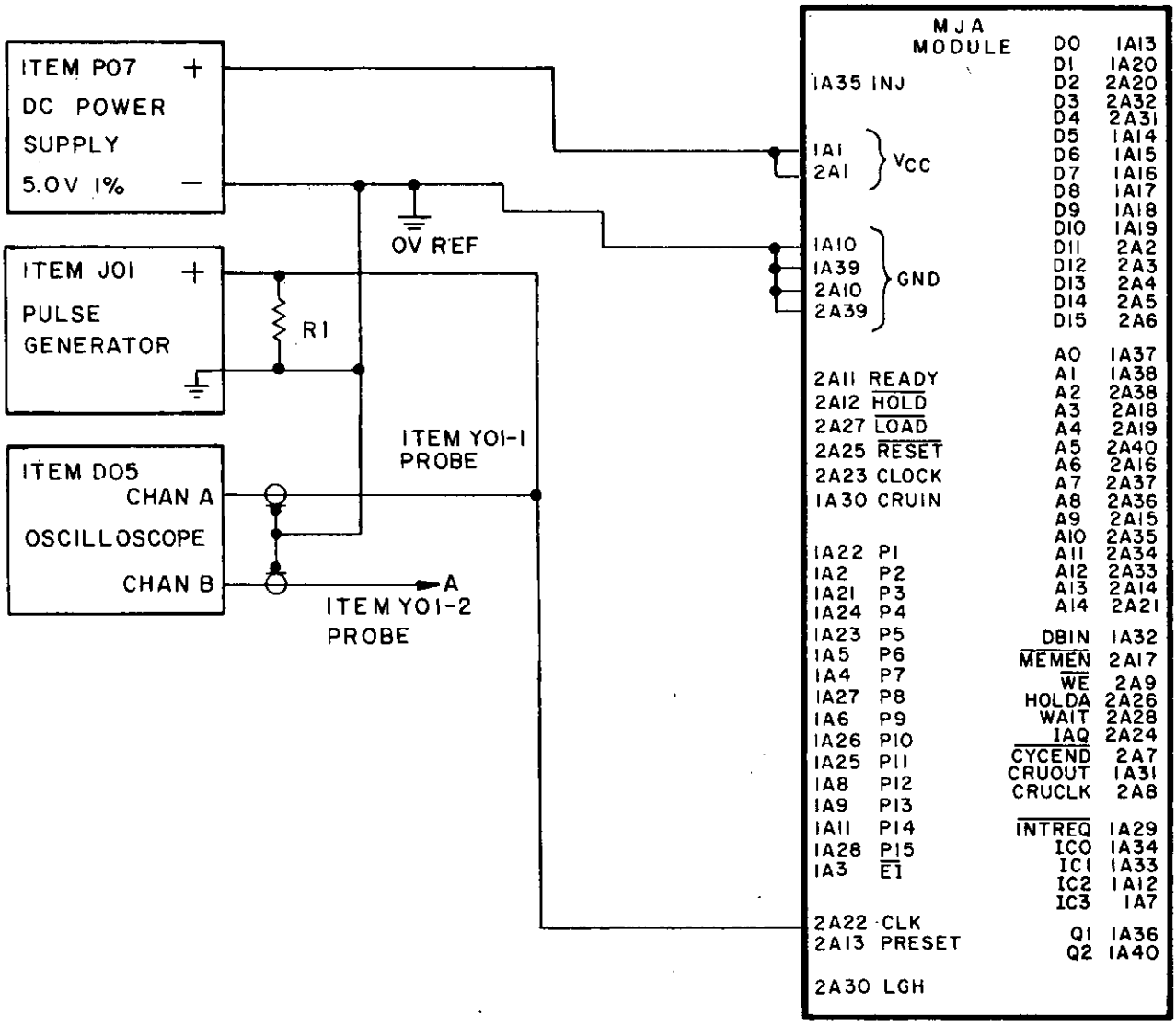
FIGURE 51. Dynamic functional (minimum and maximum injector current) test and propagation delay time (t_{PLH} and t_{PHL}) check waveforms for table XX.

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TABLE XXII. Maximum frequency (clock shaping) check test.

Test condition	Parameter	Connect A to pin	Initial limits 25°+5°C ambient Min	Initial limits 25°+5°C ambient Max	Unit	Test figure
U1	Q1 Maximum frequency (clock shaping) check, PRP2	1A36	201	207	ns	52,53
U2	Q1 Maximum frequency (clock shaping) check, PWA2	1A36	52	69	↓	↓
U3	Q2 Maximum frequency (clock shaping) check, PRP2	1A40	201	207	↓	↓
U4	Q2 Maximum frequency (clock shaping) check, PWA2	1A40	52	69	↓	↓

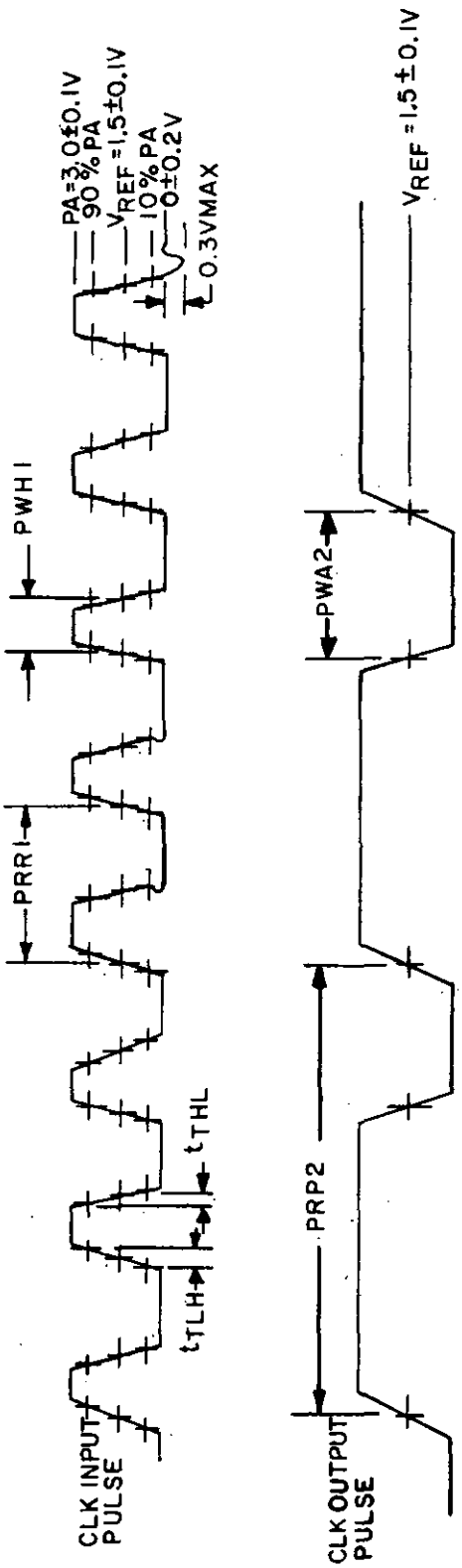
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NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. Load the outputs under test as specified on figure 22.

FIGURE 52. Maximum frequency (clock shaping) check test circuit for table XXII.

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NOTE: Input pulse parameters:
$t_{TLH} = t_{THL} = 10 \pm 2$ ns.
PRR1 = 68 +2, -0 ns.
PWH1 = 25 +2, -0 ns.

FIGURE 53. Maximum frequency (clock shaping) check test circuit waveforms for table XXII.

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TABLE XXIII. Output crosstalk (low-level and high-level) test.

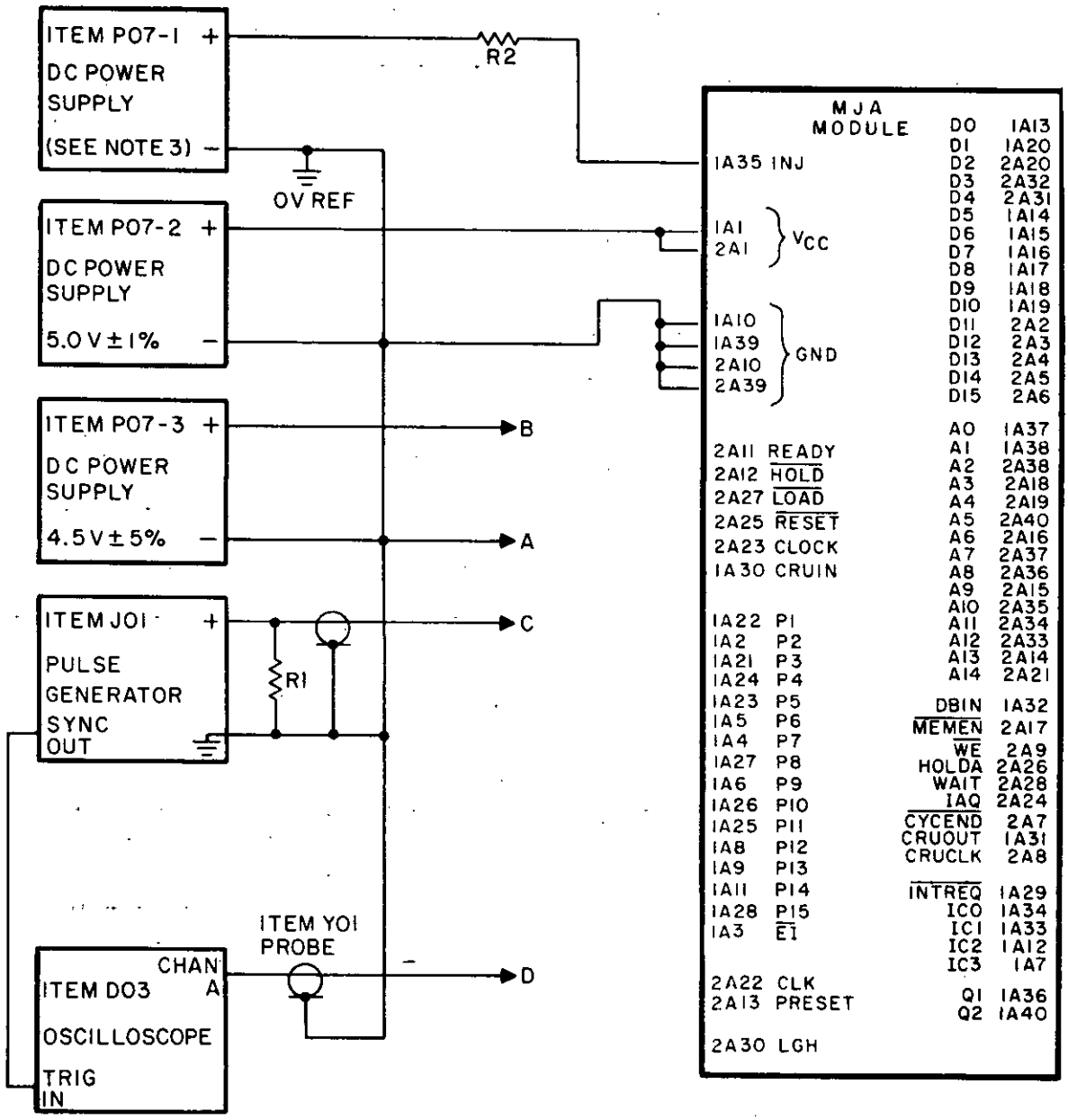
Test condition	Parameter	Connect				Initial limits 25°+5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
V1	Q1 Output crosstalk, low level	2A30	1A2, 1A4, 1A5, 1A6, 1A8, 1A9, 1A11, 1A21, 1A22, 1A23, 1A24, 1A25, 1A26, 1A27, 1A28 2A11, 2A12, 2A27	1A3	1A36	0	400	mV	54, 55
V2	Q2				1A40				54, 55
V3	DBIN				1A32				54, 55, 56
V4	HOLDA				2A26				
V5	WAIT				2A28				
V6	$\overline{\text{CYCEND}}$				2A7				
V7	IAQ				2A24				
V8	A0				1A37				
V9	A1				1A38				
V10	A2				2A38				
V11	A3				2A18				
V12	A4				2A19				
V13	A5				2A40				
V14	A6				2A16				
V15	A7				2A37				
V16	A8				2A36				
V17	A9				2A15				
V18	A10				2A35				
V19	A11				2A34				
V20	A12				2A33				
V21	A13				2A14				

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TABLE XXIII. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
V22	A14	Output crosstalk, low level	2A30	1A2,1A4,1A5,1A6, 1A8,1A9,1A11,1A21, 1A22,1A23,1A24,1A25, 1A26,1A27,1A28, 2A11,2A12,2A27	1A3	2A21	0	400	mV	54,55,56
V23	$\overline{\text{MEMEN}}$					2A17				
V24	$\overline{\text{WE}}$					2A9				
V25	CRUOUT					1A31				
V26	CRUCLK					2A8				
V27	$\overline{\text{INTREQ}}$		1A3,1A22	1A2,1A4,1A5,1A6, 1A8,1A9,1A11,1A21, 1A23,1A24,1A25,1A26, 1A27,1A28,2A30	2A22	1A29				54,55
V28	IC0					1A34				
V29	IC2					1A12				
V30	IC1					1A33				
V31	IC3		1A2,1A3	1A4,1A5,1A6,1A8,1A9, 1A11,1A21,1A22,1A23, 1A24,1A25,1A26,1A27, 1A28,2A30		1A7				
W1	$\overline{\text{INTREQ}}$	Output crosstalk, high level	Open	1A3		1A29				
W2	IC3					1A7				
W3	IC1					1A33				
W4	IC2					1A12				
W5	IC0		1A3	Open		1A34				

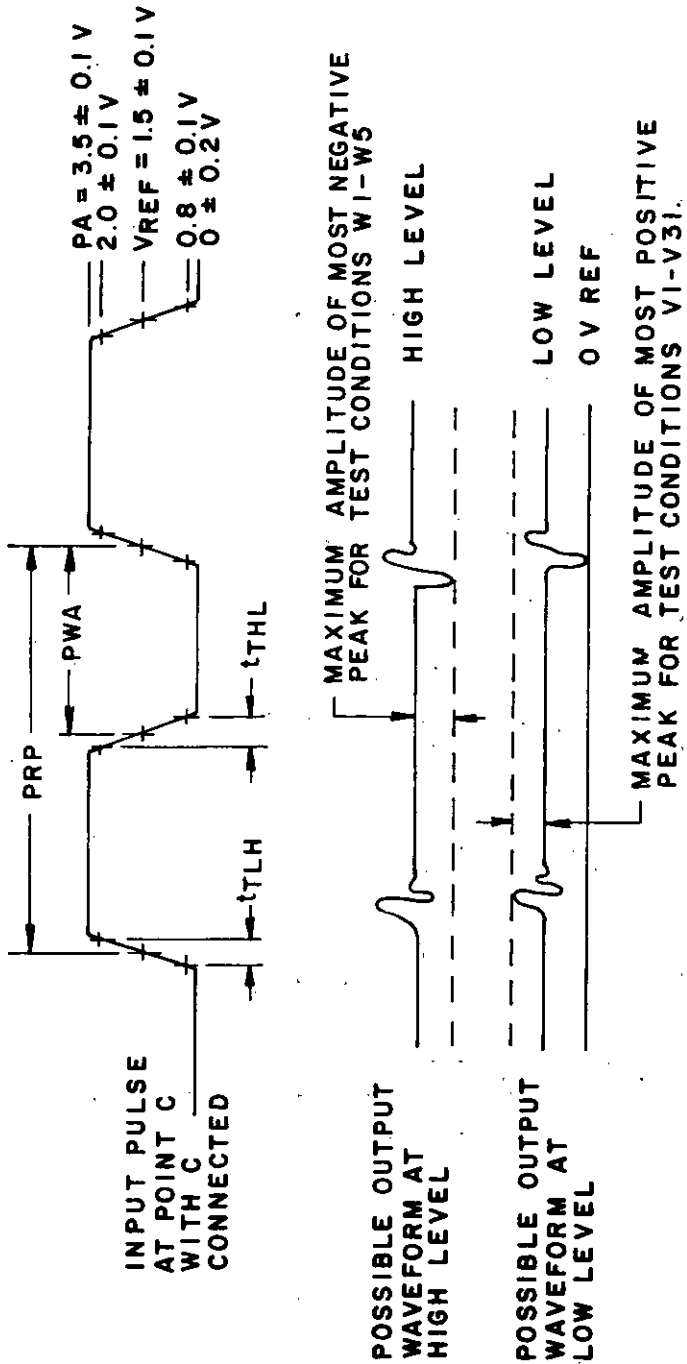
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NOTES:
1. R1 = 51Ω ±5%; 1/2 W.
2. R2 = 4Ω ±1%, 2 W.
3. Adjust item PO7-1 voltage to limit pin 1A35 current to 500 mA ±1%.

FIGURE 54. Output crosstalk (low-level and high-level) test circuit for table XXIII.

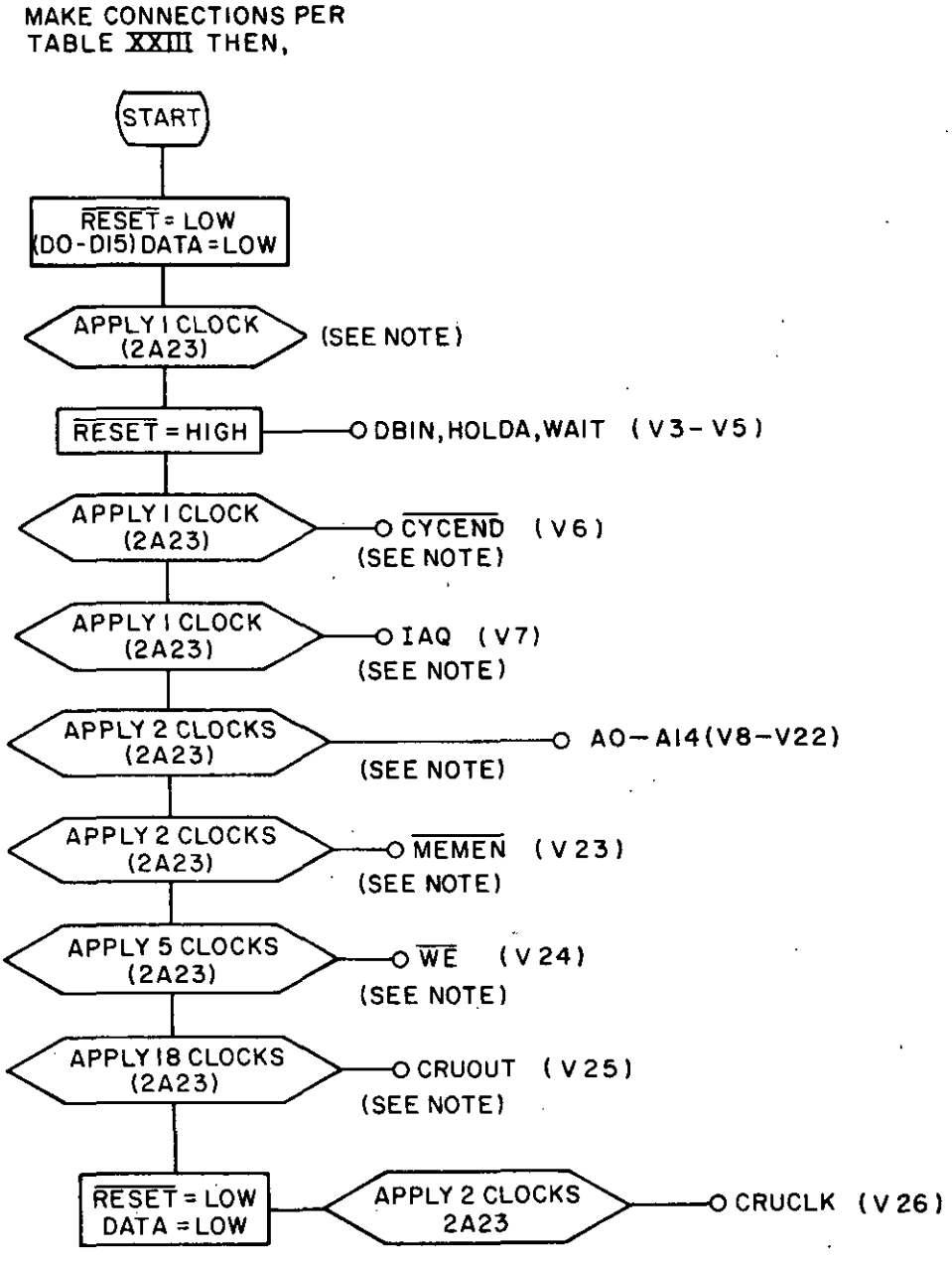
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NOTE: Input pulse parameters:
$t_{TLH} = t_{THL} = 5 \pm 5$ ns.
PWA = 100 ns ±10%.
PRP = 300 ns ±10%.

FIGURE 55. Output crosstalk (low-level and high-level) test circuit waveforms for table XXIII.

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NOTE: Use clock parameters per figure 21.

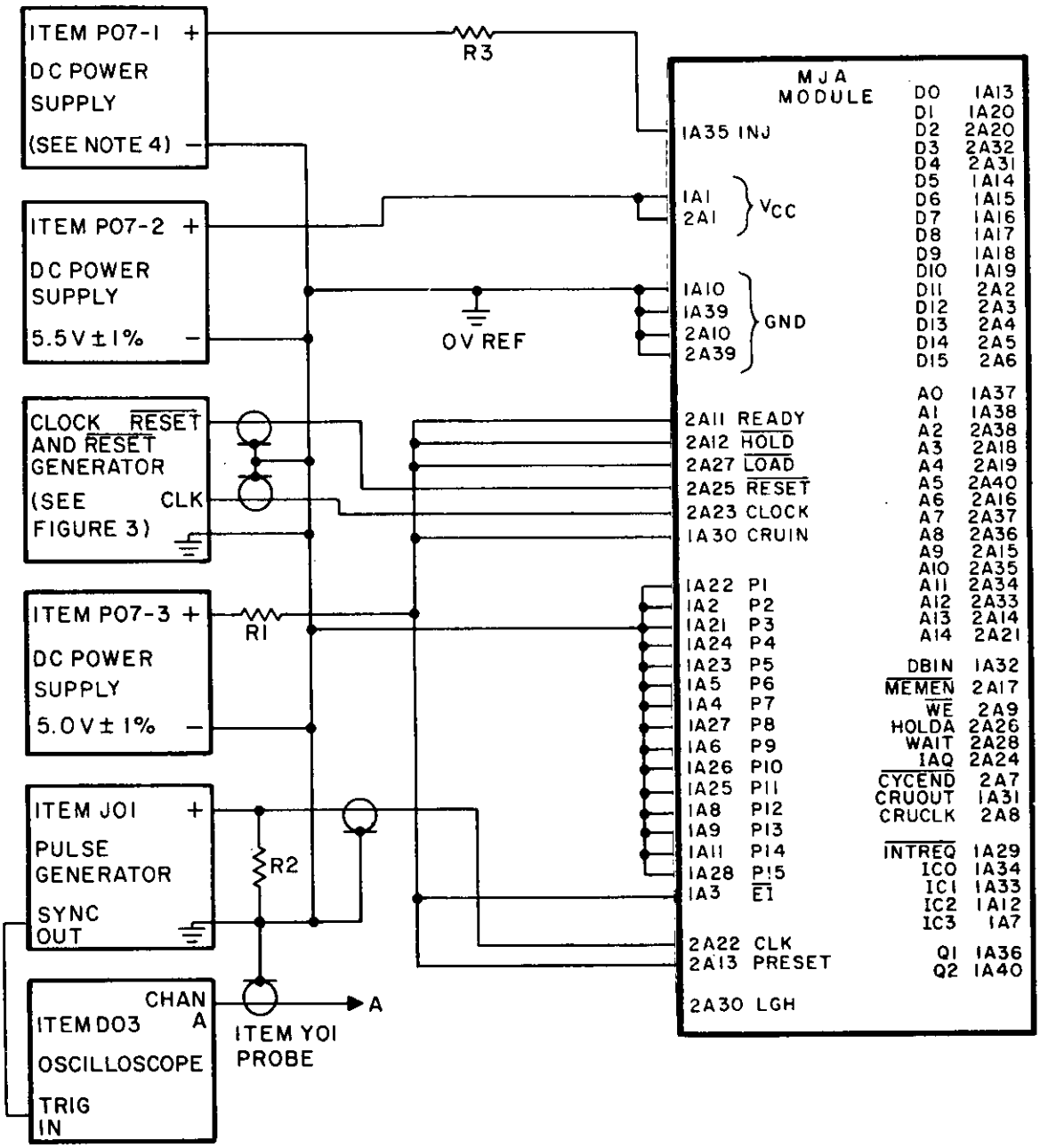
FIGURE 56. Output crosstalk (low-level and high-level) test flow chart for table XXIII.

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TABLE XXIV. Power supply transient amplitude test.

Test condition	Parameter	Connect	Initial limits 25°+5°C ambient		Unit	Test figure
		A to pin	Min	Max		
X	Power supply transient amplitude	1A1, 2A1	0	400	mV	57,58

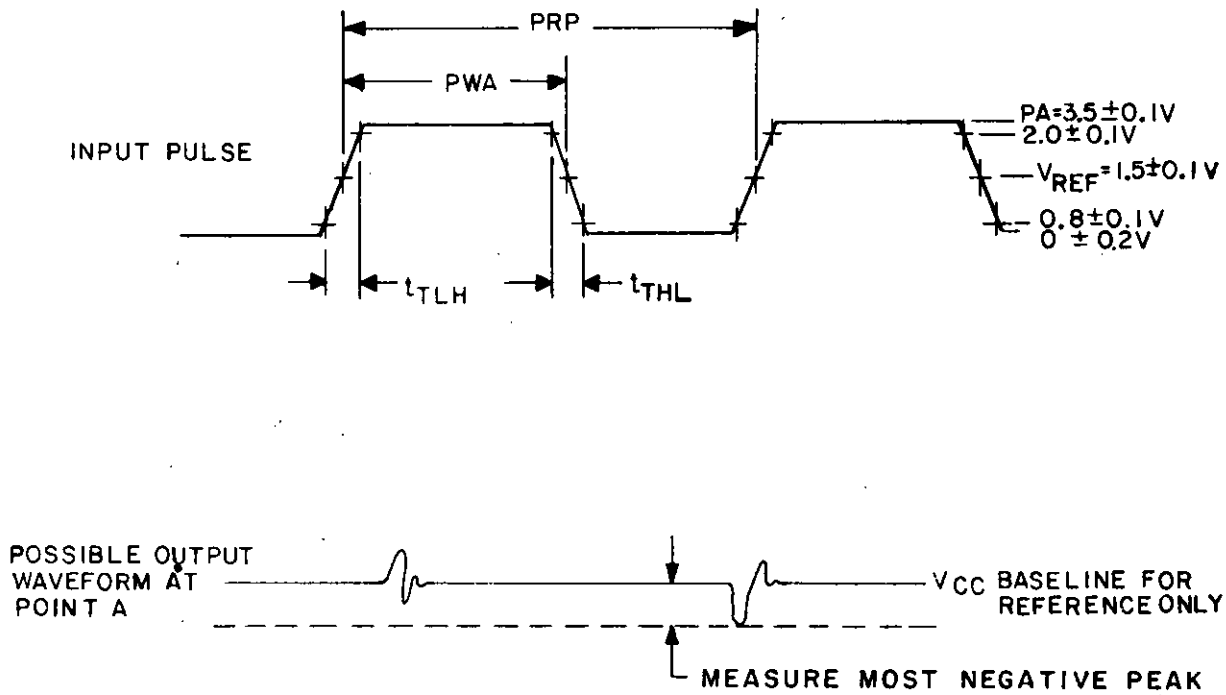
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NOTES:
1. R1 = 1.0 kΩ ±5%, 1/8 W.
2. R2 = 51Ω ±5%, 1/2 W.
3. R3 = 4Ω ±1%, 2 W.
4. Adjust item P07-1 voltage to limit pin 1A35 current to 550 mA ±1%.

FIGURE 57. Power supply transient amplitude test circuit for table XXIV.

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NOTE: Input pulse parameters:
$t_{TLH} = t_{THL}$ = 15 ±5 ns.
PWA = 100 ns ±10%.
PRP = 300 ns ±10%.

FIGURE 58. Power supply transient amplitude test circuit waveforms for table XXIV.

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TABLE XXV. Power supply and input capacitance test.

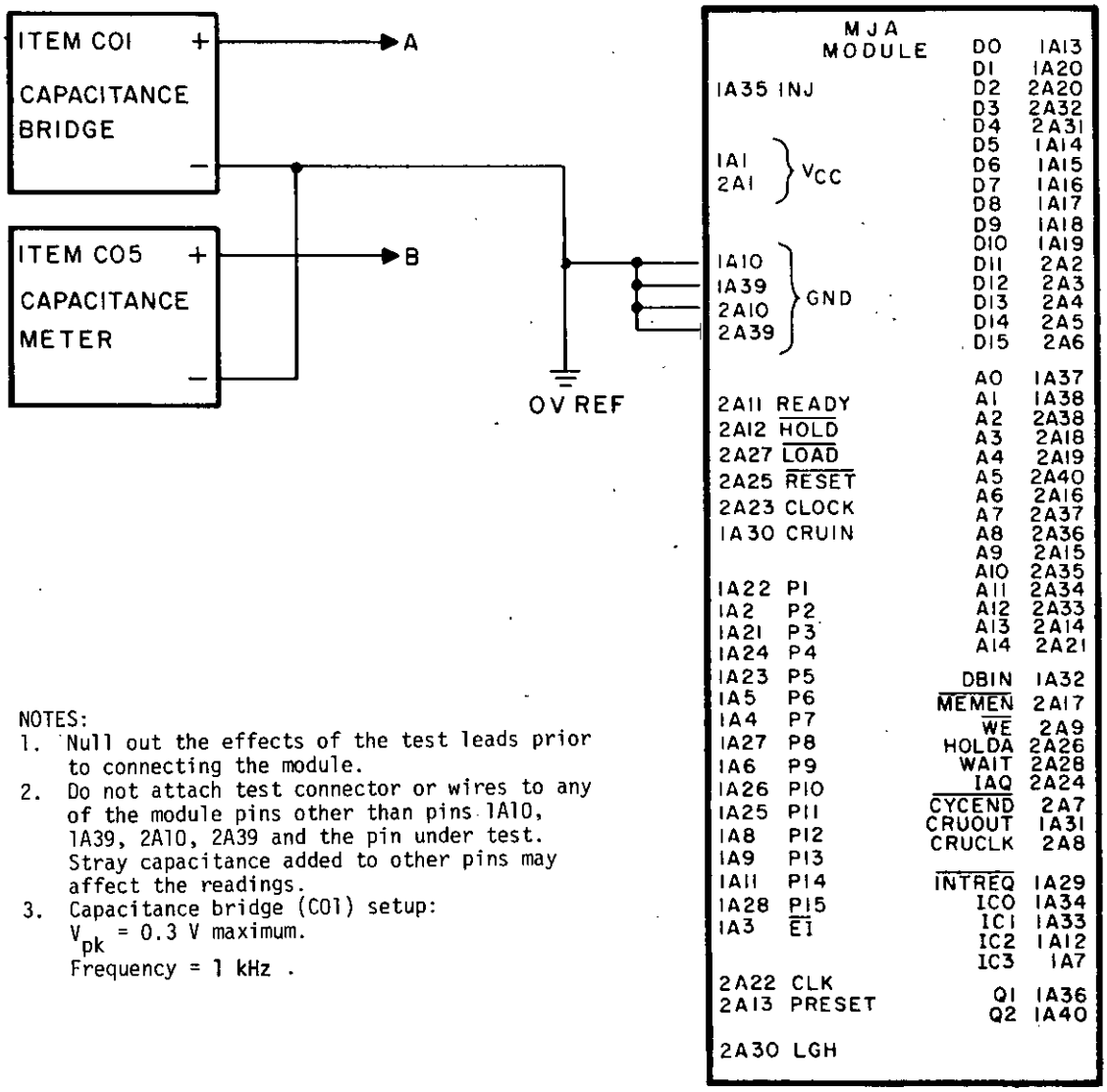
Test condition	Parameter	Connect		Initial limits 25°+5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Y1	V_{CC} Power supply capacitance	1A1, 2A1	Open	3.22	5.98	µF	59
Y2	INJ Power supply capacitance	Open	1A35	200	1500	pF	
Z1	P1 Input capacitance		1A22	2	20		
Z2	P2		1A2				
Z3	P3		1A21				
Z4	P4		1A24				
Z5	P5		1A23				
Z6	P6		1A5				
Z7	P7		1A4				
Z8	P8		1A27				
Z9	P9		1A6				
Z10	P10		1A26				
Z11	P11		1A25				
Z12	P12		1A8				
Z13	P13		1A9				
Z14	P14		1A11				
Z15	P15		1A28				
Z16	$\overline{EI}$		1A3				

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TABLE XXV. Power supply and input capacitance test - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Z17	CLK Input capacitance	Open	2A22	2	40	pF	59
Z18	PRESET		2A13		40		
Z19	LGH		2A30		40		
Z20	READY		2A11		20		
Z21	$\overline{\text{HOLD}}$		2A12				
Z22	$\overline{\text{LOAD}}$		2A27				
Z23	$\overline{\text{RESET}}$		2A25				
Z24	CLOCK		2A23				
Z25	CRUIN		1A30				

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pins 1A10, 1A39, 2A10, 2A39 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Capacitance bridge (C01) setup:
   V_{pk} = 0.3 V maximum.
   Frequency = 1 kHz .

FIGURE 59. Power supply and input capacitance test circuit for table XXV.

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